

Received 10 November 2023, accepted 20 December 2023, date of publication 22 December 2023,
date of current version 29 December 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3346299

SURVEY

Internet of Things and Wireless Sensor Networks for Smart Agriculture Applications: A Survey

MD. NAJMUL MOWLA¹, NEAZMUL MOWLA², A. F. M. SHAHEN SHAH³, (Senior Member, IEEE),
KHALED M. RABIE^{4,5}, (Senior Member, IEEE), AND
THOKOZANI SHONGWE⁵, (Senior Member, IEEE)

¹Department of Aerospace Engineering, Adana Alparslan Türkeş Science and Technology University, 1250 Adana, Turkey

²Department of Computer Science and Engineering, North South University, Dhaka 1229, Bangladesh

³Department of Electronics and Communication Engineering, Yıldız Technical University, 34349 Istanbul, Turkey

⁴Department of Engineering, Manchester Metropolitan University (MMU), M15 6F Manchester, U.K.

⁵Department of Electrical and Electronics Engineering, University of Johannesburg, Johannesburg 1809, South Africa

Corresponding author: Md. Najmul Mowla (najmulmowla01@gmail.com)

ABSTRACT The increasing food scarcity necessitates sustainable agriculture achieved through automation to meet the growing demand. Integrating the Internet of Things (IoT) and Wireless Sensor Networks (WSNs) is crucial in enhancing food production across various agricultural domains, encompassing irrigation, soil moisture monitoring, fertilizer optimization and control, early-stage pest and crop disease management, and energy conservation. Wireless application protocols such as ZigBee, WiFi, SigFox, and LoRaWAN are commonly employed to collect real-time data for monitoring purposes. Embracing advanced technology is imperative to ensure efficient annual production. Therefore, this study emphasizes a comprehensive, future-oriented approach, delving into IoT-WSNs, wireless network protocols, and their applications in agriculture since 2019. It thoroughly discusses the overview of IoT and WSNs, encompassing their architectures and summarization of network protocols. Furthermore, the study addresses recent issues and challenges related to IoT-WSNs and proposes mitigation strategies. It provides clear recommendations for the future, emphasizing the integration of advanced technology aiming to contribute to the future development of smart agriculture systems.

INDEX TERMS

Internet of Things, wireless sensor networks, wireless network protocols, smart agriculture applications.

I. INTRODUCTION

In the agricultural sector, there has been substantial technological progress, integrating advanced innovations such as the Internet of Things (IoT), Wireless Sensor Networks (WSNs), Wireless Network Protocols, Unmanned Aerial Vehicles (UAVs), Artificial Intelligence (AI), Agricultural Robotics, Big Data Analytics, and Blockchain systems. The increasing global adoption of IoT systems signifies an evolution towards innovative approaches, utilizing device-generated data to enhance productivity. IoT enables connections between machines and humans on a broader scale,

primarily facilitating real-time information sharing across independent networks. Within this framework, real-time data captured by intelligent computational sensors can be effortlessly transmitted to people worldwide via the internet, regardless of time or location.

Agriculture, a cornerstone of economic development, forms a significant portion of a developing country's gross domestic product (GDP) [1]. The imminent food crisis, exacerbated by rapid population growth, necessitates urgent measures to enhance production and meet the rising demands [2]. These challenges—food crisis and population growth—pose threats to preserving the agricultural chain. The global population is projected to reach 8.5 billion by 2030 and nearly 10 billion by 2050 [3], [4]. Researchers

The associate editor coordinating the review of this manuscript and approving it for publication was Barbara Masini¹.

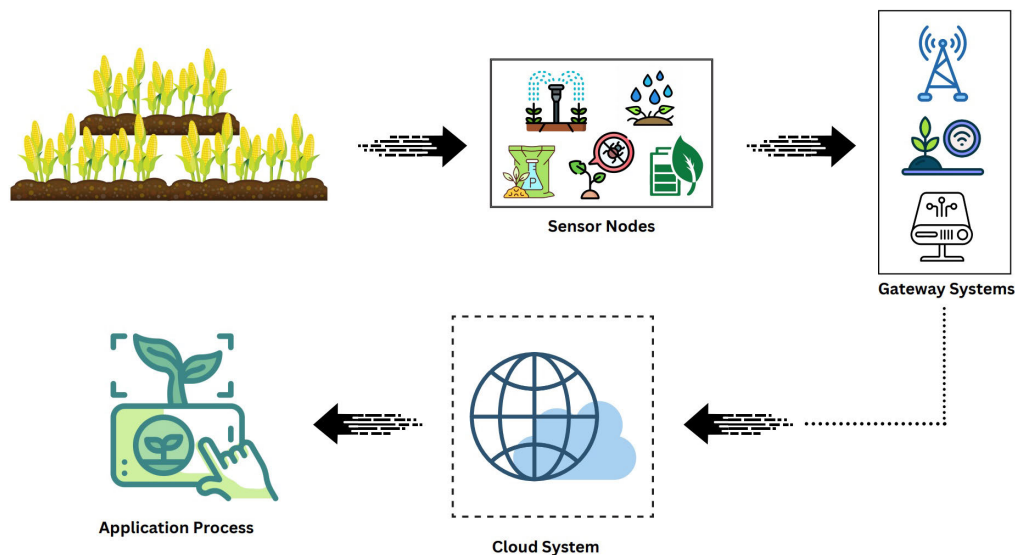


FIGURE 1. A process of IoT-based SA system.

actively explore and leverage advanced technologies such as WSNs and the IoT to boost agricultural productivity [5]. The integration of IoT with WSNs has revolutionized the agricultural sector, improving production efficiency and resource distribution, especially in SA [6]. The global IoT market is poised for significant expansion, offering new opportunities for integrating agricultural applications, including irrigation, soil monitoring, pest control, and greenhouse environmental monitoring [7], [8]. Beginning with an initial valuation of 18.12 billion in 2021, the market witnessed a remarkable surge to 91.91 billion in 2022, followed by a precipitous decline to 21.89 billion in 2023. Projections suggest a sustained upward trend, reaching a valuation of 43.37 billion by 2030. This trajectory is supported by the anticipated compound annual growth rate of 10.2% forecasted between 2022 and 2030. [9]. Sensor-based devices facilitate data collection and analysis, empowering informed decision-making for farmers, a crucial aspect in rural areas with limited power supply.

Recent WSNs have evolved, enriching human lives through numerous applications, promising essential convenience [10]. They have become a widespread and expanding network, finding broad utility in agriculture, environmental monitoring, industrial automation, and transportation systems [11]. Within the evolving landscape of WSNs, a key aspect is the global connectivity of access points, linking to many wireless sensor nodes scattered across diverse locales, diligently gathering data on a spectrum of physical parameters [5], [12]. Acquiring such data from remote locations will be as effortless as managing cellular data online [13]. At the heart of this intricate IoT framework lies its central monitoring infrastructure, WSNs [14]. Current research increasingly focuses on ensuring the sustainability of IoT-WSNs, given their inherent resource limitations [8].

In the context of SA, integration, and interaction among intelligent entities are propelled by IoT, marking a transformation in a technologically advanced era [7], [25], [26], [27]. This integration includes IoT combined with WSNs, encompassing soil-embedded sensors for environmental monitoring. It features diverse transceivers, microcontrollers, and communication protocols for efficient data transmission, fundamental for control and monitoring. A vital component of this integration is WSN, embracing many sensors and showcasing substantial advancements within SA [28]. However, the rapid proliferation of sensor devices raises concerns about energy consumption, prompting focused research [11]. Given the vast agricultural landscape, efficient and extended operation of sensor nodes is imperative to meet specific application requirements. IoT-WSNs have become crucial, facilitating data provision to other layers and technological advancements. They are designed with multiple innovative battery-powered nodes, employing wireless connectivity for seamless communication.

These nodes, enriched with advancements in Microelectromechanical systems (MEMS), now have a smaller profile, reduced cost, and enhanced energy efficiency. They are strategically positioned across the landscape, diligently collecting data from the farm and its surroundings, encompassing vital metrics such as soil moisture, temperature, humidity, and crop well-being. In addition, embedded microcontrollers within these nodes allow for essential data processing. The harvested data and insights reach a central hub or base station through direct and indirect transmission. This dynamic flow of information brings about a marked enhancement in the decision-making apparatus shown in Figure 1. WSNs offer substantial advantages and competencies, especially in elevating monitoring precision, automation prowess, and optimization finesse. This surge in potential has prompted

TABLE 1. A comparative analysis of IoT-WSNs in the present and existing survey papers in the field of SA applications.

Year	References	Architectures	WNP	Survey	AI	AGI	Robotics	UAVs	BDA	5G-6G	Blockchain	Renewable Energy	Privacy & Challenges
2019	[15]	×	✓	×	✓	×	×	✓	×	×	×	✓	✓
2019	[16]	×	×	×	×	×	×	×	×	✓	×	×	✓
2020	[17]	✓	×	×	×	×	×	×	×	×	×	×	✓
2020	[18]	✓	✓	×	✓	×	×	×	✓	✓	×	×	✓
2021	[19]	✓	✓	×	✓	×	×	✓	✓	×	×	×	✓
2021	[20]	✓	✓	×	✓	×	✓	✓	✓	×	✓	×	✓
2022	[21]	✓	×	×	✓	×	✓	✓	✓	×	✓	×	✓
2022	[22]	×	✓	×	✓	×	×	✓	✓	✓	✓	×	✓
2023	[23]	✓	✓	×	✓	×	✓	✓	✓	×	×	×	✓
2023	[24]	✓	×	×	✓	×	×	✓	✓	×	×	×	✓
2023	This study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

numerous scholars to delve into its promising applications within agriculture.

Integrating applications with WSNs in agriculture offers significant advantages, covering various aspects such as irrigation control, plant disease and pest monitoring, fertilizer optimization, autonomous agricultural machinery, and UAVs for crop monitoring. These technological advancements synergize to drive agricultural automation, with the overarching objective of enhancing the long-term sustainability of food production in the agricultural sector. Key components such as land evaluation, crop protection, and yield forecasting are crucial in ensuring global food production. Real-time monitoring of field conditions and effective agricultural field management are made possible through wireless sensors and mobile networks. Additionally, technology empowers farmers to gather critical data and create accurate yield maps, facilitating precision agriculture and the production of high-quality crops at an economical cost using WSNs.

WSNs with IoT play a fundamental role in SA by providing essential information for specific systems and applications. A condensed overview of the main aspects includes:

- **Irrigation Systems:** WSNs actively monitor water usage, enabling farmers to manage water resources effectively and prevent excessive irrigation. Furthermore, they facilitate real-time monitoring of soil moisture content, allowing farmers to identify areas with drainage issues or sub-optimal irrigation distribution.
- **Soil Moisture Monitoring Systems:** WSNs continuously measure and transmit soil moisture data at various depths, empowering farmers to optimize irrigation practices and mitigate under and over-watering challenges.
- **Fertilizer Optimization and Control:** WSNs are valuable tools for monitoring soil nutrient levels and providing real-time information regarding the soil's nutritional status. Through thorough soil data analysis, WSNs assist in fine-tuning fertilizer applications by offering precise recommendations on quantities and optimal timings.
- **Early Stage Control of Pest and Crop Diseases:** WSNs gather essential data on temperature, humidity, and other factors affecting pest and disease development. By enabling early detection and timely intervention,

this data equips farmers with the necessary insights to prevent or minimize crop damage.

- **Energy Saving and Power Consumption:** WSNs facilitate monitoring energy consumption within agricultural operations, including irrigation systems. This monitoring capability empowers farmers to identify potential areas for energy conservation and optimize power usage, leading to adopting more sustainable and efficient farming practices.

In the agricultural domain, IoT-WSNs collect precise and extensive real-time field data. Various communication protocols, including ZigBee, WiFi, SigFox, and LoRaWAN, find typical applications in SA. These technologies significantly contribute to sustainable and precise agricultural production by ensuring long-distance coverage, low data packet loss, low power consumption, and effective connectivity. In SA, leveraging IoT-WSNs and wireless communication protocols is essential. Sensors strategically positioned across the agricultural landscape collect critical soil moisture, temperature, and humidity data. These sensors connect to nodes equipped with the wireless communication protocol, forming an IoT-WSN. The collected data undergoes analysis to enable well-informed decision-making, facilitating precise resource allocation, automated irrigation, and timely actions to optimize crop growth. This integrated IoT-WSN strategy significantly enhances agricultural practices, ultimately promoting sustainable and efficient farming operations in the long run.

The IoT-WSN system represents a fusion of state-of-the-art technologies, integrating AI and its subsets to enhance operational efficiency within SA systems. Machine Learning (ML) and Deep Learning (DL) are particularly prevalent in IoT applications, especially in the SA domain. Ongoing research efforts are focused on refining and streamlining this architecture for optimal integration into SA applications. Since IoT systems operate entirely online, privacy concerns emerge as a critical consideration. A robust big data system in SA is critical, enabling data-driven decision-making for farmers and stakeholders. While blockchain systems effectively address data privacy concerns, further advancements are essential. Current research findings are significant, offering valuable insights, illuminating existing

gaps, and inspiring the conception of innovative technologies for future SA research.

Additionally, farmers are interested in adopting novel, advanced, and cost-effective systems into their agricultural operations, highlighting the importance of cost-effectiveness in IoT technology amalgamating modern systems. In this context, comprehensive reviews and survey papers are key in observing research gaps, presenting a proactive technological roadmap, and guiding researchers toward effectively advancing the SA system.

In recent years, there has been a significant increase in reviews and survey papers focusing on WSNs-based SA systems. Table 1 presents a detailed comparative analysis, comparing the present review study and existing survey-review publications. This analysis thoroughly investigates past research contributions across four key sections: IoT-WSNs and their communication protocols for SA systems, integration of AI with WSNs in SA applications, leveraging Artificial General Intelligence with IoT-WSNs in SA applications, and critical factors in WSNs-based SA applications. The review extensively delves into IoT-WSN architectures and their associated wireless network protocols, comprehensively assessing their applications across five prevalent SA domains since 2019. This exploration emphasizes privacy considerations and challenges in integrating IoT-WSNs into SA practices. Furthermore, the study evaluates future trends and potential applications of WSNs in conjunction with emerging technologies, aiming to enhance the efficiency and productivity of SA practices by leveraging IoT-WSNs alongside state-of-the-art technologies. The contribution of this research is summarized as follows:

- I. This research extensively examines existing literature related to SA, focusing on the IoT, WSNs, and wireless communication technologies. The study involves a thorough analysis, categorizing articles by publication year since 2019, IoT-WSNs application area, and network protocols. The analysis sheds light on reviews of IoT-WSN architectures and their associated network protocols for SA systems.
- II. Examination of wireless communication protocols for SA, including Zigbee, Wi-Fi, SigFox, and LoRaWAN, emphasizes their characteristics and practical applications.
- III. A comprehensive survey discusses five key applications within SA: irrigation systems, soil moisture monitoring, fertilizer optimization, pest and crop disease control, and energy optimization. Additionally, the survey explores integration with wireless communication protocols since 2019.
- IV. This study thoroughly explores the discussion regarding current challenges and open issues in SA technology. This involves addressing concerns about IoT-WSNs scalability and reliability, data privacy and confidentiality, network security and intrusion detection, data integrity and authenticity, user privacy and consent,

resilience to attacks and failures, location privacy, power consumption, and cost and standardization. Potential solutions to these challenges are also thoroughly examined.

- V. Furthermore, an extensive future recommendation encompassing advanced architectures involving AI and AGI systems, agri-robotics, Big Data, blockchain analytics, 5G and 6G, and renewable energy that can be utilized for SA applications are addressed. These advancements hold the potential to create new opportunities for sustainable, cost-effective, and user-friendly agricultural systems for future farmers.

The subsequent sections of this paper are organized as follows: Section II provides an extensive review of the existing literature. Section III discusses the architecture of IoT and WSN for SA, encompassing its layers. In addition, Section IV elucidates wireless communication protocols for the WSN SA application and their respective properties. Furthermore, Section V delves into IoT-WSNs with network protocols for SA applications. Afterword, Section VI comprehensively addresses the current challenges and limitations of IoT-WSNs. Following this, Section VII explores future trends and research opportunities entailing advanced technology integration with IoT-WSNs. Finally, Section VIII encapsulates the conclusion, summarizing the study's findings and insights.

II. RELATED WORK

The advent of the WSN produced a new direction of research. However, with a rapidly increasing number of studies based on essential data information, relevant studies must be significantly studied and reviewed. Khanna et al. conducted an in-depth review of existing challenges within agricultural operations while simultaneously projecting potential research advancements for emerging scholars. This approach aims to enhance their comprehension of the constantly evolving landscape. The article meticulously delineates its origins and evolution, explaining its growth and advancements. The study follows a systematic structure, commencing with foundational principles and progressively delving into functional intricacies while acknowledging the constraints and impediments in the agricultural domain. The findings present thorough and precise observations, examining precision agriculture using IoT and WSNs, aligning diligently with contemporary market imperatives. Furthermore, prospective research directions in IoT for precision agriculture are outlined, demonstrating thoughtful consideration of impending challenges [16]. Shi et al. presented an organized literature review on IoT research and implementations within protected agriculture spanning the last decade. Their analysis enclosed an assessment of the different inputs from various scholars and entities. The study thoroughly reviewed diverse SA applications involving WSNs and sophisticated agricultural systems. Additionally, they explored the hurdles and potential directions for future research in this domain [29].

Kumar et al. thoroughly examined and presented a study on integrating IoT and WSNs for SA applications. The study emphasized the design of intelligent techniques to enhance agricultural processes, alleviating the instrumental role of WSNs in advancing the SA. Additionally, they meticulously explored a wide range of applications of WSN and IoT within the agricultural domain, with a particular emphasis on sensors utilized for crop management in the context of precision agriculture [30]. Kour et al. proposed and reviewed the role of advancing IoT technologies in SA systems. Their research focuses on enhancing both hardware and software systems. The scholarly article thoroughly investigates various projects in the public and private sectors, aspiring to offer sustainable and intelligent solutions for agriculture. The study provided a detailed summary of the current scenario, applications, potential research areas, limitations, and prospects in this SA field [17].

Tao et al. summarised scientifically validated publications on IoT communication technologies in SA to address this matter. This study applied a detailed investigation from ScienceDirect, IEEE Xplore, and Scopus platforms. 94 research articles were inspected after the 886 titles were reviewed for relevancy [31]. Abdollahi et al. introduced a review paper and demonstrated the contribution of WSN in agriculture applications based on current academic literature. In this study, they applied bibliometric techniques; 2444 publications were extracted from the Scopus database and examined to specify the temporal distribution of WSN research, the most productive journals, the most cited authors, the most influential studies, the most relevant keywords [32].

Ayşar et al. introduced a research initiative that delves into the remarkable features of wireless communication protocols such as ZigBee, Wi-Fi, Sigfox, NB-, and LoRaWAN. The study outlines the technical attributes and real-world applications of these protocols commonly employed in IoT applications and examines variations in technical specifications reported across different sources. Furthermore, the study addresses IoT wireless communication protocol's privacy implications and future trends in SA applications [33]. Pandey et al. have comprehensively reviewed available IoT solutions, including soil health monitoring, crop health monitoring, smart irrigation, and real-time weather forecasting in the SA domain [34]. Gulati et al. reviewed the literature with a precise awareness of WSN for energy preservation and collection of data [35]. IoT and WSN constantly demand internet connection; therefore, this architecture has a security issue. Sinha et al. presented and reviewed the recent issues and challenges of IoT-based SA applications [21]. Similarly, Xu et al. reviewed and summarised the present crisis of IoT in the SA domain xu2022review. ML and DL, a subset of AI, are also commonly applied in SA. It helps to prevent the loss of agricultural yield. Rahaman et al. proposed a review paper based on WSNs and ML and their applications, issues, and challenges in SA platforms [36].

Sethi et al. discussed and reviewed the architecture of IoT and WSNs in the context of SA applications. The application of agricultural IoT in various sectors is also addressed in this study. They explored the existing challenges within agricultural IoT and provided a forecast for its future growth. The key aspects covered in their work encompassed the progress and design of IoT with WSNs, novel sensors, SA applications, and the utilization of data for screening plant and animal life [37]. Pathmudi et al. conducted an extensive literature review, focusing on essential technologies for enabling SA architectures. They precisely compared and provided insights into various components such as sensors, controllers, communication standards, and advanced machinery. These sensors continuously collected substantial data from agricultural fields, then transmitted to a centralized control unit for comprehensive analysis, addressing the specific needs for water, fertilizer, pesticides, and more. The study emphasized elucidating the architecture and significance of data analytics in agricultural IoT with WSNs. Furthermore, the research deeply analyzed critical challenges and unresolved issues about agricultural IoT technology [1]. Adli et al. undertook a comprehensive literature review on integrating AI and the IoT in the SA domain. The objective was to explain the current strides, applications, and advantages in the context of SA. The exploration involved a deep dive into AI and IoT, encompassing the utilization of smart devices within IoT frameworks and the application of AI methodologies. Eventually, the research integrated into the challenges that hinder the effective implementation of AI in IoT technology for SA, offering valuable insights into potential improvements and solutions [38].

This study's primary objective was to explore academic articles between 2019 and 2023 focusing on IoT-WSN applications in SA. The inclusion criteria encompassed openly accessible research papers, well-cited scientific publications, and the latest research-based papers. The survey covered a wide spectrum, including journal articles and conference papers. The approach involved a comprehensive literature review to provide a landscape overview. The process comprised searching, selecting relevant studies, and conducting a detailed analysis. The central research questions guiding this review were:

- I. The role of IoT-WSN architectures in SA applications.
- II. Integration of IoT-WSNs with wireless communication in SA applications.
- III. Leveraging advanced technology using IoT-WSNs for SA.
- IV. Challenges faced by IoT-WSN architecture in fulfilling this purpose.

To compile related work addressing these research questions, keyword-based searches were executed across reputable databases such as Google Scholar, Web of Science, IEEE Xplore, Scopus, ScienceDirect, Multidisciplinary Digital Publishing Institute (MDPI), and Springer. The emphasis was on journal articles and conference papers. A carefully devised search term with relevant keywords was employed.

To streamline the literature review, clear inclusion and exclusion criteria were established. Articles must be published in English, directly addressing the study objectives, and indexed in selected databases. The search engine was optimized using key phrases such as “IoT for SA,” “WSNs for SA,” “IoT-WSNs in agriculture,” and related variations, which had to appear in the title, abstract, or anywhere within the document.

III. ARCHITECTURE OF IoT AND WSNs FOR SA APPLICATIONS

WSNs integrated with IoT technologies have rapidly advanced across various agricultural domains. The IoT system serves as a network wherein physical devices, machinery, sensors, and objects communicate seamlessly without requiring human intervention. Within this framework, WSNs are crucial, extending their influence across many real-time agricultural applications. These IoT and WSNs are structured with multiple layers in the SA application, illuminated comprehensively in the subsequent subsections. The overview of IoT and WSN architectures is shown in Figure 2 and 3.

A. IoT ARCHITECTURES

The emergence of IoT is an essential feature in contemporary agriculture systems, driven by the rapid progression of agricultural technology [39], [40]. IoT encompasses several critical segments, including sensors and devices, connectivity, action and automation, and user interface and interaction [1], [41]. These elements gather data from agricultural yields, allowing farmers to make informed decisions. Additionally, these elements are connected within various layers of IoT architectures for SA applications. However, determining an architecture for -based SA poses challenges due to the extensive potential scale and specific requirements, such as soil conditions, weather dynamics, and geographical variations [13], [42], [43].

Moreover, integrating IoT devices and systems into agricultural practices necessitates diligent efforts [44]. These efforts encompass acquiring IoT devices and utilizing various protocols and standards to ensure seamless compatibility and integration. In agricultural IoT configurations, sensors and devices generate substantial data, heightening the complexity of real-time data management, processing, and analysis. The designated framework for this purpose should efficiently facilitate structured data storage, efficient processing, and robust analytical capabilities [45]. The agricultural sector emphasizes precise and timely data, highlighting the necessity of ensuring the selected architectural structure is trustworthy and robust. It must have the capacity to maintain continuous collection and processing of data, even under adverse environmental circumstances or in the presence of challenges with network connectivity [46], [47].

On the other hand, ensuring a balanced integration of envisioned architectural features, functionalities, and the designated budget is essential. This is especially critical for

small-scale farmers or financially constrained organizations. At the beginning of rapid advancements in IoT technologies, it is significant to underscore the importance of selecting an adaptable architectural framework [48], [49]. This architecture should incorporate emerging technologies and standards, ensuring the enduring relevance and effectiveness of the IoT-based SA system without interruption. Emphasizing the need for this strategic preference ensures the system's sustained applicability and efficiency.

The architecture of IoT technologies is employed based on various methodologies, varying from application to application [50], [51], [52]. This results in diverse design and deployment patterns. Instead, it must be customized based on specific needs. Typically, architecture is structured in a framework comprising three, four, and five layers for SA applications [26], [49], [50], [52], [53], [54], [55], [56]. Originating from a hierarchical framework of three to five layers, the IoT architecture is conventionally described, featuring primary and most common layers, including the perception layer, connectivity layer, and application layer for the SA application [20], [33], [50], [57]. These layers are also mentioned as the lower layer in the IoT architecture [21]. Furthermore, the other layers are the middleware and processing layer [44], [58], [59]. In the below subsection, we described the primary and main layer of IoT architecture for the SA system.

1) PERCEPTION LAYER

The perception layer, also known as the physical layer, is a crucial component in the IoT framework [13], [60]. It operates as a robust interface, enabling appropriate interaction between the physical and digital domains [33]. This layer is essential for immediately collecting diverse data from sensors and devices [44]. It encompasses critical environmental parameters such as weather conditions, wind flow, humidity, etc. For example, Zeng et al deployed an ultrasonic water level gauge for measuring the water level of a smart irrigation system [57]. The perception layer within the SA application presents typical challenges due to complex requirements during crop and environmental monitoring, especially in unfavorable conditions [60]. Enhancing the energy and communication infrastructure in agricultural fields is crucial, and using wired power and communication channels to connect IoT nodes is not practical or cost-effective. Data collection processes have progressed, integrating various tools. For example, sensors and cameras use Bluetooth, wireless networks, and short-distance wireless and wired transmission methods to transmit data to the central gateway [1], [61]. The sensing layer employs relevant devices to convert biological data into web-accessible information, constituting a foundational step in network control.

2) CONNECTIVITY LAYER

The communication layer, recognized as the network and transport layer, facilitates uninterrupted communication and data transfer across diverse devices, constituting the

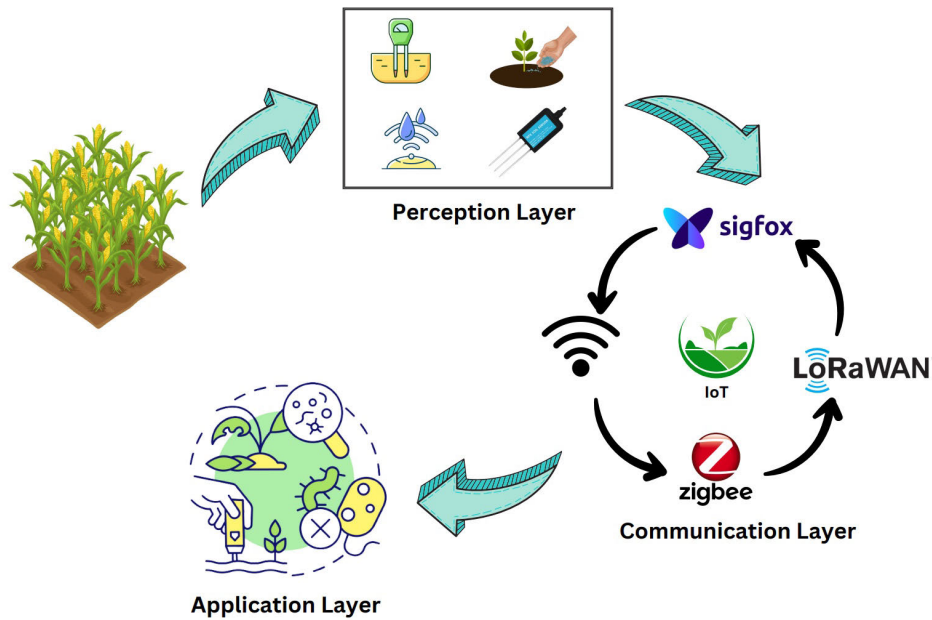


FIGURE 2. Architecture of IoT for SA application systems.

fundamental framework of IoT architectures [62], [63], [64], [65]. A profound understanding of the sophistication of the communication layer is critical to optimizing IoT networks, fostering scalability, resilience, and secure data exchange, thus propelling the potential and applications of IoT technologies across various domains. As the key element of the entire system, this layer delivers data transmission from the perception layer to the application layers [66]. The data transmission channels, encompassing wired or wireless, short or long-distance mechanisms, serve as foundational components. These channels effectively utilize both network infrastructure and wireless sensors. Attaining consistent and reliable performance in data transmission is paramount, especially in light of the substantial interference prevalent in the complex agricultural production environment and frequent climate changes that continually challenge this technology [67], [68].

3) APPLICATION LAYER

IoT has revolutionized the SA system, fundamentally transforming agriculture and agribusiness approaches. The IoT application layer, central to this transformation, drives the functionality and intelligence of IoT applications, particularly in intelligent agriculture [69], [70]. This layer integrates data from diverse sensors and devices in agricultural settings, enabling insightful analysis and informed actions [48], [66]. Advanced technologies within this layer, such as ML algorithms and predictive analytics, drive precision farming, optimizing resource allocation crop management, and fostering sustainable agricultural practices [71]. The application layer processes information and makes critical decisions [72]. It closely integrates IoT technology with

agricultural production, utilizing data analysis from the connectivity layer [73]. It also possesses related equipment to achieve SA management [9]. Given the complexity of crop data and climate change, technology proves pivotal in identifying agricultural production process issues aligning with user needs.

B. WSN ARCHITECTURES

WSNs are essential in IoT technology because they collect and transmit data from various physical phenomena and environments. They operate as the information-gathering infrastructure of IoT by capturing real-time data through diverse sensors placed across different locations [74]. This collected data is relayed to centralized systems for analysis, interpretation, and informed decision-making. Especially within the SA system, WSNs manifest as a connected network of sensor nodes utilizing wireless connections. These nodes possess diverse functionalities, encompassing processing, transmission, and sensing capabilities, empowering them for self-organization, self-configuration, and self-diagnosis [75]. The categorization of WSNs is contingent upon their deployment contexts, with notable classifications being terrestrial WSNs (TWSNs), wireless underground sensor networks (WUSNs), underwater WSNs (UWSNs), wireless multimedia sensor networks (WMSNs), and mobile wireless sensor networks (MWSNs) [76], [77], [78]. TWSNs and UWSNs are commonly utilized in SA applications [23], [79]. In contrast, WUSNs are positioned underground, requiring a higher node density due to the restricted communication range caused by soil attenuation of higher frequencies [77], [80]. Scholarly literature extensively explores the various applications of WSNs in agriculture, encompassing activities

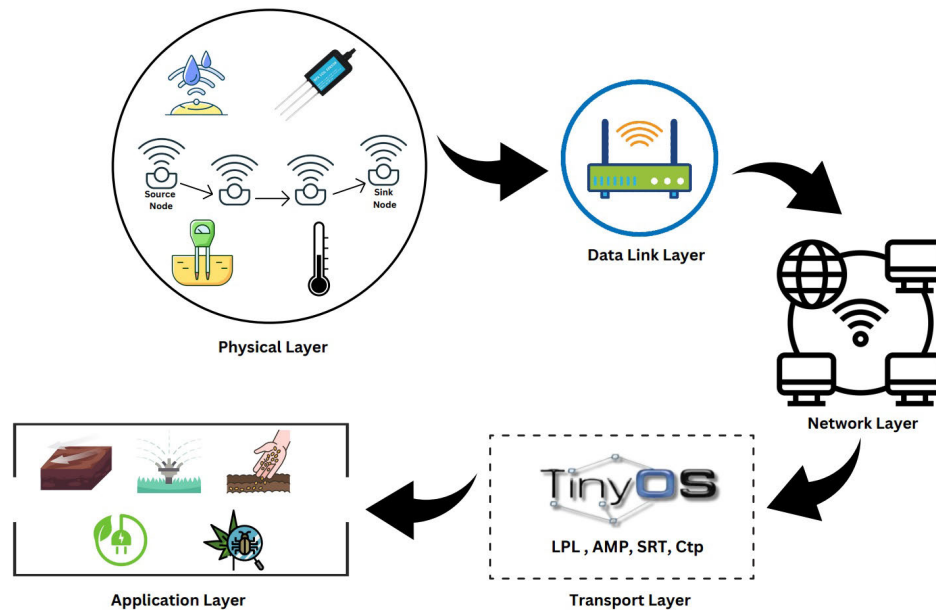


FIGURE 3. WSNs architecture for SA application.

such as managing irrigation, assessing water quality, and monitoring the environment. Exploring these applications underscores the pivotal role of WSNs in propelling advancements in agricultural practices.

A WSN typically comprises five foundational layers within its architecture: the physical, data link, network, transport, and application layers [81], [82]. The details of these layers are given below.

1) PHYSICAL LAYER

The physical layer is the foundation of a WSN, encompassing crucial hardware components and communication interfaces. In the SA domain, this layer holds various sensors meticulously designed to measure essential environmental parameters such as temperature, soil moisture, and sunlight exposure [64], [83]. These sensors play a pivotal role by converting these physical parameters into electrical signals, forming the bedrock for comprehensive data collection within the agricultural environment. For WSNs, the most relevant and widely recognized set of standards is the IEEE 802.15 family [84]. Specifically, the IEEE 802.15.4 standard defines the physical and Medium access control layer (MAC) specifications for low-rate wireless personal area networks (LR-WPANs) extensively utilized in WSNs [85], [86]. This standard is crafted to deliver low-cost, low-power, and low-data rate communication, catering to diverse applications, including WSNs.

On the other hand, in the data flow, sensor nodes actively collect data and transmit it to a centralized node, ensuring processing aligns with user requirements. The physical layer is entrusted with transmitting bitstreams, carefully selecting frequencies, generating carrier frequencies, modulating data,

encrypting data, and detecting signals [87]. This layer encompasses the definition of transmission medium specifications and intricately involves establishing the network topology, incorporating the crucial functions of encoding and decoding signals.

2) DATA LINK LAYER

The layer above the physical layer is the data link layer, which establishes reliable and secure connections between adjacent nodes in the network [81]. In the context of SA, this layer ensures error-free communication among sensors, the central base station, and sensor nodes, allowing for precise monitoring of field conditions and crop health [88], [89]. It handles multiple functions, including multiplexing data streams, frame detection, MAC, and error control implementation [90]. Additionally, the data link layer ensures the dependability of both point-to-point and multi-point channel access strategies through efficient scheduling and proficient buffer management [91].

3) NETWORK LAYER

The network layer is positioned higher within the WSN architecture and is crucial in overseeing routing and data packet progression among diverse sensor nodes [92]. It optimizes data flow significantly from sensors scattered across extensive farmlands to the central server or gateway, making efficient routing a vital aspect for well-informed decisions in SA, particularly in irrigation, pest control, and resource management. The network layer's primary function lies in routing, establishing a pathway through intermediary nodes from the source to the destination node [93]. Research in this layer primarily focuses on developing highly

efficient routing protocols that meet various constraints, encompassing energy efficiency, quality of service (QoS), and robustness. Moreover, the network layer integrates the communication network protocol selected from the existing network protocols for WSN, encompassing features relevant to precision agriculture applications.

4) TRANSPORT LAYER

The transport layer assumes a critical role, ensuring reliability and effectively managing congestion to prevent or mitigate it. Specific protocols are in place within this layer to serve these essential functions, employing either upstream or downstream techniques [94]. These protocols can be categorized as packet-driven and event-driven [95]. The collaborative capabilities demonstrated by sensor nodes are the fundamental basis for the operations of this vital layer. In addition, the transport layer is crucial in maintaining end-to-end communication and data integrity [96]. In the context of advanced agricultural applications, this layer can ensure the consistent and reliable transmission of data packets from sensors to the application layer. It efficiently manages data flow, meticulously maintains packet sequencing, and applies necessary error correction measures to ensure that crucial agricultural data reaches the applications with the highest accuracy and precision.

5) APPLICATION LAYER

In the domain of SA, the application layer holds great significance. It supports various applications crucial for optimizing agricultural practices, enhancing yields, and promoting sustainable farming methods [97]. Farmers and analysts can easily visualize field data on their mobile devices and computers through this layer, aiding in informed decision-making. This layer also plays a pivotal role in analyzing field data, providing valuable insights [83]. Moreover, the application layer controls essential management functionalities within the WSN. It efficiently manages traffic, offers software for diverse applications, and converts data into understandable formats. These functions encompass network management, query processing, communication, time synchronization, and localization [98].

IV. WIRELESS COMMUNICATION PROTOCOLS FOR IoT-WSNS IN SA

WSN developed various wireless sensing devices placed over a broad location. These appliances include distributed communication network protocols that gather data from the agricultural yield and centralize adequately to commission the collected data. The state-of-the-art WSN comprises several affordable sensing devices utilizing low-power communications network protocols, such as ZigBee, SigFox, WiFi, and LoRaWAN. WSNs are distinguished functionally from the usual sensing device collection by their network capabilities, which allow performance between sensing assets.

The network protocol is also classified into long-range and short-range protocols depending on their area coverage and properties. For this purpose, long-range network protocols can convey data over a long distance; in contrast, short ranges can transmit within a short distance [33], [99]. The essential features and attributes of these four protocols are shown in Table 1. The short-range and long-range protocols are given below.

A. SHORT-RANGE PROTOCOLS

1) ZIGBEE

The Alliance has introduced ZigBee technology based on the IEEE 802.15.4 standard, employing a wireless communication approach that conserves power and integrates a high-speed, energy-efficient standard protocol [84], [100]. ZigBee operates within multiple radio frequency bands, including 2.4 GHz, 915 MHz, and 868 MHz, achieving an impressive transmission speed of 250 kbps. Notably, ZigBee can function in a low-power sleep mode using batteries for extended periods [23]. It supports various network techniques, encompassing star, tree, and mesh topology, resulting in three primary Zigbee network types: star, tree, and wireless mesh network topology [101].

Zigbee wireless networks facilitate autonomous wireless data transmission in the agricultural IoT-WSNs domain. It ensures convenient and stable remote data transmission through successive integration with wired data transmission. Continuous advancements in IoT-WSNs microprocessor research and development are evident, especially in integrating wireless sensing, control, communication, and data processing functions within the microprocessor [27]. Concerning real-time monitoring in agricultural production, utilize intelligence systems for precise functional and monitoring tasks in field cultivation, irrigation and fertilizer, and well-established production processes. Agricultural IoT supports enriched planting experiences and precise crop management.

Remarkable strides have been taken in the development of high-precision information monitoring and diagnostic equipment, significantly advancing the application of IoT in agriculture. Currently, equipment for acquiring crop and plant information, monitoring environmental information, and tracking animal behavior plays a pivotal role in the SA system. The vital data required for crop monitoring is gathered using various Zigbee sensors in battery self-powered mode, forming a wireless sensor network [102], [103], [104]. Given the substantial number of Zigbee nodes required in production, establishing a robust network topology is essential to facilitate fast communication between network nodes [105].

2) WIFI

Wireless Fidelity (WiFi), officially introduced by the Wireless Ethernet Compatibility Alliance (WECA) in the late 1990s [106], [107], represented a significant advancement

TABLE 2. Overview and characteristics of ZigBee, WiFi, SigFox, and LoRaWAN.

Attributes	ZigBee	WiFi	SigFox	LoRaWAN
Topology	Star and mesh cluster	Star	Star	Star-of-stars
Data rate	250 kbps	150 kbps	100 bps and 600 bps	0.3–27 kbps
Network	WPAN	WLAN	LPWAN	LPWAN
Frequency	2.4 GHz	2.4/60 GHz	868/902 GHz	-
Standard	IEEE 802.15.4	802.11g	SigFox	LoRaWAN
Coverage range	10–100m	10–50m	30–50 km/3–10 km	5–15–45 km
Features	Built on PHY and MAC layers of IEEE 802.15.4	Access and availability, flexibility and cost savings	Highly efficient energy and long-range	Low power, long range

in wireless technology. Functioning as a wireless local area network (WLAN) technology, it effectively replaced Ethernet, granting devices the ability to connect to the internet without being tethered by wires or cables. Operating in compliance with the IEEE 802 communication standard, devices equipped with WiFi communicate via radio signals across the airwaves with an access point (AP), an essential piece of networking hardware connected to a wired or cellular network [108]. This technology covers a range of radio frequencies, spanning from 2.4 to 60 GHz, and precisely defines the structure of data packets [23], [33]. WiFi is widely adopted across a spectrum of devices, primarily due to its coverage range, typically 3–7 km, with a large transmitting antenna and its potential to achieve information transfer speeds of up to 700 Mbps [109], [110].

In the IoT applied in SA, integrating computing and WiFi-based long-distance networks facilitates connectivity within agricultural and farming processes, particularly in rural areas. It is used to transmit information within the system due to its high throughput and ease of integration with web-based services. WiFi integration is essential while measuring critical field parameters such as water quantity, soil humidity, and temperature [111]. It is ideal for establishing communication between the sink nodes and gateways or connecting to the cloud via the internet. WiFi finds specific applications in agricultural IoT, encompassing crop disease detection, precision greenhouse farming, and remote diagnosis [112], [113].

B. LONG-RANGE PROTOCOLS

1) SIGFOX

SigFox is a Low-Power Wide Area Network (LPWAN) technology designed to transmit minimal data volumes, typically ranging from a few bytes to hundreds of kilobytes [114]. The core modulation technique employed is Differential Binary Phase-Shift Keying (D-BPSK), operating within a fixed bandwidth of 100Hz and achieving efficient spectrum utilization [25], [33]. The transmission speed varies, offering rates of 100bps in Europe and a higher 600bps in the U.S.A. region. Operating within the unlicensed frequency spectrum below 1GHz, SigFox utilizes 868MHz in Europe and 915MHz in the U.S. [115]. This Ultra-Narrow Band modulation, known for its minimal power consumption, is ideal for establishing robust connections between nodes and the

base station, especially when combined with the Chirp Spreading Spectrum. The D-BPSK modulation is relatively straightforward to implement [116], [117]. Moreover, SigFox handles small data packets, typically composed of 12 bytes, and operates within a bandwidth of 100 Hz [118]. A key strength of SigFox lies in its outstanding power efficiency, capable of sustaining operations for up to an impressive 15 years on a single battery charge [119]. This network technology operates in a frequency band between 860 and 920 MHz [120]. The coverage range spans 10 to 40 km, and the information transfer speed can reach 600 bits per second [121].

Owing to Sigfox's advanced network protocol features, it finds extensive applications within -WSN-based SA systems [7]. It is notably used to implement cloud-based WSNs for irrigation systems, real-time agricultural data collection, and energy harvesting [118]. Moreover, SigFox has been utilized to develop solar-powered autonomous sensor nodes, effectively collecting meteorological parameters and demonstrating successful assessment, particularly in vineyard environments [118].

2) LORAWAN

LoRaWAN is a low-power, wide-area networking protocol developed to enable communication among low-energy devices in various IoT applications [101]. It operates over unlicensed radio frequencies, allowing for long-distance communication with minimal power consumption [122]. The technology uses chirp spread spectrum modulation, making it ideal for connecting devices over several kilometers in urban or rural environments [123]. This wireless communication technology consumes energy infrequently and operates in an unlicensed band [124]. Its coverage range is about 20 km, and the information transfer speed can reach 100 kbps [125], [126].

Additionally, LoRaWAN utilizes various spreading factors (SF) from SF7 to SF12, strategically managing the trade-off between transmission range and data rate [127]. Among them, SF12 achieves an extensive communication range at the cost of a lower data rate [128]. LoRa is the fundamental physical technique integrated with the LoRaWAN MAC layer, serving as a foundation for various applications [129]. The architectural design of LoRaWAN nodes is structured into three distinct classes: A, B, and C [130]. Class A nodes

exhibit minimal power consumption, efficiently transmitting a limited number of data packets to the gateway and spending most of their time idle [131]. In contrast, Class B nodes, in addition to the reception windows resembling Class A, open specific reception windows during scheduled time slots [130]. Meanwhile, Class C nodes maintain a perpetually open window for reception, except during data transmission, resulting in higher energy consumption than their counterparts in the other two classes [132].

V. IoT-WSNS IN SA APPLICATION

Agriculture is essential for any nation and a pillar of the economy. SA is a significant aspect and rising contemporary topic for all nations. The world's inhabitants are rapidly increasing, and the demand for food boosts as the population extends. The production of food and recovering the individual's fundamental needs can only be possible by concentrating on the agriculture sectors. Thus, automation should be incorporated into agriculture and reformed from traditional to SA. For this purpose, IoT-WSN is frequently utilized in SA applications. Several agricultural applications are discussed in the following subsections. WSN-integrated IoT-based SA applications utilizing wireless network protocols are summarised in Table 3.

A. IRRIGATION SYSTEM

The smart irrigation system (SIS) is a scientific domain that uses data-intensive approaches to improve agricultural productivity while decreasing environmental impact. Advanced agricultural processes generate data from different sensors, leading to a better understanding of the operational circumstances and process activities. It allows extra accurate and efficient decision-making. The SIS improves performance and is an emerging approach that automates irrigation systems and conserves water usage. This process modifies irrigation based on actual soil and weather conditions, allowing farmers to fulfill their needs with a recently adopted method that preserves the water for the irrigation process. Hence, WSN and wireless network protocols are utilized in the irrigation system to achieve this facility [133].

Routis et al. have presented an innovative IoT-based prototype system for precise crop irrigation. This system utilizes microprocessors and a Single-Board Computer (SBC) to collect sensor data, monitoring essential soil parameters, including soil moisture, humidity, temperature, and ultraviolet (UV) light. A significant feature is Raspberry devices, which incorporate powerful 4-core CPUs operating at 1.5GHz, underscoring their computational efficiency compared to the Arduino UNO's operation at 16MHz. The system integrates various sensors, including Capacitive Soil Moisture sensors, the DHT22 sensor for measuring air humidity and temperature, and the VEML6070 UV sensor, which is essential for monitoring UV radiation and its influence on crop growth. This study substantially contributes to agricultural technology and IoT applications for crop management [201].

Vandome et al. proposed low energy consumption and cost-effectiveness wireless soil moisture sensors, making them accessible both from a technical and economic perspective. This innovative sensor employs a precise calibration method based on a single parameter, enabling real-time monitoring of irrigation water requirements. Importantly, it was intentionally designed to meet the specific needs of water users and was successfully implemented within a Tunisian irrigation scheme, effectively addressing significant water use efficiency challenges. They evaluated the sensor by testing the WSN on pilot plots throughout a growing season and comparing its performance to commercial sensors. Notably, Wi-Fi technology was skillfully employed for communication within the network. Furthermore, the authors provided valuable insights by advocating for using these cost-effective sensors for real-time irrigation monitoring and as a pivotal tool for improving water resource management [283].

Fernández et al. proposed an economical cloud-based irrigation system that relies on WSN-based micro-controller ESP32-Lora and internet connectivity via the SigFox network. The results obtained validate the system's stability and robustness. This system had various sensors to measure different irrigation parameters, encompassing hydraulic network and environmental variables, including air temperature, humidity, irradiance, wind speed, precipitation, and soil variables such as humidity, temperature, pH, and matric potential. The study demonstrated the effectiveness of an IoT-based solution for irrigation control and management, offering scalability suitable for diverse agricultural contexts. The deployment of cost-effective SigFox technology addresses the connectivity and energy availability challenges of SA systems in rural areas [291].

Mathew et al. developed an IoT-based system to enable farmers to estimate irrigation water requirements. This innovative system employs sensors to detect soil moisture and temperature, with the collected data transmitted to the LoRaWAAN system for comprehensive analysis, including evapotranspiration calculation. The determination of global evapotranspiration is facilitated through Cropwat software, while the sensor data undergoes examination, enabling precise estimations tailored to microclimate conditions. The chosen Rx-MCU is the ESP32 MCU, equipped with integrated Wi-Fi connectivity to ensure uninterrupted internet gateway connection, facilitating seamless data streaming into application layer services. This approach is firmly grounded in WSNs, significantly enhancing the efficiency of both data collection and transmission [284].

B. SOIL MOISTURE MONITORING SYSTEM

Soil moisture plays an important role in an agricultural field that massively contributes to crop growth [493], [494]. It is recognized as one of the major drivers for plant ecosystems and a significant state variable for the irrigation system of the agriculture field [495]. Soil moisture is strongly variable and depends on various soil properties and terrain attributes [496].

TABLE 3. IoT-WSNs-based applications with wireless network protocols.

Applications	ZigBee	WiFi	SigFox	LoRaWAN
Irrigation system	[105], [134]–[206]	[83], [93], [105], [112], [148], [179], [180], [204], [207]–[289]	[290]–[292]	[111], [128], [129], [131], [161], [188], [230], [233], [248], [259], [267], [272], [283], [284], [289], [291], [293]–[347]
Soil moisture monitoring system	[105], [156], [182], [188], [195]–[204], [243], [348]–[359]	[83], [105], [112], [156], [204], [242], [244], [245], [247], [255], [258]–[276], [280], [282]–[289], [357], [358], [360]–[383]	[384]	[111], [122], [129], [257], [259], [267], [272], [283], [284], [289], [295], [298], [299], [306], [317], [321], [324], [332]–[339], [341], [345]–[347], [385]–[410]
Fertilizer optimization and control	[105], [137], [140], [143], [144], [157], [170], [182], [183], [190], [195], [198], [200], [353], [411]–[416]	[105], [112], [210], [234], [244], [261], [267], [273]–[275], [285], [287], [288], [365], [368], [375], [380], [417]–[422]		[122], [267], [297], [298], [340], [347], [385], [391], [405], [423]–[434]
Early-stage control of pest and crop diseases	[105], [149], [157], [176], [190], [198], [200], [348], [359], [435]–[440]	[83], [105], [261], [267], [273], [288], [375], [403], [422], [441]–[446]		[161], [267], [305], [324], [391], [403], [430], [446]–[461]
Energy saving and power consumption	[102], [105], [190], [195], [196], [198], [200], [201], [203]–[206], [357]–[359], [438], [440], [462]–[473]	[128], [241], [242], [247], [249], [258], [271], [275], [276], [283], [288], [357], [362], [379], [382], [464], [474]–[477]	[290]	[111], [128], [129], [259], [267], [283], [306], [324], [334]–[336], [338], [341], [399], [402]–[404], [410], [438], [464], [468], [473], [476], [478]–[492]

Therefore, a modern, sustainable, automatic, low-cost, and power-efficient soil moisture monitoring process is required. Soil moisture monitoring systems can be performed with various procedures. In these terms, WSN is one of the most utilized approaches to monitoring soil condition and moisture [84], [497]. Based on this, several works have been done to monitor the soil nutrient, PH, and moisture monitoring by using WSN.

Mohammed et al. introduced a real-time, fully automated WSN prototype for irrigation systems. This innovative system relies on automated WSN technology to respond to soil conditions, particularly soil moisture, for informed irrigation decisions. It utilizes the ZigBee protocol integrated into the XBee module and incorporates a cost-effective capacitive soil moisture sensor to measure soil moisture levels at each ZigBee node. Integrated with the low-cost soil moisture sensor, this system demonstrated promise in enhancing agricultural practices and conserving water resources [203].

Patrizi et al. submitted WSN architecture that leverages Long Short-Term Memory (LSTM) technology to develop

a virtual soil moisture sensor. This virtual sensor utilizes data collected by other transducers on the same node. They utilized the ESP32 system-on-a-chip microcontroller to efficiently process and transmit data to a centralized gateway through the WiFi protocol. The WSN consists of ten independent sensor nodes, each equipped with an array of environmental sensors for air temperature, humidity, soil temperature, soil moisture, and radiation. An essential challenge addressed in this work was measuring soil humidity. In response, the study introduced a sophisticated soft sensing algorithm based on the DL algorithm, resulting in a virtual soil moisture sensor capable of overcoming the limitations of physical sensors and enhancing precision in soil moisture measurements [379].

Wu et al. presented a soil quality monitoring system to enhance agricultural practices. This system designated users to conduct real-time monitoring of their farmland via a mobile application, providing a convenient means to establish and customize soil parameter thresholds. Their research offered a novel approach to integrating IoT technology into agricultural soil measurements, incorporating multiple

sensors for temperature and moisture, a microprocessor, a microcomputer, a cloud platform, and a dedicated mobile application. The wireless sensors efficiently collect and transmit real-time soil information, with the mobile app serving as a central monitoring hub through the cloud platform. Data transmission facilitated by the LoRa module ensures precise measurements that closely align with those obtained through calibration equipment. Field experiments have showcased the system's ability to predict soil moisture and temperature with enhanced accuracy, drawing upon data from various soil layers. This system equipped users with the necessary tools to promptly assess soil conditions, enabling routine checks for changes in soil quality [406].

C. FERTILISER OPTIMIZATION AND CONTROL

The world faces a food supply crisis, with less food production than population growth [498]. Besides, there needs to be more integration and utilization of state-of-the-art technology in agricultural applications [499]. The annual food production is also damaged due to a lack of soil fertility, moisture, and fundamental nutrients of NPK [500]. On the other hand, conventional methods are even applied to measure the soil nutrients and apply a fertilizer that is also harmful to crops and soil. The excessive volume of fertilizer can damage the standard scale of PH and the soil nutrition of agricultural land [501]. To address this challenge, it becomes essential to form a precision agriculture exercise through IoT with the involvement of IoT and WSN. Soil nutrient and fertilizer investigation using WSN allows different applications, including remote soil fertility monitoring.

Contreras et al. introduced SAgric-IoT, an SA system integrating IoT and DL technologies. This sophisticated system aims to monitor environmental conditions, swiftly detect diseases, and automate irrigation and fertilization processes within greenhouses. It comprises four key components: WSN for environmental monitoring, cameras for visual data, a gateway for centralized control, and a processing and storage unit. The gateway operates using three distinct communication protocols—ZigBee for sensor connectivity, Wi-Fi for cameras, and a cellular interface for transmitting data to the central unit. Notably, SAgric-IoT demonstrates remarkable efficiency in minimizing packet loss, thereby significantly conserving energy [105].

Senapaty et al. introduced the IoTSNA-CR model, utilizing IoT technology to analyze soil nutrients and suggest optimal crops, aiming to enhance productivity while reducing fertilizer usage. The model begins with IoT sensors collecting data in cultivation areas, followed by real-time storage in cloud services, accessible via an Android app, and undergoing subsequent data processing and analysis. A cost-effective WSN sensory system integrated various sensors to monitor soil attributes—temperature, moisture, pH, GPS, and color—efficiently gathering diverse data, including NPK values, timestamps, and geolocation specifics. Research advised

using Agrinex NPK soil testing tablets and an LDR color sensor for soil sample analysis, with results stored in Firebase cloud storage. Interconnected through the ESP8266 Wi-Fi module and Arduino microcontroller, sensor nodes formed a low-power, efficient wireless network. The pivotal FC28 soil moisture sensor significantly enhanced the IoTSNA-CR model's effectiveness by evaluating soil water content [112].

Doan et al. proposed a WSN framework employing LoRa technology for remote monitoring within agricultural settings. This network consisted of three strategically placed LoRa sensor nodes within separate rice fields, tailored to different crops and utilizing various tillage and fertilization methods. These nodes gathered essential environmental indicators for effective monitoring, encompassing temperature, air humidity, soil moisture, water pH, CH₄, and NH₃ levels. Research findings underscored the system's effectiveness in monitoring the rice cultivation environment, ensuring robust and comprehensive data transmission, network security, and impressive long-range signal transmission of up to 3.5 km at a reasonable cost. The framework demonstrated proficiency in long-distance data transmission, maintaining secure data transfer from source nodes to the central gateway, ensuring high reliability, and facilitating seamless deployment for extensive agricultural monitoring [347].

D. EARLY STAGE CONTROL OF PEST AND CROP DISEASES

Damaging crop production due to various factors, pests, and crop diseases is one significant issue, as a large quantity of crop is wasted every production cycle by affecting pests and diseases [84], [502]. Thus, an early-stage sustainable decision support system is mandatory to enable the farmer to carry out proper actions in profitable harvesting [503]. A smart monitoring system should employ state-of-the-art sensor technology to solve plant diseases and pest-related concerns. WSN systems can collect data and store them in a cloud platform with the help of network protocols and process them, which helps to make early decisions to prevent pest and crop diseases [36], [47].

Additionally, Crop disease detection is a massive challenge in precision agriculture applications since crop diseases cannot be accurately predicted by analyzing individual disease causes. Azfar et al. introduced an IoT sensor system explicitly designed for pest detection and control, real-time detection of Cotton Flying Moths. This system experienced rigorous testing in controlled and uncontrolled environments to evaluate its accuracy and efficiency. The innovative prototype integrated a series of precise infrared sensors, a communication module based on Zigbee technology, an Arduino 2560 Mega board, a lithium polymer battery to power the sensor, a gateway device, and a UAV configured to act as a pesticide sprayer upon detecting pests. The pest detection algorithm, embedded within the system, monitors changes in reflected light to identify the presence of flying insects. Upon detection, the system sends an alert to the gateway device, which transmits the detection coordinates to

the drone/UAV. Subsequently, the drone responds by spraying pesticide in the identified pest-infested area [440].

Strawberry cultivation is a significant agricultural venture, offering considerable advantages over various vegetable crops [504], [505]. Despite its prominence, the sensitivity of strawberries renders them highly susceptible to diverse pests and diseases. Resorting to chemicals and pesticides for protection significantly hampers production efficiency due to this sensitivity. To address this challenge, integrating deep computer vision architectures has been pivotal. Cruz and colleagues introduced a novel approach, employing the Yolo V5 computer vision architecture within an IoT system designed for strawberry disease detection. This innovative system utilizes LoRaWAN and WiFi protocols for internal data transmission and employs the Message Queuing Telemetry Transport protocol for data uploading to the internet [403].

Hnatiuc et al. developed an innovative IoT sensor network paired with a LoRaWAN-based system designed precisely for intelligently detecting grapevine diseases and collecting essential environmental and plant-related data. This technology was deployed within the experimental plots of the Research Station for Viticulture and Enology (SDV) to offer early insights into grapevine health. Data transmission across the Wi-Fi network was facilitated through the LoRaWAN-EU868 protocol. The study is balanced to extend its investigation to analyze results from IoT sensors trialed across vineyards in diverse regions [446].

E. ENERGY SAVING AND POWER CONSUMPTION

Advanced agriculture uses new processes, such as precision agriculture, to optimize the workflow under environmental aspects. For this purpose, WSN is incorporated with IoT, and network protocols combined with systems are used. Various required WSN nodes and actuators are also equipped. Such equipment requires a continuous power supply to achieve non-stop services [506]. In the PA-based WSN, energy consumption may differ due to additional parameters, i.e., active computational overload or sensor density deviations [507]. The existing conventional energy harvesting strategy cannot serve under such a requirement while harvesting and thus may reduce the overall lifespan of the network and entire system. To meet the energy requirements, a significant energy harvesting design is required [508].

Recent advancements and development of technology with low-power consumption, IoT, and WSN have been tremendously utilized and deployed for various SA applications. WSN has improved agricultural productivity and efficiency in agriculture yield. However, the energy and power shortage of WSNs is a major issue as instantly charging batteries is usually demanded [509], [510], [511], [512]. Sadowski et al. proposed an agricultural monitoring system with energy harvesting regarding these factors. The study presented a comparison arrangement between ZigBee, WiFi, and LoRaWAN wireless network protocols. It has

been demonstrated that LoRaWAN demanded less power than other protocols within agricultural monitoring systems where power consumption and legibility of the network are considered. The experimental results recommended selecting wireless technology for future agricultural monitoring applications [464].

Similarly, Arshad et al. introduced a study based on a smart sensor module with an advanced irrigation system and supervised fertilizer architectures. The system has incorporated WSN, cloud-utilizing decision support layers, and networking-based DSS to recommend cautiousness for optimum sustainable agricultural field and production. For this purpose, A WSN node is equipped with an MCU LoRaWAN wireless module. The sensor node with LoRaWAN is connected to solar panels for stored energy since LoRaWAN can achieve enormous distances with low energy [267], [403].

VI. ISSUES AND CHALLENGES OF IoT-WSNS-BASED SA SYSTEMS

Despite the remarkable progress in integrating -WSN-based methods within SA, significant limitations must be carefully considered and addressed through proactive mitigation strategies. These obstacles manifest in various aspects, encompassing the system's infrastructure, the devices utilized, the network structure, and the crucial element of securing the data [64]. Constructing an adequate architecture for IoT-WSNs-based SA systems is challenging, requiring the continuous integration of diverse devices, sensors, actuators, and communication interfaces into a unified and efficient framework [513]. Striking the right balance between integration, system reliability, and scalability presents a considerable challenge [514].

Moreover, the diverse range of devices employed in IoT-WSNs-based SA systems, such as sensors for measuring soil moisture, drones for aerial surveillance, and automated irrigation systems, continuously pose challenges related to careful selection, setup, and maintenance. Factors such as device compatibility, power efficiency, durability, and cost-effectiveness are critical considerations in their deployment, representing ongoing challenges within this technological framework. Addressing the complexities of the network architecture in IoT-WSNs-based SA setups is equally essential, requiring the establishment of a strong, low-latency, and high-bandwidth network capable of managing the constant flow of data generated by numerous devices. Deliberate planning for scalability and reliability is vital to meet the heightening demand for real-time data processing and decision-making in profound SA applications.

A. SCALABILITY AND RELIABILITY

Substantial dual challenges exist within the domain of SA propelled by -WSN technology. These challenges encompass ensuring scalability and defending reliability, significantly influencing the system's efficacy and enduring sustainability. Scalability emerges as a central concern due to the increasing

TABLE 4. A summary of issues and challenges of IoT-WSNs based SA domain.

References	Critical Issues and Challenges	Recommended Steps and Resolving Actions
[9], [9], [21]–[23], [47], [515]–[519]	Scalability and Reliability: data traffic, potentially causing network congestion, delays in data transmission	Well-optimized network blueprints, resilient communication protocols, effective power management techniques, and dependable fail-over mechanisms.
[13], [21], [21], [515], [517]–[521]	Data Privacy and Confidentiality: unauthorized access, breaches, data misuse, identifying individuals farmers or agriculture stakeholder through data analysis, and robust encryption, effective authentication, and appropriate access control measures	Encryption protocols, secure data transmission methods, routine security assessments, and educational initiatives on privacy best practices.
[1], [64], [72], [521]–[527]	Location Privacy: track the location of farming equipment, livestock, Unauthorized access to precise location data	Robust privacy-preserving mechanisms and compliance with privacy regulations, ethical use of location data.
[13], [33], [61], [69], [131], [498], [525], [528]–[541]	Network Security and Intrusion Detection: security risks including node compromise, tampering, malicious attack, unauthorized access, data breaches, cyber-attacks, and adapting suitable IDS	Robust encryption techniques, secure communication protocols, efficient access, formidable security measures and energy-efficient operations.
[13], [93], [542]–[553]	Data Integrity and Authenticity: manipulation and tampering with sensor data, unauthorized alterations, introducing fabricated data	Considering massive amounts of data generated and transmitted across these networks, maintaining data integrity ensuring, review of data authenticity, secure authentication processes, and cryptographic techniques, digital signatures, message authentication codes, and integrity checks.
[13], [28], [526], [550], [554]–[559]	User Privacy and Consent: securing from hacking and data stolen, encompassing personal and operational data	Implementing transparent data usage policies, employing robust anonymization techniques, empowering users, effective communication of the purpose of data collection.
[49], [97], [560]–[564]	Resilience to Attacks and Failures: failure and disrupt agricultural operations	Implementing redundancy, fault tolerance mechanisms and backup systems.
[33], [68], [205], [334], [565]–[572]	Mitigation of Power Consumption: fixed battery life, high hardware costs, energy shortages, necessitating regular battery recharging	Energy-efficient routing protocol, implementation of renewable energy in SA systems
[21], [23], [27], [523], [573]–[579]	Optimizing hardware and software costs, expenses associated with imported agricultural devices, costs remain a notable gap in current research, lack of standardized data representation and operational procedures	Collaborative efforts among researchers and stockholders, establishment of necessary standards for effective IoT implementation

implementation of -WSN devices, resulting in a burgeoning network interconnecting various intelligent devices, sensors, and actuators [9], [23]. This growth presents challenges in effectively managing the expanding array of devices and the resulting surge in data traffic, potentially causing network congestion, delays in data transmission, and diminished overall system performance [22], [47]. Simultaneously, guaranteeing reliability is pivotal, ensuring a consistent and precise data flow, an indispensable component for well-informed agricultural decision-making. Aspects such as irregular network connections, device malfunctions, and power disruptions can undermine the system's trustworthiness. Effectively addressing these hurdles necessitates a comprehensive and integrated strategy encompassing well-optimized network blueprints, resilient communication protocols, effective power management techniques, and dependable failover mechanisms [21], [515], [517], [518], [519]. Achieving a meticulous balance between scalability and reliability becomes critical, fostering a sustainable and resilient -WSN-based SA system capable of adeptly meeting the evolving requisites of contemporary agriculture [9], [516].

B. DATA PRIVACY AND CONFIDENTIALITY

Incorporating -WSN technology into the sphere of SA introduces significant barriers concerning the security and

privacy of data [98]. As SA systems become more interconnected, significant volumes of sensitive data are gathered, encompassing information on crop yields, weather patterns, and details about farmers and consumers [520]. Ensuring this data is covered from unauthorized access, breaches, or misuse is essential [13]. Privacy concerns emerge due to the potential risk of identifying individuals or businesses through data analysis [21]. Furthermore, the distributed deployment of -WSN devices, often across remote agricultural sites, challenges establishing robust encryption, effective authentication, and appropriate access control measures [521]. Insufficient security measures may lead to unauthorized data access, tampering, or intentional harm. Striking a balance between leveraging data analytics to improve agricultural productivity and upholding individual privacy and data security is a multifaceted yet vital objective. Addressing these challenges necessitates a comprehensive approach encompassing encryption protocols, secure data transmission methods, routine security assessments, and educational initiatives on privacy best practices for all IoT-WSNs-based SA ecosystem stakeholders [46], [68], [513], [522], [557], [580], [581], [582].

C. LOCATION PRIVACY

The advent of -WSN technology in SA has significantly enhanced data collection and decision-making processes.

However, this integration has introduced a crucial concern about location privacy. IoT-WSNs often necessitate the deployment of sensor nodes in agricultural fields to monitor various parameters and activities. These nodes can precisely track the location of farming equipment, livestock, and even the farmers themselves [1], [64], [522], [523]. While this location data is indispensable for optimizing farming practices, it raises substantial privacy challenges. Unauthorized access to precise location data can violate the privacy of farmers and stakeholders, potentially leading to misuse or security breaches [521], [524], [525]. Robust privacy-preserving mechanisms and compliance with privacy regulations are essential to address these concerns and ensure the responsible and ethical use of location data in IoT-WSNs-based SA [72], [526], [527].

D. NETWORK SECURITY AND INTRUSION DETECTION

The inherent interconnectivity of IoT-WSNs renders them susceptible to a range of security risks, including node compromise, tampering, and malicious attacks [525], [528], [529]. Threats such as unauthorized access, data breaches, and cyber-attacks pose significant data integrity and confidentiality risks as data transits these networks [13], [530], [531]. Addressing these challenges involves developing robust encryption techniques, secure communication protocols, and efficient access controls for safeguarding sensitive data from unauthorized access [33], [61], [532], [533], [534], [535], [536], [537]. Additionally, the limited resources of sensor nodes in WSNs add complexity to implementing robust security measures while maintaining energy efficiency. Intrusion detection systems (IDS) are crucial in identifying and mitigating potential security breaches [69], [538], [539], [540]. However, adapting IDS to suit the specific characteristics and limitations of IoT-WSNs remains a significant challenge. The right balance between formidable security measures and energy-efficient operations is paramount for establishing a secure and reliable IoT-WSNs-based SA architecture [131], [498], [541].

E. DATA INTEGRITY AND AUTHENTICITY

Maintaining data integrity and authenticity is critical for reliable decision-making in SA. Manipulation or tampering with sensor data can lead to erroneous conclusions and potentially harmful actions. Considering the massive amounts of data generated and transmitted across these networks, ensuring the accuracy and reliability of the data becomes imperative [542], [543], [544]. Potential challenges stem from attempts at data tampering, unauthorized alterations, or introducing fabricated data into the system [545], [546], [547]. Maintaining data integrity, ensuring it remains accurate and unaltered throughout its lifecycle, presents a formidable challenge [548], [549]. Equally critical is validating data authenticity, confirming that the data originates from a credible and genuine source [548], [550]. Effective solutions necessitate the development of robust data validation methods, secure authentication processes, and cryptographic

techniques to ensure the integrity and authenticity of the data [13], [93], [550]. Techniques such as digital signatures, message authentication codes, and integrity checks can be implemented to verify the authenticity and integrity of the data collected from sensor nodes [548], [551], [552], [553].

F. USER PRIVACY AND CONSENT

In IoT-WSNs-based SA, preserving user privacy and obtaining informed consent is essential. These systems gather a wide range of user data, encompassing personal and operational data, raising valid concerns regarding data privacy [554], [555]. Users have a rightful expectation that their data will be handled ethically, securely, and in compliance with privacy regulations [556]. Obtaining informed consent poses a significant challenge due to the diverse and often distributed nature of data collection in SA. Users must be fully informed about the data being collected, its intended use, and who will have access to it. Striking a balance between optimizing agricultural practices through data utility and safeguarding user privacy is crucial [28]. Implementing transparent data usage policies, employing robust anonymization techniques, and empowering users with control over their data are vital steps in addressing these privacy and consent challenges [526], [550], [557], [558]. Effective communication of the purpose of data collection, the intended use, and potential risks to stakeholders is essential to obtain informed consent from users [13], [550], [559].

G. RESILIENCE TO ATTACKS AND FAILURES

SA depends significantly on the uninterrupted operation of IoT-WSNs [97], [560]. The reliability of this domain is crucial, as any compromise or failure can severely disrupt agricultural operations and impact productivity. Implementing redundancy, fault tolerance mechanisms and backup systems are vital to mitigate these risks [561], [562]. Redundancy ensures the maintenance of critical functions by providing backup paths or components in case of failures, ensuring continuous system operation. Fault tolerance mechanisms detect and manage failures, enabling quick recovery and continued functioning without significant performance drops [97], [563]. Backup systems act as safety nets, preserving critical data and functionalities accessible in case of failure [49], [564]. These actions collectively enhance the resilience of IoT-WSNs against potential attacks and failures, ensuring continuous monitoring and data availability for optimal agricultural outcomes.

H. MITIGATION OF POWER CONSUMPTION

Efficient power management is essential in IoT-WSNs-based SA. These systems are crucial for applications such as in-field monitoring and real-time tracking of field conditions, significantly reducing yield losses caused by unforeseen circumstances. However, despite their importance in agricultural contexts, persistent challenges stem from fixed battery life, high hardware costs, and limited bandwidth [33], [68],

[205], [565]. A significant aspect of these challenges is energy shortages, necessitating regular battery recharging to maintain an uninterrupted power supply [334], [566], [567]. Insufficient power provision can disrupt the entire system, causing interruptions in seamless farmland monitoring. IoT-WSNs comprise numerous sensors with limited energy resources deployed based on specific requirements. Therefore, in developing an energy-efficient routing protocol, renewable energy implementation in SA can be an essential requirement [568], [569], [570], [571], [572]. Current routing systems often feature complex architectures, underscoring the need for proactive approaches to address this concern and enhance power efficiency.

I. COST EFFECTIVENESS AND STANDARDIZATION

In IoT and WSNs applied in SA, a primary global research focus is optimizing hardware and software costs to improve system efficiency [23]. This focus is particularly vital in developing nations seeking to reduce expenses associated with imported agricultural devices [573]. While international farming entities have made technological advancements, the challenge of further reducing costs remains a notable gap in current research [523], [574]. Standardization is important in IoT as numerous studies need more standardized data representation and operational procedures, hindering seamless integration [575]. Emphasizing standardization becomes pivotal for driving IoT advancements alongside cost-effectiveness. Its potential lies in reducing initial barriers, addressing interoperability issues, and fostering healthy competition among various products and services. The evolution of security, communication, and identification standards within IoT demands a suitable approach to developing new technologies [21], [27]. Collaborative efforts among researchers are essential to define industry-specific guidelines and establish necessary standards for effective IoT implementation [576], [577]. This collective endeavor will pave the way for a more robust and efficient IoT framework formed specifically for SA applications [576], [578], [579].

VII. FUTURE PROSPECTS AND OPPORTUNITIES

State-of-the-art digital technologies have significantly progressed, integrating seamlessly with IoT-WSNs to augment the sustainability of SA applications. This integration substantially optimizes the entire scope of agricultural processes, from cultivation to harvest, encompassing the entire agricultural sector. These visionary initiatives stand on the point of a substantial transformation in the agricultural landscape.

The customization of digitization necessitates a substantial financial expenditure to align with the specific needs of individual farmers. To enhance the reliability of this digitization, embracing government-backed initiatives, grants, strategic public-private partnerships, and open data policies becomes essential. These initiatives should be accompanied by regionally focused research efforts to reinforce their effectiveness. Precision in composing digitization to unique

needs mandates a significant financial commitment to organize to the specific demands of individual farmers. Transparent data policies are equally vital and assurance reinforcement through regional research endeavors. A systematic approach involves the meticulous implementation of a well-structured roadmap for the development of SA systems. This journey initiates with establishing a foundational architecture comprising essential components and more streamlined functionalities.

A. AI IN SA APPLICATION

The agriculture landscape is rapidly expanding, presenting various future trends and prospects. At the forefront of this expansion is the digital strategy, which encompasses a range of advanced technologies. AI is a computer system advancement standard aiming to replicate human intelligence and facilitate intricate decision-making processes. It assumes a critical position within SA, where the IoT-WSNs play a prominent role. Its capabilities extend into meticulous sensor data analysis, empowering stakeholders to make well-informed decisions regarding crop management, resource optimization, and predictions concerning future agricultural trends.

1) ML

As a subset of AI, ML is intricately focused on nurturing computer systems capable of learning and evolving through experiences. When applied to IoT-WSNs in agriculture, ML algorithms meticulously analyze historical and real-time data [583]. The objective is to optimize irrigation patterns, predict and proactively address crop diseases, and automate various agricultural processes [36], [584]. This integration significantly enhances efficiency and overall productivity within the agricultural landscape [36], [522].

2) DL

DL, an advanced ML technique, employs neural networks with multiple layers to replicate the intricate structure of the human brain. DL algorithms emerge as indispensable tools in IoT-WSNs tailored for SA [27], [585]. They engage in sophisticated data analysis, excel in image recognition, and unravel complexities in NLP. The application of DL leads to a comprehensive understanding of agricultural processes. Computer Vision is pivotal in empowering machines to interpret and comprehend visual information extracted from images or videos. Its integration within IoT-WSNs for agricultural purposes amplifies the potential for monitoring crop health, swiftly detecting diseases, and accurately assessing growth stages through meticulous analysis of imagery data [586], [587]. This, in turn, enables timely and well-informed interventions in agricultural practices, promoting efficiency and yield.

In recent years, integrating state-of-the-art AI technologies with the -WSN has emerged as a beacon of transformative potential within the agricultural sector. This integration

presents an unparalleled opportunity to optimize farming practices, enhance resource utilization, and significantly boost agricultural output. The subsequent sections comprehensively delve into several AI-based technologies that can be effectively utilized to further the cause of sustainable agriculture, discussing their potential impact and contributions to the agricultural landscape.

3) FEDERATED LEARNING (FL)

A federated learning approach is promising for AI based on -WSN. It allows models to be trained across multiple edge devices or servers while keeping data localized [588]. In agriculture, where data privacy is crucial, this technique enables collaborative model training without centralized data storage [589]. Farmers can contribute to a global model without sharing sensitive data, improving model accuracy for various agricultural tasks. The federated deep learning approach enhances resource usage and data privacy, leading to classification results comparable to the fundamental ML setup. Applying this sophisticated learning method involves incorporating IoT technology to identify crop diseases precisely [590], [591]. Furthermore, encryption techniques can be employed when sharing trained models to address privacy issues in the federated setting [592].

4) EXPLAINABLE AI (XAI)

Transparency in AI is fundamental for establishing trust and comprehensibility of AI systems, especially within the domain of SA [593]. XAI integration is key in ensuring that AI models provide clear and understandable explanations for their decisions. Within SA, XAI demonstrates diverse applications, encompassing critical tasks such as monitoring crop growth, assessing crop health, and efficiently managing pests and diseases through integration with IoT-WSNs [594], [595]. Consequently, this framework becomes imperative for farmers and stakeholders, enabling them to learn AI-driven insights and recommendations concerning crucial aspects such as crop management and resource allocation within agricultural processes. Moreover, the fusion of XAI-based models with state-of-the-art feature optimization techniques can significantly enhance real-time malware detection in SA applications [596]. By explaining the insights derived from AI model data and demystifying the indistinct nature of black box predictions, XAI effectively bridges the understanding gap by shedding light on the rationale behind these predictions—an aspect often elusive in conventional AI models [597].

5) REINFORCEMENT LEARNING (RL)

RL involves training AI agents to make sequential decisions to maximize rewards, making it a viable solution for data-deprived scenarios. In SA, RL opens up extensive opportunities. Particularly, when integrated with IoT-WSN, RL can optimize crucial crop management decisions, including irrigation schedules, pesticide application, and harvest timing [14]. Applying Deep Reinforcement Learning within

crop classification systems for precision agriculture is a promising approach to tackling farmers' challenges. DRL-based advanced agricultural techniques effectively filter out suboptimal choices, significantly enhancing crop production within the crop recommendation system [598]. Furthermore, RL is essential in addressing the area coverage problem related to monitoring crop health in semi-structured farm settings [639]. RL agents adeptly learn from environmental feedback provided by sensors, ultimately enhancing precision and efficiency in crop management [499].

6) GRAPH NEURAL NETWORKS (GNNs)

GNNs demonstrate high effectiveness in unraveling complex relationships within agricultural data. Particularly in the IoT-WSNs in SA, GNNs excel in capturing and modeling the interconnectivity among various factors that significantly impact crop health and yield. Leveraging data from multiple sources, such as sensors, weather patterns, and soil conditions, GNNs deliver valuable insights for optimized crop analysis and management. Additionally, GNNs find practical application in field-road classification methods based on GNSS recordings from agricultural machinery [599]. These methods enhance revolution classification accuracy, designed explicitly for farming machines [600]. Another significant application involves employing GNN models to predict the concentration of heavy metals in both soil and crops, including staple crops such as rice [601].

7) GENERATIVE ADVERSARIAL NETWORKS (GANs)

GANs demonstrate the capability to simulate the growth and development of crops under a diverse range of environmental conditions. This simulation proves invaluable for farmers as it equips them with the foresight to anticipate crop responses in various circumstances, aiding in making informed decisions concerning planting strategies, resource allocation, and risk management [602], [603]. Furthermore, AI leveraging DL is fundamentally transforming the analysis and modeling of agricultural images. Image augmentation is essential in enhancing the precision of DL models while reducing the need for manual efforts in image collection and annotation [604]. This augmentation is accomplished through the algorithmic generation and expansion of datasets. Moving beyond traditional data augmentation methods, Generative Adversarial Networks (GANs) in Computer Vision introduce innovative approaches to acquire effective data representations and generate highly realistic samples [605]. The potential of GANs to synthesize authentic and diverse images presents novel opportunities to enhance the performance of DL models tailored for agricultural applications. These advancements are particularly advantageous in scenarios where extensive labeled image datasets are inaccessible. Additionally, utilizing GAN-based data augmentation techniques enhances imaging classification tasks by generating artificial images, effectively doubling the training data for existing classes, thereby improving classification performance [606].

TABLE 5. A summary of future perspective and recommendation for IoT-WSNs-based SA system.

References	System/Architecture	Future Prospects and Recommendation
[14], [27], [36], [499], [522], [582]–[609]	AI	Integrate cost-effective sensors, ML, DL, XAI, GNNs, GANs, and self-supervised learning for advanced SA with improved resource management and transparency. Prospects involve sustainability via XAI, GNN complex modeling, GAN data generation, and self-supervised learning insights. Recommendations include XAI and self-supervised learning research, exploring GNNs and GANs integration, prioritizing reinforcement learning for autonomy, addressing data security, promoting farmer education, and ensuring cost-effective scalability.
[610]–[616]	Artificial General Intelligence (AGI)	AGI enhances real-time communication, streamlines farm management, fosters knowledge exchange, and improves crop monitoring and market predictions. Recommendations comprise ongoing R&D, user-friendly interfaces, data security, farmer training, IoT-WSN integration, emotional intelligence exploration, sustainability, and scalability for wider farming community adoption.
[7], [515], [617]–[619]	Agri-robotics	Autonomous, sustainable farming with real-time decision-making. Continuous robust data analysis, farmer training, data security, scalability, environmental impact reduction, predictive maintenance, fostering collaboration, and cost-efficiency to drive agriculture sector adoption and sustainability.
[431], [620]–[623]	UAVs	Real-time data collection for precision farming encompasses technology advancements, robust data platforms, training, data security, scalability, environmental sustainability, regulatory compliance, and cost-efficiency to promote adoption.
[14], [23], [74], [624]–[626]	Big Data Analytics (BDA)	Unlocks potential for data-driven insights, predictive modelling, and sustainability. The roadmap includes advanced data processing, machine learning, data security, farmer training, scalability, collaboration, and cost-efficiency to facilitate extensive utilization.
[538], [627]–[633]	5G and 6G for SA	It holds immense potential for real-time data transmission, enabling precision farming and remote monitoring. Key focus areas encompass infrastructure development, regulatory support, farmer training, data security, and integration to unlock the full capabilities of these technologies for sustainable and efficient agriculture.
[65], [627], [634]–[636]	Block Chain Systems	It offers transparent, decentralized data management. The path forward involves investing in infrastructure, standardizing data, empowering farmers, ensuring data security, and fostering collaborations to maximize the technology's potential, reinforcing data integrity and trust in agriculture.
[63], [567], [623], [637], [638]	Renewable Energy Integration	It promises sustainability and efficiency, encompassing advanced energy storage, AI-driven optimization, distributed grids, seamless integration, and collaborative sharing for greener, more productive farming.

8) SELF-SUPERVISED LEARNING

Self-supervised learning is gaining prominence in SA, especially when access to labeled data is scarce. This methodology empowers AI models to determine and extract meaningful features from extensive collections of unlabeled data, presenting a notable advantage in situations where annotating data demands substantial resources [607]. Integrating self-supervised learning techniques proves especially advantageous in -WSN-based systems, emphasizing efficiency and optimal resource utilization [608]. This avenue allows for robust data processing, extracting valuable insights without heavy reliance on extensive labeled datasets [609]. Consequently, it amplifies the capacity for well-informed decision-making in precision agriculture.

B. ARTIFICIAL GENERAL INTELLIGENCE IN SA APPLICATION

Artificial General Intelligence (AGI) is positioned at the cutting edge, demonstrating the substantial potential to wield a profound influence across various sectors, and agriculture is a prominent domain in this regard [610], [611].

AGI, especially in Natural Language Processing (NLP) and Agri-Robotics, can enhance crop yields, reduce waste, and promote sustainable farming practices [613], [640], [641]. This prowess uniquely positions AGI as a promising solution to the intricate challenges faced by the agricultural sector.

1) HUMAN-IN-THE-LOOP AI

Human-in-the-loop AI is a symbolic integration of human expertise with AI systems, facilitating collaborative decision-making [612]. Human experts contribute by providing input, validating AI-generated insights, and guiding AI models to ensure precise and dependable outcomes. This collaboration significantly enhances the accuracy and relevance of AI-driven insights within SA [613], [614].

2) CONVERSATIONAL AI AND AGRI-CHAT APPLICATIONS

Conversational AI, representing a pinnacle of technological advancement, entails the development of AI systems highly proficient in engaging with humans using natural language. Its specific application in SA is exemplified through Agri-Chat Applications, effectively interacting with farmers [642].

These applications provide real-time assistance, share invaluable insights, and offer essential crop management and pest control guidance. Particularly, technologies such as ChatGPT have the capability to analyze agricultural data, presenting immense potential for various agricultural applications such as crop forecasting, soil analysis, crop disease and pest identification, precision farming, and efficient irrigation scheduling [615]. The profound importance of these applications lies in their significant contribution to knowledge dissemination and the facilitation of seamless communication within the agricultural community.

3) VOICE ASSISTANTS FOR FARM MANAGEMENT

Voice-activated assistants, designed explicitly for farm management, herald a transformative stride in operational efficiency. They empower farmers to seamlessly structure their schedules, oversee equipment, and stay instantly informed through simple voice commands. These tools not only enhance accessibility to information but can also play a crucial role in promptly notifying farmers about any issues with the crops. Such immediate notifications enable farmers to take timely actions, potentially leading to a successful harvest [643], [644]. Consequently, this hands-free approach substantially amplifies productivity and streamlines the day-to-day management of farms. Farmers can now redirect their precious time and resources towards other critical agricultural tasks, advancing efficiency in the agricultural landscape.

4) COLLABORATIVE AI PLATFORMS FOR KNOWLEDGE SHARING

Innovative, collaborative AI platforms serve as sophisticated mediums to cultivate a culture of knowledge sharing and collaboration among farmers, researchers, and experts within the agricultural domain. These platforms facilitate collective problem-solving, exchanging worthwhile insights, and disseminating best practices. By leveraging the collective expertise of the agricultural community, these platforms empower individuals to optimize processes and elevate overall productivity. This concerted effort leads to implementing sustainable agricultural practices, aligning with ecological and agricultural sustainability principles.

5) EMOTION ANALYSIS FOR CROP MONITORING

Emotion Analysis technology marks a significant advancement in monitoring the health of crops. It thoroughly assesses plant well-being and stress levels through a detailed analysis of physiological and growth patterns. This analytical method provides crucial insights into how crops respond to their environment, such as understanding human emotions. It allows a more profound comprehension of their health and reactions to environmental factors. With such insights, agricultural practitioners can implement timely interventions to optimize growth conditions, promptly

diagnose diseases, and tailor the agricultural environment to meet the specific needs of the crops. Ultimately, this innovative technology substantially improves crop yield and the overall quality of agricultural produce, presenting a promising path for sustainable and efficient farming practices.

6) AI-DRIVEN MARKET ANALYSIS AND PRICE PREDICTION

The application of AI-driven Market Analysis represents a significant advancement, employing complex algorithms to analyze market dynamics such as trends, consumer behavior, and economic indicators [645], [646]. Predictive models are central in foreseeing market prices, estimating demand-supply dynamics, and predicting trade fluctuations [616]. These data-driven insights empower farmers and stakeholders, enabling them to make strategic decisions and adapt to the dynamic nature of the market effectively. This approach enhances their resilience and ability to thrive in a constantly evolving market landscape.

C. AGRI-ROBOTICS

Agri-robotics involves the integration of robotic systems and automation technologies into agricultural practices. Combined with IoT-WSNs, these robotic systems are further enhanced by integrating sensor data optimizing tasks such as precise planting, monitoring, and harvesting [7]. This ultimately enhances efficiency and productivity in agriculture. These robots possess precise supratemporal resolutions due to their specialized sensing and actuation capabilities, potentially reducing labor while improving agricultural processes [515]. Additionally, drones are crucial in pesticide spraying, irrigation, crop harvesting, seed sowing, and soil cultivation, essential in transforming traditional farming practices [617].

D. UAVS

UAVs, commonly referred to as drones, are equipped with a suite of cameras and sensors, establishing them as essential tools for data collection and monitoring within the agricultural sector. When integrated with IoT-WSNs, UAVs facilitate aerial data collection, providing a real-time and in-depth understanding of the agricultural landscape [431], [620]. This integration significantly contributes to the precise assessment of crops and efficient agricultural management. Considerable research studies have investigated the potential of UAVs in data collection for agriculture, underscoring the efficacy of UAV-WSN systems in SA. For instance, researchers have explored a UAV-enabled agricultural system, wherein the UAV acts as a decode-and-forward relay, enhancing communication between controllers and multiple robots. Additionally, UAVs have been utilized to design optimal flight paths, generating UAV trajectories for efficient data collection from sensor nodes with non-uniform distributions, thus enhancing the overall effectiveness of data gathering [621], [622], [623].

E. BIG DATA ANALYTICS (BDA)

Big data analytics in IoT-based agriculture leverages advanced data analysis methodologies to extract valuable insights from the extensive data generated by IoT devices within agricultural settings [14], [74], [624]. These IoT devices, furnished with various sensors, actively gather diverse data concerning soil health, prevailing weather conditions, crop growth patterns, equipment performance, and more. The process of big data analytics involves discerning patterns, trends, and correlations within this data, all of which are critical for informed, data-driven decision-making aimed at enhancing agricultural practices. This analytical approach anticipates future trends and conditions by analyzing historical and real-time data, empowering proactive decision-making. It optimizes the usage of vital resources, such as water, fertilizers, and pesticides, by precisely tailoring their application to the specific requirements of the crops. The role of big data analytics in assisting farmers in making informed choices regarding the cultivation of diverse crops, considering seasonal variations such as winter and summer, cannot be overstated [23], [625], [626]. This analytical approach has demonstrated high cost-effectiveness, especially for small-scale farmers, and exhibits potential for indoor application within households.

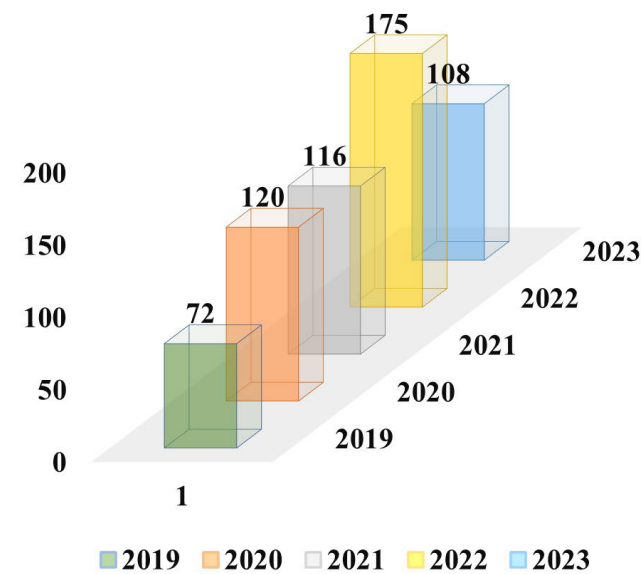


FIGURE 4. Progression of IoT-WSNs and network protocols in SA systems.

F. 5G AND 6G FOR SA

The advancement of 5G and the promising revolution of 6G within IoT-WSNs bear substantial implications for the agricultural sector. 5G, constituting the fifth generation of wireless technology, represents a remarkable leap forward, offering significant enhancements such as accelerated data speeds, diminished latency, and improved connectivity. These enhancements translate to heightened real-time monitoring and faster transmission of sensor data, ultimately

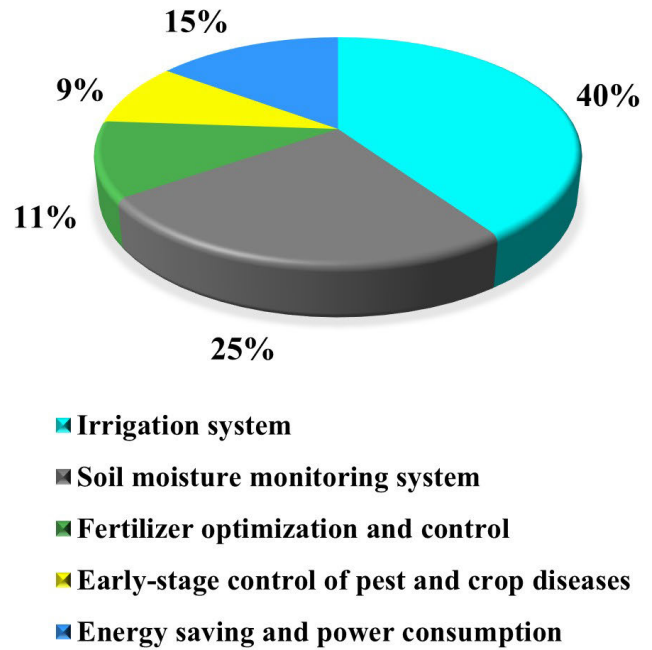


FIGURE 5. Yearly evolution of research contributions in IoT-WSN within SA applications.

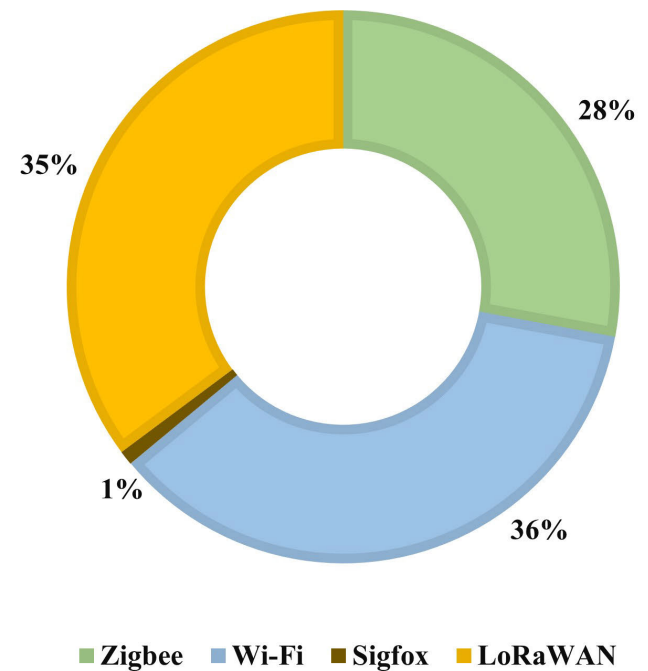


FIGURE 6. Research contribution distribution in IoT-WSN with their network protocols in SA applications.

refining coordination among devices in agricultural applications [538], [627]. Looking ahead, 6G technology, still in its developmental phase, holds the potential to revolutionize data collection and analysis, potentially achieving speeds in the range of terabits per second, thereby ensuring exceptionally timely and precise insights for farmers [628], [631]. Incorporating 5G and the envisioned capabilities

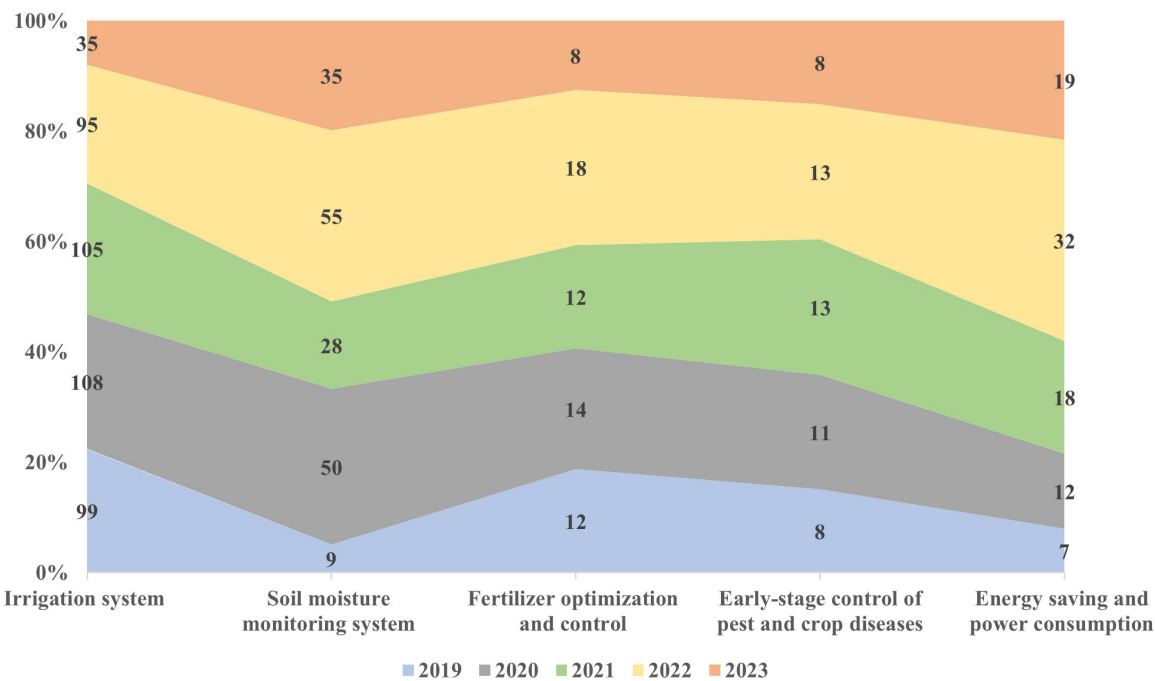


FIGURE 7. Annual research contribution rates in SA application areas.

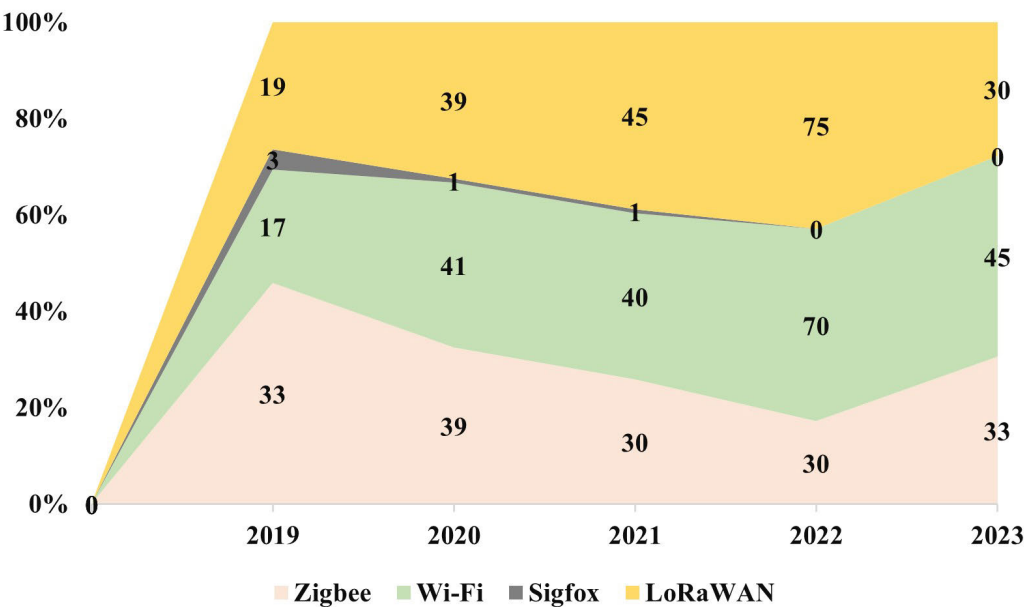


FIGURE 8. Annual wireless network protocol rates in IoT-WSN research within SA Applications.

of 6G within IoT-WSNs for agriculture holds immense promise. This combination facilitates real-time monitoring of critical factors such as soil conditions, crop health, weather patterns, and equipment performance, empowering optimized decision-making. The present generation of intelligent agricultural applications, relying on a relatively restrained number of wireless sensors, necessitates improved accuracy and effectiveness. However, the ongoing development of

6G- communication technologies sets the groundwork for the future of intelligent and sustainable agriculture [629]. 6G-technology pledges to enable the interconnection of various extensive sensors, granting farmers the capacity to gather intricate, plant-specific information. This transformative potential fueled by 6G- technology is poised to revolutionize SA, ensuring precise data collection, advanced robotics, and precision agriculture in remote locations, rendering

agriculture significantly more efficient, sustainable, and cost-effective.

G. BLOCK CHAIN SYSTEMS

Blockchain technology, a decentralized and highly secure digital ledger, plays a crucial role in documenting transactions across a network. When integrated with IoT-WSNs, it ensures secure and transparent record-keeping within the domain of SA, particularly benefiting supply chains, traceability, and transactions. This integration significantly strengthens trust and operational efficiency throughout the agricultural ecosystem by assuring data integrity and fortifying defense against potential threats. One notable application involves countering data manipulation attacks, where crucial information about crops with minimal pesticide usage is securely stored within the blockchain's immutable ledger [627], [634], [635]. This advancement ensures the integrity and reliability of essential agricultural data, ultimately contributing to enhanced decision-making processes.

H. RENEWABLE ENERGY INTEGRATION

Incorporating renewable energy sources such as solar and wind power into IoT-WSNs for agriculture is crucial, aiming to establish sustainable and environmentally friendly energy solutions for powering devices and systems in remote agricultural regions [63]. Integrating renewable energy contributes to energy efficiency and environmental sustainability within the agricultural landscape. By incorporating energy from renewable sources into the IoT-based SA observation systems, sustainability and cost-effectiveness are markedly enhanced [637]. Using solar or wind power to energize the system diminishes reliance on conventional power sources, consequently reducing energy expenses and decreasing carbon emissions. Additionally, the scalability of this integrated system makes it viable for adoption by small-scale farmers who may lack access to conventional power sources.

I. IoT-WSNs AND THEIR PROTOCOLS IN FUTURE SA APPLICATIONS AREAS

In recent years, the integration of the IoT has led to significant advancements in agriculture, as shown in Table 3. Notably, there has been a 2.4% increase in IoT's growth, currently rising by 1.5%, as shown in Figure 4, which has captured substantial attention within the agricultural community. Farmers have extensively adopted IoT frameworks throughout the agricultural process, particularly in irrigation systems. They are renowned for their precision, efficiency, remote monitoring, and data-driven decision-making, as shown in Figure 5.

IoT has notably enhanced agricultural efficiency by linking devices, instruments, and stakeholders, reducing labor costs and increasing overall productivity. Farmers seek reliable, cost-effective, and power-efficient IoT devices seamlessly integrating with WSNs. The landscape of wireless network protocols is evolving, witnessing a significant shift towards

low-power alternatives such as LoRaWAN, outperforming Wi-Fi due to its efficiency, network coverage, and minimal power requirements in SA, as shown in Figure 6.

Moreover, SigFox, operating on narrowband technology and offering extensive coverage with fewer base stations, remains an area of interest for IoT research in agricultural systems. Although its usage has maintained a 1% rate since 2019, as shown in Figure 6, SigFox and LoRaWAN, both with low power consumption, differ in flexibility. LoRaWAN, due to its greater flexibility, is better suited for specific agricultural setups requiring increased control or distinct coverage patterns.

A detailed analysis of research contributions emphasizes a significant focus on irrigation systems, soil monitoring, and energy optimization. Soil monitoring saw a slight decrease post-2021, while by the end of 2022, as shown in Figure 7, there was an increase in fertilizer optimization through IoT, along with applications for energy savings and plant pest and disease control in SA. Regarding network protocols, the rise of LoRaWAN significantly impacted the dominance of ZigBee and Wi-Fi from 2020 to 2022, securing a 35% share over the past five years, as shown in Figure 8. This positions LoRaWAN as a promising protocol for efficient resource utilization in SA. Simultaneously, Wi-Fi retains its popularity due to widespread availability and short-range capabilities.

VIII. CONCLUSION

Integrating IoT with WSNs in SA aims to maximize yield and optimize agricultural processes. This comprehensive IoT with WSNs architecture facilitates effective monitoring and control of agricultural fields, enabling valuable data collection to address current challenges. Wireless communication protocols are essential in efficiently transmitting the collected data to the system.

Integrating WSNs with IoT and wireless network protocols has proven cost-effective and efficient in SA applications, minimizing agricultural expenses. Notably, these integrated systems exhibit low cost, low power consumption, and enhanced efficiency. This survey presents a detailed review of the state-of-the-art IoT technology and the deployment of WSNs with wireless network protocols for SA applications since 2019.

The study extensively explores the architecture of IoT-WSN integration with network protocols in the SA domain, shedding light on prominent network protocols such as ZigBee, WiFi, SigFox, and LoRaWAN. These protocols have found significant utility in IoT-WSNs SA applications, including irrigation systems, soil moisture monitoring, fertilizer optimization and control, early-stage pest and crop disease management, and energy-saving measures. The study deliberates on current challenges and issues in this domain, aiming to provide insights and potential solutions for further advancements in integrating IoT-WSNs for the future of SA systems.

REFERENCES

- [1] V. R. Pathmudi, N. Khatri, S. Kumar, A. S. H. Abdul-Qawy, and A. K. Vyas, "A systematic review of IoT technologies and their constituents for smart and sustainable agriculture applications," *Sci. Afr.*, vol. 19, Mar. 2023, Art. no. e01577.
- [2] N. Alam, "Opportunity assessment and feasibility study of IoT-based smart farming in Bangladesh for meeting sustainable development goals," in *The Fourth Industrial Revolution and Beyond*. Cham, Switzerland: Springer, 2023, pp. 723–736.
- [3] S. Basnet, A. Wood, E. Röös, T. Jansson, I. Fetzer, and L. Gordon, "Organic agriculture in a low-emission world: Exploring combined measures to deliver sustainable food system in Sweden," *Sustainability Sci.*, vol. 18, no. 1, pp. 501–519, 2023.
- [4] A. Nandi, N. Counts, S. Chen, B. Seligman, D. Tortorice, D. Vigo, and D. E. Bloom, "Global and regional projections of the economic burden of Alzheimer's disease and related dementias from 2019 to 2050: A value of statistical life approach," *eClinicalMedicine*, vol. 51, Sep. 2022, Art. no. 101580.
- [5] H. Alloui and Y. Mourdi, "Exploring the full potentials of IoT for better financial growth and stability: A comprehensive survey," *Sensors*, vol. 23, no. 19, p. 8015, Sep. 2023.
- [6] S. P. Singh, G. Dhiman, S. Juneja, W. Viriyasitavat, G. Singal, N. Kumar, and P. Johri, "A new QoS optimization in IoT-smart agriculture using rapid adaption based nature-inspired approach," *IEEE Internet Things J.*, 2023, doi: [10.1109/JIOT.2023.3306353](https://doi.org/10.1109/JIOT.2023.3306353).
- [7] F. K. Shaikh, S. Karim, S. Zeadally, and J. Nebhen, "Recent trends in Internet-of-Things-enabled sensor technologies for smart agriculture," *IEEE Internet Things J.*, vol. 9, no. 23, pp. 23583–23598, Dec. 2022.
- [8] R. Chataut, A. Phoummalayvane, and R. Akl, "Unleashing the power of IoT: A comprehensive review of IoT applications and future prospects in healthcare, agriculture, smart homes, smart cities, and Industry 4.0," *Sensors*, vol. 23, no. 16, p. 7194, Aug. 2023.
- [9] A. Morchid, R. El Alami, A. A. Raedah, and Y. Sabbar, "Applications of Internet of Things (IoT) and sensors technology to increase food security and agricultural sustainability: Benefits and challenges," *Ain Shams Eng. J.*, vol. 15, no. 3, Mar. 2024, Art. no. 102509.
- [10] S. Anand and A. Sharma, "Comprehensive analysis of services towards enhancing security in IoT-based agriculture," *Meas., Sensors*, vol. 24, Dec. 2022, Art. no. 100599.
- [11] H. Dadhaneeya, P. K. Nema, and V. K. Arora, "Internet of Things in food processing and its potential in Industry 4.0 era: A review," *Trends Food Sci. Technol.*, vol. 139, Sep. 2023, Art. no. 104109.
- [12] T. Anees, Q. Habib, A. S. Al-Shamayleh, W. Khalil, M. A. Obaidat, and A. Akhunzada, "The integration of WoT and edge computing: Issues and challenges," *Sustainability*, vol. 15, no. 7, p. 5983, Mar. 2023.
- [13] S. Rudrakar and P. Rughani, "IoT based agriculture (Ag-IoT): A detailed study on architecture, security and forensics," *Inf. Process. Agricult.*, 2023, doi: [10.1016/j.inpa.2023.09.002](https://doi.org/10.1016/j.inpa.2023.09.002).
- [14] A. Bhola, S. Srivastava, A. Noonina, B. Sharma, and S. K. Narang, "A status quo of machine learning algorithms in smart agricultural systems employing IoT-based WSN: Trends, challenges and futuristic competences," in *Machine Intelligence, Big Data Analytics, and IoT in Image Processing: Practical Applications*. Wiley, 2023, pp. 177–195, doi: [10.1002/9781119865513.ch8](https://doi.org/10.1002/9781119865513.ch8).
- [15] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, and E. M. Aggoune, "Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk," *IEEE Access*, vol. 7, pp. 129551–129583, 2019.
- [16] A. Khanna and S. Kaur, "Evolution of Internet of Things (IoT) and its significant impact in the field of precision agriculture," *Comput. Electron. Agricult.*, vol. 157, pp. 218–231, Feb. 2019.
- [17] V. P. Kour and S. Arora, "Recent developments of the Internet of Things in agriculture: A survey," *IEEE Access*, vol. 8, pp. 129924–129957, 2020.
- [18] K. L. Raju and V. Vijayaraghavan, "IoT technologies in agricultural environment: A survey," *Wireless Pers. Commun.*, vol. 113, no. 4, pp. 2415–2446, Aug. 2020.
- [19] T. Ojha, S. Misra, and N. S. Raghuwanshi, "Internet of Things for agricultural applications: The state of the art," *IEEE Internet Things J.*, vol. 8, no. 14, pp. 10973–10997, Jul. 2021.
- [20] R. K. Singh, R. Berkvens, and M. Weyn, "AgriFusion: An architecture for IoT and emerging technologies based on a precision agriculture survey," *IEEE Access*, vol. 9, pp. 136253–136283, 2021.
- [21] B. B. Sinha and R. Dhanalakshmi, "Recent advancements and challenges of Internet of Things in smart agriculture: A survey," *Future Gener. Comput. Syst.*, vol. 126, no. 4, pp. 169–184, Jan. 2022.
- [22] S. Qazi, B. A. Khawaja, and Q. U. Farooq, "IoT-equipped and AI-enabled next generation smart agriculture: A critical review, current challenges and future trends," *IEEE Access*, vol. 10, pp. 21219–21235, 2022.
- [23] C. Prakash, L. P. Singh, A. Gupta, and S. K. Lohan, "Advancements in smart farming: A comprehensive review of IoT, wireless communication, sensors, and hardware for agricultural automation," *Sens. Actuators A, Phys.*, vol. 362, Nov. 2023, Art. no. 114605.
- [24] S. Dhal, B. M. Wyatt, S. Mahanta, N. Bhattarai, S. Sharma, T. Rout, P. Saud, and B. S. Acharya, "Internet of Things (IoT) in digital agriculture: An overview," *Agronomy J.*, vol. 1, p. 20, 2023, doi: [10.1002/agj.2.21385](https://doi.org/10.1002/agj.2.21385).
- [25] A. Pawar and S. B. Deosarkar, "IoT-based smart agriculture: An exhaustive study," *Wireless Netw.*, vol. 29, no. 6, pp. 2457–2470, Aug. 2023.
- [26] V. K. Quy, N. V. Hau, D. V. Anh, N. M. Quy, N. T. Ban, S. Lanza, G. Randazzo, and A. Muzirafuti, "IoT-enabled smart agriculture: Architecture, applications, and challenges," *Appl. Sci.*, vol. 12, no. 7, p. 3396, Mar. 2022.
- [27] D. A. Gzar, A. M. Mahmood, and M. K. A. Al-Adilee, "Recent trends of smart agricultural systems based on Internet of Things technology: A survey," *Comput. Electr. Eng.*, vol. 104, Dec. 2022, Art. no. 108453.
- [28] T. A. Shaikh, T. Rasool, and F. R. Lone, "Towards leveraging the role of machine learning and artificial intelligence in precision agriculture and smart farming," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107119.
- [29] X. Shi, X. An, Q. Zhao, H. Liu, L. Xia, X. Sun, and Y. Guo, "State-of-the-art Internet of Things in protected agriculture," *Sensors*, vol. 19, no. 8, p. 1833, Apr. 2019.
- [30] N. Kumar and B. Sharma, "Opportunities and challenges with WSN's in smart technologies: A smart agriculture perspective," in *Handbook wireless Sensor Networks: Issues Challenges Current Scenario's*. Cham, Switzerland: Springer, 2020, pp. 441–463.
- [31] W. Tao, L. Zhao, G. Wang, and R. Liang, "Review of the Internet of Things communication technologies in smart agriculture and challenges," *Comput. Electron. Agricult.*, vol. 189, Oct. 2021, Art. no. 106352.
- [32] A. Abdollahi, K. Rejeb, A. Rejeb, M. M. Mostafa, and S. Zailani, "Wireless sensor networks in agriculture: Insights from biometric analysis," *Sustainability*, vol. 13, no. 21, p. 12011, Oct. 2021.
- [33] E. Avsar and M. N. Mowla, "Wireless communication protocols in smart agriculture: A review on applications, challenges and future trends," *Ad Hoc Netw.*, vol. 136, Nov. 2022, Art. no. 102982.
- [34] A. K. Pandey and A. Mukherjee, "A review on advances in IoT-based technologies for smart agricultural system," *Internet Things Analytics Agricult.*, vol. 3, pp. 29–44, Jan. 2022.
- [35] K. Gulati, R. S. K. Boddu, D. Kapila, S. L. Bangare, N. Chandnani, and G. Saravanan, "A review paper on wireless sensor network techniques in Internet of Things (IoT)," *Mater. Today, Proc.*, vol. 51, pp. 161–165, Jan. 2022.
- [36] M. M. Rahaman and M. Azharuddin, "Wireless sensor networks in agriculture through machine learning: A survey," *Comput. Electron. Agricult.*, vol. 197, Jun. 2022, Art. no. 106928.
- [37] S. S. Sethi and P. Sharma, "New developments in the implementation of IoT in agriculture," *Social Netw. Comput. Sci.*, vol. 4, no. 5, p. 503, Jun. 2023.
- [38] H. K. Adli, M. A. Remli, K. N. S. W. S. Wong, N. A. Ismail, A. González-Briones, J. M. Corchado, and M. S. Mohamad, "Recent advancements and challenges of AIoT application in smart agriculture: A review," *Sensors*, vol. 23, no. 7, p. 3752, Apr. 2023.
- [39] R. Zhou and Y. Yin, "Digital agriculture: Mapping knowledge structure and trends," *IEEE Access*, vol. 11, pp. 103863–103880, 2023.
- [40] X. Li, B. Hou, R. Zhang, and Y. Liu, "A review of RGB image-based Internet of Things in smart agriculture," *IEEE Sensors J.*, vol. 23, no. 20, pp. 24107–24122, Oct. 2023.
- [41] Y. Jaraarweh, S. Fatima, M. Jarrah, and S. AlZu'bi, "Smart and sustainable agriculture: Fundamentals, enabling technologies, and future directions," *Comput. Electr. Eng.*, vol. 110, Sep. 2023, Art. no. 108799.
- [42] A. Balkrishna, R. Pathak, S. Kumar, V. Arya, and S. K. Singh, "A comprehensive analysis of the advances in Indian digital agricultural architecture," *Smart Agricult. Technol.*, vol. 5, Oct. 2023, Art. no. 100318.
- [43] A. Gupta and P. Nahar, "Classification and yield prediction in smart agriculture system using IoT," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 8, pp. 10235–10244, Aug. 2023.

- [44] R. Manikandan, G. Ranganathan, and V. Bindhu, "Deep learning based IoT module for smart farming in different environmental conditions," *Wireless Pers. Commun.*, vol. 128, no. 3, pp. 1715–1732, Feb. 2023.
- [45] K. Bakthavatchalam, "IoT framework for measurement and precision agriculture: Predicting the crop using machine learning algorithms," *Technologies*, vol. 10, p. 13, Nov. 2022.
- [46] E. M. B. M. Karunathilake, A. T. Le, S. Heo, Y. S. Chung, and S. Mansoor, "The path to smart farming: Innovations and opportunities in precision agriculture," *Agriculture*, vol. 13, no. 8, p. 1593, Aug. 2023.
- [47] R. Akhter and S. A. Sofi, "Precision agriculture using IoT data analytics and machine learning," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 8, pp. 5602–5618, Sep. 2022.
- [48] M. R. Kabir and S. Ray, "Virtual prototyping for modern Internet-of-Things applications: A survey," *IEEE Access*, vol. 11, pp. 31384–31398, 2023.
- [49] E. E. K. Senoo, E. Akansah, I. Mendonça, and M. Aritsugi, "Monitoring and control framework for IoT, implemented for smart agriculture," *Sensors*, vol. 23, no. 5, p. 2714, Mar. 2023.
- [50] J. Xu, B. Gu, and G. Tian, "Review of agricultural IoT technology," *Artif. Intell. Agricult.*, vol. 6, pp. 10–22, Jan. 2022.
- [51] C. Verdouw, H. Sundmaeker, B. Tekinerdogan, D. Conzon, and T. Montanaro, "Architecture framework of IoT-based food and farm systems: A multiple case study," *Comput. Electron. Agricult.*, vol. 165, Oct. 2019, Art. no. 104939.
- [52] V. Ramachandran, R. Ramalakshmi, B. Kavim, I. Hussain, A. Almaliki, A. Almaliki, A. Elnaggar, and E. Hussein, "Exploiting IoT and its enabled technologies for irrigation needs in agriculture," *Water*, vol. 14, no. 5, p. 719, Feb. 2022.
- [53] T. Saranya, C. Deisy, S. Sridevi, and K. S. M. Anbananthan, "A comparative study of deep learning and Internet of Things for precision agriculture," *Eng. Appl. Artif. Intell.*, vol. 122, Jun. 2023, Art. no. 106034.
- [54] L. A. Nguyen, P. T. Kiet, S. Lee, H. Yeo, and Y. Son, "Comprehensive survey of sensor data verification in Internet of Things," *IEEE Access*, vol. 11, pp. 50560–50577, 2023.
- [55] A. Jahangeer, S. U. Bazai, S. Aslam, S. Marjan, M. Anas, and S. H. Hashemi, "A review on the security of IoT networks: From network layer's perspective," *IEEE Access*, vol. 11, pp. 71073–71088, 2023, doi: [10.1109/ACCESS.2023.3246180](https://doi.org/10.1109/ACCESS.2023.3246180).
- [56] X. Hernández-Cruz, J. R. Villalobos, G. Runger, and G. Neal, "Building an intelligent system to identify trends in agricultural markets," *J. Cleaner Prod.*, vol. 425, Nov. 2023, Art. no. 138956.
- [57] Y.-F. Zeng, C.-T. Chen, and G.-F. Lin, "Practical application of an intelligent irrigation system to Rice paddies in Taiwan," *Agricult. Water Manage.*, vol. 280, Apr. 2023, Art. no. 108216.
- [58] Q. Zheng, N. Lin, D. Fu, T. Liu, Y. Zhu, X. Feng, and J. Ruan, "Smart contract-based agricultural service platform for drone plant protection operation optimization," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21363–21376, 2023, doi: [10.1109/JIOT.2023.3288870](https://doi.org/10.1109/JIOT.2023.3288870).
- [59] G. S. Hundal, C. M. Laux, D. Buckmaster, M. J. Sutton, and M. Langemeier, "Exploring barriers to the adoption of Internet of Things-based precision agriculture practices," *Agriculture*, vol. 13, no. 1, p. 163, Jan. 2023.
- [60] Q. Liu, X. Zhao, and K. Shi, "The analysis of agricultural Internet of Things product marketing by deep learning," *J. Supercomput.*, vol. 79, no. 4, pp. 4602–4621, Mar. 2023.
- [61] A. Parnian, M. Mahbod, C. K. Mitra, H. Beyrami, and M. N. V. Prasad, "Internet of Things (IoT) in soil management for achieving smart agriculture," in *Agroecological Approaches for Sustainable Soil Management*, 2023, pp. 457–486. [Online]. Available: <https://doi.org/10.1002/9781119911999.ch20>
- [62] A. Pal and V. Kumar, "AgriDet: Plant leaf disease severity classification using agriculture detection framework," *Eng. Appl. Artif. Intell.*, vol. 119, Mar. 2023, Art. no. 105754.
- [63] I. Roussaki, K. Doolin, A. Skarmeta, G. Routis, J. A. Lopez-Morales, E. Claffey, M. Mora, and J. A. Martinez, "Building an interoperable space for smart agriculture," *Digit. Commun. Netw.*, vol. 9, no. 1, pp. 183–193, Feb. 2023.
- [64] A. Vangala, A. K. Das, V. Chamola, V. Korotaev, and J. J. P. C. Rodrigues, "Security in IoT-enabled smart agriculture: Architecture, security solutions and challenges," *Cluster Comput.*, vol. 26, no. 2, pp. 879–902, Apr. 2023.
- [65] H. Zeng, G. Dhiman, A. Sharma, A. Sharma, and A. Tselykh, "An IoT and blockchain-based approach for the smart water management system in agriculture," *Expert Syst.*, vol. 40, Jan. 2023, Art. no. e12892.
- [66] H. Zhang and L. Zhang, "A reliable data-driven control method for planting temperature in smart agricultural systems," *IEEE Access*, vol. 11, pp. 38182–38193, 2023.
- [67] H. Rastegari, F. Nadi, S. S. Lam, M. Ikhwanuddin, N. A. Kasan, R. F. Rahmat, and W. A. W. Mahari, "Internet of Things in aquaculture: A review of the challenges and potential solutions based on current and future trends," *Smart Agricult. Technol.*, vol. 4, Aug. 2023, Art. no. 100187.
- [68] N. Chamara, M. D. Islam, G. Bai, Y. Shi, and Y. Ge, "Ag-IoT for crop and environment monitoring: Past, present, and future," *Agricult. Syst.*, vol. 203, Dec. 2022, Art. no. 103497.
- [69] A. J. Simla, R. Chakravarthi, and L. M. Leo, "Agricultural intrusion detection (AID) based on the Internet of Things and deep learning with the enhanced lightweight M2M protocol," *Soft Comput.*, vol. 27, pp. 1–12, 2023. [Online]. Available: <https://doi.org/10.1007/s00500-023-07935-1>
- [70] R. Fu, X. Ren, Y. Li, Y. Wu, H. Sun, and M. A. Al-Absi, "Machine learning-based UAV assisted agricultural information security architecture and intrusion detection," *IEEE Internet Things J.*, vol. 10, no. 21, pp. 18589–18598, 2023, doi: [10.1109/JIOT.2023.3236322](https://doi.org/10.1109/JIOT.2023.3236322).
- [71] Á. L. P. Gómez, P. E. López-de-Teruel, A. Ruiz, G. García-Mateos, G. B. García, and F. J. G. Clemente, "FARMIT: Continuous assessment of crop quality using machine learning and deep learning techniques for IoT-based smart farming," *Cluster Comput.*, vol. 25, no. 3, pp. 2163–2178, Jun. 2022.
- [72] M. Abbasi, M. Plaza-Hernández, and Y. Mezquita, "Security of IoT application layer: Requirements, threats, and solutions," in *Proc. Int. Symp. Ambient Intell.* Cham, Switzerland: Springer, 2022, pp. 86–100.
- [73] F. Masood, W. U. Khan, S. U. Jan, and J. Ahmad, "AI-enabled traffic control prioritization in software-defined IoT networks for smart agriculture," *Sensors*, vol. 23, no. 19, p. 8218, Oct. 2023.
- [74] R. Verschae, "Smart technologies in agriculture," in *Encyclopedia of Smart Agriculture Technologies*. Cham, Switzerland: Springer, 2023, pp. 1–11.
- [75] N. Jihani, M. N. Kabbaj, and M. Benbrahim, "Kalman filter based sensor fault detection in wireless sensor network for smart irrigation," *Results Eng.*, vol. 20, Dec. 2023, Art. no. 101395.
- [76] T. Sharma, A. Singh, N. Kumar, and G. Chauhan, "Digital technologies and tools for the ensuing digital era," in *Applying Drone Technologies and Robotics for Agricultural Sustainability*. Hershey, PA, USA: IGI Global, 2023, pp. 1–11.
- [77] R. Abbasi, P. Martinez, and R. Ahmad, "The digitization of agricultural industry—A systematic literature review on agriculture 4.0," *Smart Agricult. Technol.*, vol. 2, Dec. 2022, Art. no. 100042.
- [78] G. Singh, N. Kalra, N. Yadav, A. Sharma, and M. Saini, "Smart agriculture: A review," *Siberian J. Life Sci. Agricult.*, vol. 14, no. 6, pp. 423–454, 2022.
- [79] M. Alrizq, S. Stalin, S. Alyami, V. Roy, A. Mishra, A. K. Chandanan, N. A. Awad, and P. Venkatesh, "Optimization of sensor node location utilizing artificial intelligence for mobile wireless sensor network," *Wireless Netw.*, vol. 29, pp. 1–13, 2023, doi: [10.1007/s11276-023-03469-4](https://doi.org/10.1007/s11276-023-03469-4).
- [80] K. Radhakrishnan, D. Ramakrishnan, O. I. Khalaf, M. Uddin, C.-L. Chen, and C.-M. Wu, "A novel deep learning-based cooperative communication channel model for wireless underground sensor networks," *Sensors*, vol. 22, no. 12, p. 4475, Jun. 2022.
- [81] M. Z. Hasan and Z. M. Hanapi, "Efficient and secured mechanisms for data link in IoT WSNs: A literature review," *Electronics*, vol. 12, no. 2, p. 458, Jan. 2023.
- [82] M. Z. Hasan, Z. M. Hanapi, and M. Z. Hussain, "Wireless sensor security issues on data link layer: A survey," *Comput., Mater. Continua*, vol. 75, no. 2, pp. 4065–4084, 2023.
- [83] K. L. Raju and V. Vijayaraghavan, "Architecture development with measurement index for agriculture decision-making system using Internet of Things and machine learning," *Multimedia Tools Appl.*, vol. 82, no. 23, pp. 36119–36142, Sep. 2023.
- [84] R. P. Sharma, R. Dharavath, and D. R. Edla, "IoT-FIS: Internet of Farm Things based prediction for crop pest infestation using optimized fuzzy inference system," *Internet Things*, vol. 21, Apr. 2023, Art. no. 100658.
- [85] A. Narwaria and A. P. Mazumdar, "Software-defined wireless sensor network: A comprehensive survey," *J. Netw. Comput. Appl.*, vol. 215, Jun. 2023, Art. no. 103636.
- [86] A. Hamza and T. Tripp, "Optical wireless communication for the Internet of Things: Advances, challenges, and opportunities," *Authorea Preprints*, vol. 1, p. 27, 2023, doi: [10.36227/techrxiv.12659789.v2](https://doi.org/10.36227/techrxiv.12659789.v2).

- [87] L. Alhoraibi, D. Alghazzawi, R. Alhebshi, and O. B. J. Rabie, "Physical layer authentication in wireless networks-based machine learning approaches," *Sensors*, vol. 23, no. 4, p. 1814, Feb. 2023.
- [88] A. Akbas and S. Buyrukoglu, "Stacking ensemble learning-based wireless sensor network deployment parameter estimation," *Arabian J. Sci. Eng.*, vol. 48, no. 8, pp. 9739–9748, Aug. 2023.
- [89] G. V. Gurram, N. C. Shariff, and R. L. Biradar, "A secure energy aware meta-heuristic routing protocol (SEAMHR) for sustainable IoT-wireless sensor network (WSN)," *Theor. Comput. Sci.*, vol. 930, pp. 63–76, Sep. 2022.
- [90] I. Dey and N. Marchetti, "Space-time- and frequency-spreading for interference minimization in dense IoT," *IEEE Internet Things Mag.*, vol. 6, no. 1, pp. 148–153, Mar. 2023.
- [91] S. Lata, S. Mehruz, and S. Urooj, "Secure and reliable WSN for Internet of Things: Challenges and enabling technologies," *IEEE Access*, vol. 9, pp. 161103–161128, 2021.
- [92] N. R. Patel, S. Kumar, and S. K. Singh, "Energy and collision aware WSN routing protocol for sustainable and intelligent IoT applications," *IEEE Sensors J.*, vol. 21, no. 22, pp. 25282–25292, Nov. 2021.
- [93] C. Fathy and H. M. Ali, "A secure IoT-based irrigation system for precision agriculture using the expeditious cipher," *Sensors*, vol. 23, no. 4, p. 2091, Feb. 2023.
- [94] F. Ojeda, D. Mendez, A. Fajardo, and F. Ellinger, "On wireless sensor network models: A cross-layer systematic review," *J. Sensor Actuator Netw.*, vol. 12, no. 4, p. 50, Jun. 2023.
- [95] D. W. Wajgi and J. V. Tembhurne, "Localization in wireless sensor networks and wireless multimedia sensor networks using clustering techniques," *Multimedia Tools Appl.*, vol. 82, pp. 1–51, 2023, doi: 10.1007/s11042-023-15956-z.
- [96] R. W. Anwar, F. Outay, K. N. Qureshi, S. Iqbal, and K. Z. Ghafoor, "State-based energy calculation scheme for Internet of Things networks," *IEEE Access*, vol. 11, pp. 106967–106979, 2023.
- [97] S. Atalla, S. Tarapiah, A. Gawanmeh, M. Daradkeh, H. Mukhtar, Y. Himeur, W. Mansoor, K. F. B. Hashim, and M. Daadoo, "IoT-enabled precision agriculture: Developing an ecosystem for optimized crop management," *Information*, vol. 14, no. 4, p. 205, Mar. 2023.
- [98] H. Benyezza, M. Bouhedda, R. Kara, and S. Rebouh, "Smart platform based on IoT and WSN for monitoring and control of a greenhouse in the context of precision agriculture," *Internet Things*, vol. 23, Oct. 2023, Art. no. 100830.
- [99] A. A. Bahashwan, M. Anbar, N. Abdullah, T. Al-Hadhrani, and S. M. Hanshi, "Review on common IoT communication technologies for both long-range network (LPWAN) and short-range network," in *Advances on smart and Soft Computing*. Cham, Switzerland: Springer, 2021, pp. 341–353.
- [100] J. L. Avinash, K. N. S. Kumar, G. B. A. Kumar, G. R. Poornima, R. Gatti, and S. S. Kumar, "A wireless sensor network based precision agriculture," in *Proc. Int. Conf. Recent Trends Electron., Inf., Commun. Technol. (RTEICT)*, Nov. 2020, pp. 413–417.
- [101] M. S. Farooq, S. Riaz, M. A. Helou, F. S. Khan, A. Abid, and A. Alvi, "Internet of Things in greenhouse agriculture: A survey on enabling technologies, applications, and protocols," *IEEE Access*, vol. 10, pp. 53374–53397, 2022.
- [102] R. Tang, N. K. Aridas, and M. S. Abu Talib, "Design of wireless sensor network for agricultural greenhouse based on improved zigbee protocol," *Agriculture*, vol. 13, no. 8, p. 1518, Jul. 2023.
- [103] G. V. K. Rayalu, P. N. Chowdary, M. Nadella, D. Harsha, P. Sathvika, and B. G. Gowri, "A ZigBee based cost-effective home monitoring system using WSN," 2023, *arXiv:2309.09332*.
- [104] A. Romputtal and C. Phongcharoenpanich, "T-slot antennas-embedded ZigBee wireless sensor network system for IoT-enabled monitoring and control systems," *IEEE Internet Things J.*, vol. 10, no. 23, pp. 20834–20845, 2023, doi: 10.1109/JIOT.2023.3284005.
- [105] J. Contreras-Castillo, J. A. Guerrero-Ibañez, P. C. Santana-Mancilla, and L. Anido-Rifón, "SAgric-IoT: An IoT-based platform and deep learning for greenhouse monitoring," *Appl. Sci.*, vol. 13, no. 3, p. 1961, Feb. 2023.
- [106] M. Aslam, W. Liu, X. Jiao, J. Haxhibeqiri, G. Miranda, J. Hoebeke, J. Marquez-Barja, and I. Moerman, "Hardware efficient clock synchronization across Wi-Fi and Ethernet-based network using PTP," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 3808–3819, Jun. 2022.
- [107] S. G. Sankaran and S. R. Gulasekaran, *Wi-Fi 6: Protocol and Network*. Norwood, MA, USA: Artech House, 2021.
- [108] V. Diya, P. Nandan, and R. R. Dhote, "IoT-based precision agriculture: A review," in *Proc. Emerg. Trends Technol. Intell. Syst. (ETTIS)*, 2022, pp. 373–386.
- [109] Y. Tang, S. Dananjayan, C. Hou, Q. Guo, S. Luo, and Y. He, "A survey on the 5G network and its impact on agriculture: Challenges and opportunities," *Comput. Electron. Agricult.*, vol. 180, Jan. 2021, Art. no. 105895.
- [110] S. B. Damsgaard, N. J. H. Marcano, M. Nørremark, R. H. Jacobsen, I. Rodriguez, and P. Mogensen, "Wireless communications for internet of farming: An early 5G measurement study," *IEEE Access*, vol. 10, pp. 105263–105277, 2022.
- [111] D. K. Singh, R. Sobti, A. Jain, P. K. Malik, and D. Le, "LoRa based intelligent soil and weather condition monitoring with Internet of Things for precision agriculture in smart cities," *IET Commun.*, vol. 16, no. 5, pp. 604–618, Mar. 2022.
- [112] M. K. Senapaty, A. Ray, and N. Padhy, "IoT-enabled soil nutrient analysis and crop recommendation model for precision agriculture," *Computers*, vol. 12, no. 3, p. 61, Mar. 2023.
- [113] I. Ünal, Ö. Kabaş, O. Eceoglu, and G. Moiceanu, "Adaptive multi-robot communication system and collision avoidance algorithm for precision agriculture," *Appl. Sci.*, vol. 13, no. 15, p. 8602, Jul. 2023.
- [114] L. Anchidin, A. Lavric, P.-M. Mutescu, A. I. Petriariu, and V. Popa, "The design and development of a microstrip antenna for Internet of Things applications," *Sensors*, vol. 23, no. 3, p. 1062, Jan. 2023.
- [115] E. Garcia-Villegas, A. Lopez-Garcia, and E. Lopez-Aguilera, "Genetic algorithm-based grouping strategy for IEEE 802.11ah networks," *Sensors*, vol. 23, no. 2, p. 862, Jan. 2023.
- [116] S.-Y. Wang, J.-E. Chang, H. Fan, and Y.-H. Sun, "Comparing the performance of NB-IoT, LTE Cat-M1, Sigfox, and LoRa for IoT end devices moving at high speeds in the air," *J. Signal Process. Syst.*, vol. 94, no. 1, pp. 81–99, Jan. 2022.
- [117] D. K. Singh and R. Sobti, "Wireless communication technologies for Internet of Things and precision agriculture: A review," in *Proc. 6th Int. Conf. Signal Process., Comput. Control (ISPCC)*, Oct. 2021, pp. 765–769.
- [118] H. Klaina, I. P. Guembe, P. Lopez-Iturri, M. Á. Campo-Bescós, L. Azpilicueta, O. Aghzout, A. V. Alejos, and F. Falcone, "Analysis of low power wide area network wireless technologies in smart agriculture for large-scale farm monitoring and tractor communications," *Measurement*, vol. 187, Jan. 2022, Art. no. 110231.
- [119] A. Alzuhaier and A. Alghaihab, "The design and optimization of an acoustic and ambient sensing AIoT platform for agricultural applications," *Sensors*, vol. 23, no. 14, p. 6262, Jul. 2023.
- [120] M. Farmani, S. Farnam, M. Khani, Z. Torabi, and Z. Shirmohammadi, "Comprehensive review of modern computing paradigms architectures for intelligent agriculture," *J. Electr. Comput. Eng. Innov. (JECEI)*, 2023.
- [121] N. Islam, M. M. Rashid, F. Pasandideh, B. Ray, S. Moore, and R. Kadel, "A review of applications and communication technologies for Internet of Things (IoT) and unmanned aerial vehicle (UAV) based sustainable smart farming," *Sustainability*, vol. 13, no. 4, p. 1821, Feb. 2021.
- [122] Y. N. Goh, D. Jamaludin, H. H. Harith, A. Y. Alkhaled, N. A. A. Latiff, and S. A. Aziz, "Long range wide area network (LoRaWAN) for oil palm soil monitoring," in *IoT and AI in Agriculture: Self-Sufficiency in Food Production to Achieve Society 5.0 and SDG's Globally*. Cham, Switzerland: Springer, 2023, pp. 97–124.
- [123] D. Botturi, A. Depari, P. Ferrari, A. Flammini, S. Pasinetti, M. Soprani, and E. Sisinni, "LoRaVine: Using LoRaWAN for smart vineyards microclimate monitoring," in *Proc. IEEE Sensors Appl. Symp. (SAS)*, Jul. 2023, pp. p1–6.
- [124] A. Pagano, D. Croce, I. Tinnirello, and G. Vitale, "A survey on LoRa for smart agriculture: Current trends and future perspectives," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 3664–3679, Feb. 2023.
- [125] S. Mythili, K. Nithya, M. Krishnamoorthi, and M. Kalamani, "Standards and protocols for Agro-IoT," in *Cloud IoT Systems for Smart Agricultural Engineering*. Boca Raton, FL, USA: CRC Press, 2022, pp. 187–208.
- [126] M. A. M. Almuhaaya, W. A. Jabbar, N. Sulaiman, and S. Abdulmalek, "A survey on LoRaWAN technology: Recent trends, opportunities, simulation tools and future directions," *Electronics*, vol. 11, no. 1, p. 164, 2022.
- [127] H. Taleb, G. Andrieux, A. Nasser, and N. Charara, "Energy efficient selection of spreading factor in LoRaWAN-based WBAN medical systems," *Internet Things*, vol. 24, Dec. 2023, Art. no. 100896.
- [128] M. Swain, D. Zimon, R. Singh, M. F. Hashmi, M. Rashid, and S. Hakak, "LoRa-LBO: An experimental analysis of LoRa link budget optimization in custom build IoT test bed for agriculture 4.0," *Agronomy*, vol. 11, no. 5, p. 820, Apr. 2021.

- [129] M. Saban, M. Bakkour, I. Amdaouch, J. El Gueri, B. A. Ahmed, M. Z. Chaari, J. Ruiz-Alzola, A. Rosado-Muñoz, and O. Aghzout, "A smart agricultural system based on PLC and a cloud computing web application using LoRa and LoRaWan," *Sensors*, vol. 23, no. 5, p. 2725, Mar. 2023.
- [130] G. Yascariyab, M. Huerta, M. Silva, and R. Clotet, "Performance evaluation of communication systems used for Internet of Things in agriculture," *Agriculture*, vol. 12, no. 6, p. 786, May 2022.
- [131] F. Kuntke, V. Romanenko, S. Linsner, E. Steinbrink, and C. Reuter, "LoRaWAN security issues and mitigation options by the example of agricultural IoT scenarios," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 5, May 2022, Art. no. e4452.
- [132] F. P. Correia, S. R. D. Silva, F. B. S. D. Carvalho, M. S. D. Alencar, K. D. R. Assis, and R. M. Bacurau, "LoRaWAN gateway placement in smart agriculture: An analysis of clustering algorithms and performance metrics," *Energies*, vol. 16, no. 5, p. 2356, Mar. 2023.
- [133] A. Mhaned, S. Mouatassim, M. El Haji, and J. Benhra, "Low-cost smart irrigation system based on Internet of Things and fuzzy logic," in *Proc. Int. Conf. Smart Appl. Data Anal.*, Marrakesh, Morocco, Sep. 2022, pp. 78–89.
- [134] L. Patroo, K. Thacooree, and A. Mungur, "A smart precision irrigation and monitoring system," in *Proc. Int. Conf. Emerg. Trends Elect., Electron. Commun. Eng.* Cham, Switzerland: Springer, Nov. 2018, pp. 105–114.
- [135] D. J. Hemanth, V. A. Kumar, S. Malathi, O. Castillo, and B. Patrut, "Emerging trends in computing and expert technology," in *Lecture Notes on Data Engineering and Communications Technologies Impact Factor & Key Scientometrics*, vol. 35, 2019. [Online]. Available: <https://doi.org/10.1007/978-3-030-32150-5>
- [136] O. P. Bodunde, U. C. Adie, O. M. Ikumapayi, J. O. Akinyoola, and A. A. Aderoba, "Architectural design and performance evaluation of a ZigBee technology based adaptive sprinkler irrigation robot," *Comput. Electron. Agricult.*, vol. 160, pp. 168–178, May 2019.
- [137] Y. Peng, Y. Xiao, Z. Fu, Y. Dong, Y. Zheng, H. Yan, and X. Li, "Precision irrigation perspectives on the sustainable water-saving of field crop production in China: Water demand prediction and irrigation scheme optimization," *J. Cleaner Prod.*, vol. 230, pp. 365–377, Sep. 2019.
- [138] W. Ruiz Martínez, Y. Díaz-Gutiérrez, R. Ferro-Escobar, and L. Pallares, "Application of the Internet of Things through a network of wireless sensors in a coffee crop for monitoring and control its environmental variables," *Tecnológicas*, vol. 22, no. 46, pp. 155–170, Sep. 2019.
- [139] A. Izaddoost, E. Ogodo, and S. Prasai, "Enhanced data transmission platform in smart farms," in *Proc. ACM Int. Conf. Omni-Layer Intell. Syst. (COINS)*, New York, NY, USA, 2019, pp. 58–61.
- [140] X. Jia, Y. Huang, Y. Wang, and D. Sun, "Research on water and fertilizer irrigation system of tea plantation," *Int. J. Distrib. Sensor Netw.*, vol. 15, no. 3, Mar. 2019, Art. no. 155014771984018.
- [141] J. John, V. S. Palaparthi, G. S. Kasbekar, and M. S. Baghini, "A multi-hop wireless sensor network for in-situ agricultural applications," in *Proc. URSI Asia-Pacific Radio Sci. Conf. (AP-RASC)*, Mar. 2019, pp. 1–4.
- [142] M. Subathra, C. J. Blessing, S. Thomas George, A. Thomas, A. D. Raj, and V. Edwards, "Automated intelligent wireless drip irrigation using ANN techniques," in *Advances in Big Data and Cloud Computing*. Cham, Switzerland: Springer, 2019, pp. 555–568.
- [143] C. Jinbo, C. Xiangliang, F. Han-Chi, and A. Lam, "Agricultural product monitoring system supported by cloud computing," *Cluster Comput.*, vol. 22, no. S4, pp. 8929–8938, Jul. 2019.
- [144] A. N. Harun, N. Mohamed, R. Ahmad, A. R. A. Rahim, and N. N. Ani, "Improved Internet of Things (IoT) monitoring system for growth optimization of brassica chinensis," *Comput. Electron. Agricult.*, vol. 164, Sep. 2019, Art. no. 104836.
- [145] B. Keswani, A. G. Mohapatra, A. Mohanty, A. Khanna, J. J. P. C. Rodrigues, D. Gupta, and V. H. C. de Albuquerque, "Adapting weather conditions based IoT enabled smart irrigation technique in precision agriculture mechanisms," *Neural Comput. Appl.*, vol. 31, no. S1, pp. 277–292, Jan. 2019.
- [146] C. Lijun and Z. Yubo, "Analysis of intelligent agricultural system and control mode based on fuzzy control and sensor network," *J. Intell. Fuzzy Syst.*, vol. 37, no. 5, pp. 6325–6336, Nov. 2019.
- [147] F. Miao, X. Lu, B. Tao, K. Liu, and D. Liu, "Design of Rice traceability system based on WSN and RFID," in *Proc. Int. Conf. Intell. Interact. Syst. Appl.* Cham, Switzerland: Springer, 2019, pp. 123–130.
- [148] R. Y. Rao, J. J. Koola, N. D. Mehta, and A. M. Haque, "Design and implementation of adaptive control algorithm for IoT based domestic irrigation system," in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, pp. 1–6.
- [149] M. M. Maha, S. Bhuiyan, and M. Masduzzaman, "Smart board for precision farming using wireless sensor network," in *Proc. Int. Conf. Robot., Elect. Signal Process. Techn. (ICREST)*, Jan. 2019, pp. 445–450.
- [150] G. Urkude and M. Pandey, "AgriSense: Automatic irrigation utility system using wireless sensor network and web of things," in *Proc. 2nd Int. Conf. Adv. Comput. Commun. Paradigms (ICACCP)*, Feb. 2019, pp. 1–6.
- [151] P. Varalakshmi, B. Y. Sivashakthivadhani, and B. L. Sakthiram, "Automatic plant escalation monitoring system using IoT," in *Proc. 3rd Int. Conf. Comput. Commun. Technol. (ICCCCT)*, Feb. 2019, pp. 212–216.
- [152] D. Thakur, Y. Kumar, and S. Vijendra, "Smart irrigation and intrusions detection in agricultural fields using IoT," *Proc. Comput. Sci.*, vol. 167, pp. 154–162, Jan. 2020.
- [153] T. Talaviya, D. Shah, N. Patel, H. Yagnik, and M. Shah, "Implementation of artificial intelligence in agriculture for optimisation of irrigation and application of pesticides and herbicides," *Artif. Intell. Agricult.*, vol. 4, pp. 58–73, Jan. 2020.
- [154] S. A. O'Shaughnessy, M. Kim, M. A. Andrade, P. D. Colaizzi, and S. R. Evett, "Site-specific irrigation of grain sorghum using plant and soil water sensing feedback—Texas high plains," *Agricult. Water Manage.*, vol. 240, Oct. 2020, Art. no. 106273.
- [155] C. Yuanyuan and Z. Zuozhuang, "Research and design of intelligent water-saving irrigation control system based on WSN," in *Proc. IEEE Int. Conf. Artif. Intell. Comput. Appl. (ICAICA)*, Jun. 2020, pp. 695–697.
- [156] Y.-L. Huang, J.-Y. Tian, and T.-Q. Shang, "Design of monitoring system in smart agriculture environment," *DEStech Trans. Social Sci., Educ. Hum. Sci.*, 2020, doi: [10.12783/dtssehs/icesd2020/34560](https://doi.org/10.12783/dtssehs/icesd2020/34560).
- [157] H. Ningombam and O. Roy, "A systematic approach on design of automation system for tea farming based on embedded wireless sensor network," in *Inventive Computation Technologies 4*. Cham, Switzerland: Springer, 2020, pp. 286–296.
- [158] M. W. Rasooli, B. Bhushan, and N. Kumar, "Applicability of wireless sensor networks & IoT in saffron & wheat crops: A smart agriculture perspective," *Int. J. Sci. Technol. Res.*, vol. 9, no. 2, pp. 2456–2461, 2020.
- [159] S. Borah, R. Kumar, and S. Mukherjee, "Low-cost IoT framework for irrigation monitoring and control," *Int. J. Intell. Unmanned Syst.*, vol. 9, no. 1, pp. 63–79, Jun. 2020.
- [160] A. Kalam, K. R. Niazi, A. Soni, S. A. Siddiqui, and A. Mundra, *Intelligent Computing Techniques for Smart Energy Systems: Proceedings of ICTSES 2018*. Cham, Switzerland: Springer, 2020.
- [161] Y. Yang, "Design and application of intelligent agriculture service system with LoRa-based on wireless sensor network," in *Proc. Int. Conf. Comput. Eng. Appl. (ICCEA)*, Mar. 2020, pp. 712–716.
- [162] I. Dasgupta, J. Saha, P. Venkatasubbu, and P. Ramasubramanian, "AI crop predictor and weed detector using wireless technologies: A smart application for farmers," *Arabian J. Sci. Eng.*, vol. 45, no. 12, pp. 11115–11127, Dec. 2020.
- [163] J. Freeda and J. Josepha menandas, "IoT based innovation schemes in smart irrigation system with pest control," in *Emerging Trends in Computing and Expert Technology*. Cham, Switzerland: Springer, 2020, pp. 657–669.
- [164] D. D. Dasig, "Implementing IoT and wireless sensor networks for precision agriculture," in *Internet of Things and Analytics for Agriculture (Studies in Big Data)*, vol. 67, P. K. Pattnaik, R. Kumar, and S. Pal, Eds. Singapore: Springer, 2020, pp. 23–44.
- [165] G. Li, D. Li, W. Chen, Y. Zhang, and S. Xu, "Design and application of special sensors and Internet of Things (IoT)-based wireless system for agricultural information monitor," *J. Phys., Conf. Ser.*, vol. 1646, no. 1, Sep. 2020, Art. no. 012130.
- [166] X. Lou, L. Zhang, X. Zhang, J. Fan, X. Hu, and C. Li, "Design of intelligent farmland environment monitoring system based on wireless sensor network," *J. Phys., Conf. Ser.*, vol. 1635, no. 1, Nov. 2020, Art. no. 012031.
- [167] K. V. K. Kishore, B. Y. Kumar, and S. Venkatramaphanikumar, "Optimized water scheduling using IoT sensor data in smart farming," in *Smart Technologies in Data Science and Communication*. Cham, Switzerland: Springer, 2020, pp. 205–219.

- [168] R. Singh, A. Gehlot, A. K. Thakur, M. Swain, and S. V. Akram, "Wireless sensor network with power management system for water level regulation in paddy fields," *Int. J. Innov. Technol. Exploring Eng.*, vol. 9, no. 7, pp. 1243–1246, May 2020.
- [169] S. Prakash, "Zigbee based wireless sensor network architecture for agriculture applications," in *Proc. 3rd Int. Conf. Smart Syst. Inventive Technol. (ICSSIT)*, Aug. 2020, pp. 709–712.
- [170] L. Ren, X. Zhai, Y. Yang, and J. Xu, "Design of horticultural wireless intelligent maintenance system based on STM32 and Android," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 474, no. 3, Apr. 2020, Art. no. 032016.
- [171] S. K. Roy, S. Misra, N. S. Raghuvanshi, and S. K. Das, "AgriSens: IoT-based dynamic irrigation scheduling system for water management of irrigated crops," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 5023–5030, Mar. 2021.
- [172] D. Sharma and G. Singh Tomar, "Comparative energy evaluation of LEACH protocol for monitoring soil parameter in wireless sensors network," *Mater. Today, Proc.*, vol. 29, pp. 381–396, Jan. 2020.
- [173] T. Thaher and I. Ishaq, "Cloud-based Internet of Things approach for smart irrigation system: Design and implementation," in *Proc. Int. Conf. Promising Electron. Technol. (ICPET)*, Dec. 2020, pp. 32–37.
- [174] A. J. Ramadhan, "Smart glasshouse system supported by global system for mobile communications and Internet of Things: Case study: Tomato plant," *J. Eng. Sci. Technol.*, vol. 15, pp. 3067–3081, Jan. 2020.
- [175] A. Srivastava, D. K. Das, and R. Kumar, "Monitoring of soil parameters and controlling of soil moisture through IoT based smart agriculture," in *Proc. IEEE Students Conf. Eng. Syst. (SCES)*, Jul. 2020, pp. 1–6.
- [176] R. P. Sharma, D. Ramesh, P. Pal, S. Tripathi, and C. Kumar, "IoT-enabled IEEE 802.15.4 WSN monitoring infrastructure-driven fuzzy-logic-based crop pest prediction," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 3037–3045, Feb. 2022.
- [177] E. Gomez, J. E. Duque, A. J. Rojas, C. C. Jaik, and J. A. Pertuz, "Agro-smart Caribe: Soil moisture measurement system," in *Proc. Workshop Eng. Appl. Cham, Switzerland: Springer*, 2021, pp. 422–434.
- [178] M. A. Guillén-Navarro, R. Martínez-España, B. López, and J. M. Cecilia, "A high-performance IoT solution to reduce frost damages in stone fruits," *Concurrency Comput., Pract. Exper.*, vol. 33, no. 2, Jan. 2021, Art. no. e5299.
- [179] M. Hamdi, A. Rehman, A. Alghamdi, M. A. Nizamani, M. M. S. Missen, and M. A. Memon, "Internet of Things (IoT) based water irrigation system," *Int. J. Online Biomed. Eng.*, vol. 17, no. 5, pp. 69–80, 2021.
- [180] B. Umar, E. M. Dogo, B. K. Nuhu, A. K. Haq, and P. T. Olaleye, "The design and performance evaluation of a wireless sensor network based irrigation system on different soil types," *J. Digit. Food, Energy Water Syst.*, vol. 2, no. 2, pp. 1–27, Dec. 2021.
- [181] N. Kuchekar and R. Pagare, "Design & implementation of automatic irrigation system using wireless sensor network & ZigBee module," *Int. J. Innov. Eng. Res. Technol.*, pp. 1–8, 2021. [Online]. Available: <https://repo.ijert.org/index.php/ijert/article/view/669>
- [182] P. Gao, J. Xie, M. Yang, P. Zhou, W. Chen, G. Liang, Y. Chen, X. Han, and W. Wang, "Improved soil moisture and electrical conductivity prediction of citrus orchards based on IoT using deep bidirectional LSTM," *Agriculture*, vol. 11, no. 7, p. 635, Jul. 2021.
- [183] A. Z. A. Naji and A. M. Salman, "Water saving in agriculture through the use of smart irrigation system," in *Proc. 4th Int. Conf. Data Storage Data Eng.*, Feb. 2021, pp. p153–160.
- [184] A. Eraliev and G. Bracco, "Design and implementation of ZigBee based low-power wireless sensor and actuator network (WSAN) for automation of urban garden irrigation systems," in *Proc. IEEE Int. IoT, Electron. Mechatronics Conf. (IEMTRONICS)*, Apr. 2021, pp. 1–7.
- [185] D. Qiong and P. Hao, "Design and implementation of irrigation water saving control system based on WSN," in *Proc. Int. Conf. Intell. Transp., Big Data Smart City (ICITBS)*, Mar. 2021, pp. 75–78.
- [186] P. Rajkumar, C. Vengatesh, M. Palpandi, T. Gopu, R. Jayapandiprathap, T. Kalimuthu, and Y. Harold Robinson, "Smart control and monitoring of irrigation system using Internet of Things," in *Multimedia Technol. Internet Things Environment*. Cham, Switzerland: Springer, 2021, pp. 1–11.
- [187] H. Zhou and Z. Liu, "Design of irrigation system based on ZigBee," in *Proc. 3rd Int. Conf. Artif. Intell. Adv. Manuf.*, Oct. 2021, pp. 940–944.
- [188] T. Sai, B. Proeung, S. Tep, S. Chhorn, R. Pec, V. Nall, P. Ket, C. Oeuring, and C. Hel, "Prototyping of smart irrigation system using IoT technology," in *Proc. 7th Int. Conf. Electr., Electron. Inf. Eng. (ICEEIE)*, Oct. 2021, pp. 1–5.
- [189] L. Liu, S. Hua, and Q. Lai, "Automatic control system of balancing agricultural stereo cultivation based on wireless sensors," *IEEE Sensors J.*, vol. 21, no. 16, pp. 17517–17524, Aug. 2021.
- [190] G. Oussama, A. Rami, F. Tarek, A. S. Alanazi, and M. Abid, "Fast and intelligent irrigation system based on WSN," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–13, Jul. 2022.
- [191] H. Alqahtani, "Role of wireless sensor network in precision agriculture," *Comput. Algorithms Numer. Dimensions*, vol. 1, no. 2, pp. 84–88, 2022.
- [192] E. P. Kho, S. N. D. Chua, S. F. Lim, L. C. Lau, and M. T. N. Gani, "Development of young sago palm environmental monitoring system with wireless sensor networks," *Comput. Electron. Agricult.*, vol. 193, Feb. 2022, Art. no. 106723.
- [193] L. Wei, Z. Gao, Y. Liu, W. Ma, N. Li, and K. Ma, "ZigBee based intelligent bird driving system design," in *Proc. IEEE 5th Int. Conf. Electron. Technol. (ICET)*, May 2022, pp. 998–1002.
- [194] B. Motamedi and B. Villányi, "Design of a smart irrigation using wireless communication protocols in greenhouse," in *Proc. IEEE 22nd Int. Symp. Comput. Intell. Informat. 8th IEEE Int. Conf. Recent Achievements Mechatronics, Autom., Comput. Sci. Robot. (CINTI-MACRo)*, Nov. 2022, pp. 000179–000184.
- [195] L. Zhao, "Design of intelligent water-saving irrigation system based on Internet of Things," *Wireless Eng. Technol.*, vol. 13, no. 3, pp. 33–40, 2022.
- [196] S. Choudhury, R. Singh, A. Gehlot, P. Kuchhal, S. V. Akram, N. Priyadarshi, and B. Khan, "Agriculture field automation and digitization using Internet of Things and machine learning," *J. Sensors*, vol. 2022, pp. 1–17, Nov. 2022.
- [197] H. Liu, "Agricultural water management based on the Internet of Things and data analysis," *Acta Agriculturae Scandinavica, Sect. B Soil Plant Sci.*, vol. 72, no. 1, pp. 300–311, Dec. 2022.
- [198] O. Ghorbel, T. Frikha, A. Hajji, R. Alabdal, R. Ayadi, and M. A. Elmasry, "Blockchain-based supply chain system for olive fields using WSNs," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–11, Sep. 2022.
- [199] S. Premkumar and A. Sigappi, "IoT-enabled edge computing model for smart irrigation system," *J. Intell. Syst.*, vol. 31, no. 1, pp. 632–650, May 2022.
- [200] D. M. N. A. Ashir, M. T. Ahad, M. Talukder, and T. Rahman, "Internet of Things (IoT) based smart agriculture aiming to achieve sustainable goals," 2022, *arXiv:2206.06300*.
- [201] G. Routis and I. Roussaki, "Low power IoT electronics in precision irrigation," *Smart Agricult. Technol.*, vol. 5, Oct. 2023, Art. no. 100310.
- [202] M. Alagarsam, S. R. Devakadacham, H. Subramani, S. Viswanathan, J. Johnmathew, and K. Suriyan, "Automation irrigation system using Arduino for smart crop field productivity," *Int. J. Reconfigurable Embedded Syst. (IJRES)*, vol. 12, no. 1, p. 70, Mar. 2023.
- [203] E. Mohammed, "Implementation of WSN based smart irrigation system," *Przegląd Elektrotechniczny*, vol. 1, no. 6, pp. 29–33, Jun. 2023.
- [204] N. J. Pierre, B. Sefu, S. Venuste, M. J. D'Amour, K. Daniel, N. J. Pierre, K. J. Bosco, and H. Felix, "Smart crops irrigation system with low energy consumption," *Academic Soc. Appropriate Technol.*, vol. 9, no. 1, pp. 9–19, Apr. 2023.
- [205] V. Rajagopal, B. Velusamy, M. Krishnan, and S. Rathinasamy, "Energy efficient data gathering using mobile sink in IoT for reliable irrigation," *Sustain. Comput., Informat. Syst.*, vol. 40, Dec. 2023, Art. no. 100916.
- [206] D. Srivastava, J. Divya, A. Sudarshanam, M. Praveen, U. Mutheeswaran, and R. Krishnamoorthy, "Wireless sensor network and Internet of Things-based smart irrigation system for farming," in *Proc. Int. Conf. Inventive Comput. Technol. (ICICT)*, Apr. 2023, pp. 1246–1250.
- [207] M. F. M. Jaafar, H. Hussin, R. Rosman, T. Y. Kheng, and M. J. Hussin, "Smart cocoa nursery monitoring system using IRT for automatic drip irrigation," in *Proc. IEEE 13th Int. Conf. Telecommun. Syst., Services, Appl. (TSSA)*, Oct. 2019, pp. 108–113.
- [208] Z. Wan, Y. Song, and Z. Cao, "Environment dynamic monitoring and remote control of greenhouse with ESP8266 NodeMCU," in *Proc. IEEE 3rd Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, Mar. 2019, pp. 377–382.
- [209] F. Kamaruddin, N. N. N. A. Malik, N. A. Murad, N. M. A. Latiff, S. K. S. Yusof, and S. A. Hamzah, "IoT-based intelligent irrigation management and monitoring system using Arduino," *TELKOMNIKA (Telecommun. Comput. Electron. Control)*, vol. 17, no. 5, p. 2378, Oct. 2019.

- [210] A. H. Adam, R. Tamilkodi, and K. V. Madhavi, "Low-cost green power predictive farming using IoT and cloud computing," in *Proc. Int. Conf. Vis. Towards Emerg. Trends Commun. Netw. (ViTECoN)*, Mar. 2019, pp. 1–5.
- [211] J. L. García-Navas, M. Parra-Boronat, L. Parra-Boronat, J. Rocher-Morant, S. Sendra, and J. Lloret, "Practical study of the temperature effect in soil moisture measurements," in *Proc. 8th Int. Conf. Commun., Comput., Netw. Technol.*, 2019, pp. 7–13.
- [212] J. D. F. Selvaraj, P. M. Paul, and I. D. J. Jingle, "Automatic wireless water management system (SPWMS) for smart vineyard irrigation using IoT technology," *Int. J. Oceans Oceanogr.*, vol. 13, no. 1, pp. 211–218, 2019.
- [213] D. D. K. Rathinam, D. Surendran, A. Shilpa, A. S. Grace, and J. Sherin, "Modern agriculture using wireless sensor network (WSN)," in *Proc. 5th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Mar. 2019, pp. 515–519.
- [214] R. Muley and V. Bhonge, "Internet of Things for irrigation monitoring and controlling," in *Computing, Communication and Signal Processing*. Cham, Switzerland: Springer, 2019, pp. 165–174.
- [215] H. Sharma, A. Haque, and Z. A. Jaffery, "Smart agriculture monitoring using energy harvesting Internet of Things (EH-IoT)," *World Sci. News*, vol. 121, pp. 17–21, Jan. 2019.
- [216] A. Gloria, C. Dionisio, G. Simões, P. Sebastião, and N. Souto, "WSN application for sustainable water management in irrigation systems," in *Proc. IEEE 5th World Forum Internet Things (WF-IoT)*, Apr. 2019, pp. 833–836.
- [217] S. Kumar and N. Ch, "IoT based multi-sensor data acquisition system for the application of smart agriculture," in *Proc. Int. Conf. Comput. Intell., Secur. Internet of Things*, Agartala, India, Dec. 2019, pp. 329–342.
- [218] K. N. Bhanu, H. S. Mahadevaswamy, and H. J. Jasmine, "IoT based smart system for enhanced irrigation in agriculture," in *Proc. Int. Conf. Electron. Sustain. Commun. Syst. (ICESC)*, Jul. 2020, pp. 760–765.
- [219] M. Bogdanoff and S. Tayeb, "An ISM-band automated irrigation system for agriculture IoT," in *Proc. IEEE Int. IoT, Electron. Mechatronics Conf. (IEMTRONICS)*, Sep. 2020, pp. 1–6.
- [220] A. Umarov, M. Kunelbayev, M. Satymbekov, G. Turken, B. Alimbayeva, K. Imanzhanova, and L. Duisembayeva, "Micro climate monitoring system for a home greenhouse as part of esp32," *Test Eng. Manage.*, vol. 82, nos. 1–2, pp. 4564–4573, 2020.
- [221] P. Visconti, N. I. Giannoccaro, R. D. Fazio, S. Strazzella, and D. Cafagna, "IoT-oriented software platform applied to sensors-based farming facility with smartphone farmer app," *Bull. Electr. Eng. Informat.*, vol. 9, no. 3, pp. 1095–1105, Jun. 2020.
- [222] B. Alhasnawi, B. Jasim, and B. Issa, "Internet of Things (IoT) for smart precision agriculture," *Iraqi J. Electr. Electron. Eng.*, vol. 16, no. 1, pp. 1–11, Jun. 2020.
- [223] L. García, L. Parra, J. M. Jimenez, J. Lloret, P. V. Mauri, and P. Lorenz, "DronAway: A proposal on the use of remote sensing drones as mobile gateway for WSN in precision agriculture," *Appl. Sci.*, vol. 10, no. 19, p. 6668, Sep. 2020.
- [224] A. J. López Rivero, C. A. Martínez Alayón, R. Ferro, D. Hernández de la Iglesia, and V. Alonso Secades, "Network traffic modeling in a Wi-Fi system with intelligent soil moisture sensors (WSN) using IoT applications for potato crops and ARIMA and SARIMA time series," *Appl. Sci.*, vol. 10, no. 21, p. 7702, Oct. 2020.
- [225] A. Salam, U. Raza, "Autonomous irrigation management in decision agriculture," in *Signals in the Soil: Developments in Internet of Underground Things*, 2020, pp. 379–398. [Online]. Available: https://doi.org/10.1007/978-3-030-50861-6_12
- [226] A. Barman, B. Neogi, and S. Pal, "Solar-powered automated IoT-based drip irrigation system," in *IoT and Analytics for Agriculture*. Cham, Switzerland: Springer, 2020, pp. 27–49.
- [227] K. S. Krishnan, K. Jerusha, P. Tanwar, and S. Singhal, "Self-automated agriculture system using IoT," *Int. J. Recent Technol. Eng.*, vol. 8, pp. 758–762, 2020.
- [228] B. Karaduman, M. Challenger, R. Eslampanah, J. Denil, and H. Vangheluwe, "Platform-specific modeling for RIOT based IoT systems," in *Proc. Int. Conf. Softw. Eng. (ICSE) Workshops*, 2020, pp. 639–646.
- [229] K. Anil Kumar and A. D, "An Internet of thing based agribot (IoT-Agribot) for precision agriculture and farm monitoring," *Int. J. Educ. Manage. Eng.*, vol. 10, no. 4, pp. 33–39, Aug. 2020.
- [230] J. A. Coelho, A. Glória, and P. Sebastião, "Precise water leak detection using machine learning and real-time sensor data," *IoT*, vol. 1, no. 2, pp. 474–493, Nov. 2020.
- [231] T. Rajesh, Y. Thrinayana, and D. Srinivasulu, "IoT based smart agriculture monitoring system," *Int. J. Innov. Technol. Explor. Eng.*, vol. 9, pp. 325–328, Jan. 2020.
- [232] R. K. Patil and S. S. Patil, "Cognitive intelligence of Internet of Things in smart agriculture applications," in *Proc. IEEE Pune Sect. Int. Conf. (PuneCon)*, Dec. 2020, pp. 129–132.
- [233] A. Glória, C. Dionisio, G. Simões, J. Cardoso, and P. Sebastião, "Water management for sustainable irrigation systems using Internet-of-Things," *Sensors*, vol. 20, no. 5, p. 1402, Mar. 2020.
- [234] P. Visconti, R. de Fazio, R. Velázquez, C. Del-Valle-Soto, and N. I. Giannoccaro, "Development of sensors-based agri-food traceability system remotely managed by a software platform for optimized farm management," *Sensors*, vol. 20, no. 13, p. 3632, Jun. 2020.
- [235] A. Sagheer, M. Mohammed, K. Riad, and M. Alhajhoj, "A cloud-based IoT platform for precision control of soilless greenhouse cultivation," *Sensors*, vol. 21, no. 1, p. 223, Dec. 2020.
- [236] N. M. Tiglaio, M. Alipio, J. V. Balanay, E. Saldivar, and J. L. Tiston, "Agrinex: A low-cost wireless mesh-based smart irrigation system," *Measurement*, vol. 161, Sep. 2020, Art. no. 107874.
- [237] C. J. H. Pornillos, M. S. O. Billones, J. D. Leonidas, E. M. A. Reyes, B. J. J. Esguerra, D. P. Bolima, and R. Concepcion, "Smart irrigation control system using wireless sensor network via Internet-of-Things," in *Proc. IEEE 12th Int. Conf. Humanoid, Nanotechnol., Inf. Technol., Commun. Control, Environ., Manage. (HNICEM)*, Dec. 2020, pp. 1–6.
- [238] F. B. Poyen, A. Ghosh, P. Kundu, S. Hazra, and N. Sengupta, "Prototype model design of automatic irrigation controller," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–17, 2021.
- [239] R. Madhumathi, T. Arumuganathan, and R. Shruthi, "Soil NPK and moisture analysis using wireless sensor networks," in *Proc. 11th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2020, pp. 1–6.
- [240] K. Muruganandam and U. Chauhan, "Soil moisture sensor nodes in IoT-based drip irrigation system for water conservation," in *Smart Sensors for Industrial Internet of Things*. Cham, Switzerland: Springer, 2021, pp. 195–205.
- [241] M. E. Karar, F. Alotaibi, A. AL Rasheed, and O. Reyad, "A pilot study of smart agricultural irrigation using unmanned aerial vehicles and IoT-based cloud system," 2021, *arXiv:2101.01851*.
- [242] M. Jiménez-Buendía, F. Soto-Valles, P. J. Blaya-Ros, A. Toledo-Moreo, R. Domingo-Miguel, and R. Torres-Sánchez, "High-density Wi-Fi based sensor network for efficient irrigation management in precision agriculture," *Appl. Sci.*, vol. 11, no. 4, p. 1628, Feb. 2021.
- [243] M. Bhattacharya, A. Roy, and J. Pal, "Smart irrigation system using Internet of Things," in *Applications of Internet of Things*. Cham, Switzerland: Springer, 2021, pp. 119–129.
- [244] V. Parashar and B. Mishra, "Designing efficient soil resistivity measurement technique for agricultural wireless sensor network," *Int. J. Commun. Syst.*, vol. 34, no. 8, May 2021, Art. no. e4785.
- [245] A. K. Podder, A. A. Bukhari, S. Islam, S. Mia, M. A. Mohammed, N. M. Kumar, K. Cengiz, and K. H. Abdulkareem, "IoT based smart agrotech system for verification of urban farming parameters," *Microprocessors Microsyst.*, vol. 82, Apr. 2021, Art. no. 104025.
- [246] A. Sengupta, B. Debnath, A. Das, and D. De, "FarmFox: A quad-sensor-based IoT box for precision agriculture," *IEEE Consum. Electron. Mag.*, vol. 10, no. 4, pp. 63–68, Jul. 2021.
- [247] L. García, "Deployment strategies of soil monitoring WSN for precision agriculture irrigation scheduling in rural areas," *Sensors*, vol. 21, no. 5, p. 1693, Mar. 2021.
- [248] M. Siddiqui, F. Akther, G. M. E. Rahman, M. M. Elahi, R. Mostafa, and K. A. Wahid, "Dimensioning of wide-area alternate wetting and drying (AWD) system for IoT-based automation," *Sensors*, vol. 21, no. 18, p. 6040, Sep. 2021.
- [249] J. Lloret, S. Sendra, L. Garcia, and J. M. Jimenez, "A wireless sensor network deployment for soil moisture monitoring in precision agriculture," *Sensors*, vol. 21, no. 21, p. 7243, Oct. 2021.
- [250] A. Shufian, M. R. Haider, and M. Hasibuzzaman, "Results of a simulation to propose an automated irrigation & monitoring system in crop production using fast charging & solar charge controller," *Cleaner Eng. Technol.*, vol. 4, Oct. 2021, Art. no. 100165.
- [251] A. Das, Y. Gupta, N. V. Wedhane, and M. R. Islam, "Smart water management in irrigation system using IoT," in *Trends in Wireless Communication and Information Security*, 2021, pp. 341–352. [Online]. Available: https://doi.org/10.1007/978-981-33-6393-9_35

- [252] A. Laha, B. Saha, A. Banerjee, P. Karmakar, D. Mukherjee, and A. Mukherjee, "IoT-based automatic irrigation scheduling using MQTT protocol," in *ICT Analysis and Applications: Proceedings of ICT4SD*. Cham, Switzerland: Springer, 2022, pp. 573–584.
- [253] J. H. Yousif and K. Abdalgader, "Experimental and mathematical models for real-time monitoring and auto watering using IoT architecture," *Computers*, vol. 11, no. 1, p. 7, Jan. 2022.
- [254] V. K. Burugari, P. Selvaraj, K. P. and C. P. Kaliappan, "An automated cloud based water utilization management system," in *Proc. Int. Conf. Comput. Commun. Informat. (ICCCI)*, Jan. 2022, pp. 1–5.
- [255] Y. A. Jeevanantham, A. Saravanan, V. Vanitha, S. Isaac, J. S. Boopathi, and D. P. Kumar, "Implementation of Internet-of Things (IoT) in soil irrigation system," in *Proc. Int. Conf. Power, Energy, Control Transmiss. Syst. (ICPECTS)*, Dec. 2022, pp. 1–5.
- [256] D. Kumar and U. Choudhury, "Agriculture-IoT-based sprinkler system for water and fertilizer conservation and management," in *Design and Development of Efficient Energy Systems*. Wiley, 2022, pp. 229–244. [Online]. Available: <https://doi.org/10.1002/9781119761785.ch13>
- [257] A. S. Kilaru, P. Madishetty, H. V. N. Yamala, and C. V. Giriraja, "Automatic remote farm irrigation system with WSN and weather forecasting," *J. Phys., Conf. Ser.*, vol. 2161, no. 1, Jan. 2022, Art. no. 012075.
- [258] J. N. Ndunagu, K. E. Ukhurebor, M. Akaaza, and R. B. Onyancha, "Development of a wireless sensor network and IoT-based smart irrigation system," *Appl. Environ. Soil Sci.*, vol. 2022, pp. 1–13, Jun. 2022.
- [259] A. B. A. Mohlisin and S. F. S. Adnan, "Palm, rubber plantation and paddy field soil monitoring system using LoRa," in *Proc. IEEE Ind. Electron. Appl. Conf. (IEACon)*, Oct. 2022, pp. 223–228.
- [260] T. D. Obasanya, I. B. Oluwafemi, O. O. Bello, and T. A. Lawal, "An Internet of Things-based irrigation and tank monitoring system," *Int. J. Informat. Commun. Technol. (IJ-ICT)*, vol. 11, no. 1, p. 65, Apr. 2022.
- [261] C. Murugamani, S. Shitharth, S. Hemalatha, P. R. Kshirsagar, K. Riyazuddin, Q. N. Naveed, S. Islam, S. P. M. Ali, and A. Batu, "Machine learning technique for precision agriculture applications in 5G-based Internet of Things," *Wireless Commun. Mobile Comput.*, vol. 2022, Jun. 2022, Art. no. 6534238.
- [262] A. Rajput, S. Chaudhary, L. Varshney, and D. Singh, "IoT based smart agriculture monitoring using node MCU AND BLYNK app," in *Proc. Int. Conf. Mach. Learn., Big Data, Cloud Parallel Comput. (COM-IT-CON)*, vol. 1, May 2022, pp. 448–451.
- [263] E. Adetiba, A. H. Ifijeh, V. Oguntosin, T. Odunuga, D. Iweala, A. Akindele, A. Abayomi, O. Obiyemi, and S. Thakur, "Development of an IoT based data acquisition and automatic irrigation system for precision agriculture," in *Proc. IEEE Nigeria 4th Int. Conf. Disruptive Technol. Sustain. Develop. (NIGERCON)*, Apr. 2022, pp. 1–5.
- [264] A. Innocent, E. Ezeji, and E. Prince, "Design and construction of solar powered automated irrigation control system using internet of thing (IoT)," *Eng. Res. J.*, vol. 2, pp. 61–76, 2022, doi: [10.46654/ERJ](https://doi.org/10.46654/ERJ).
- [265] T. Hanumann, N. V. V. S. N. Swamy, P. Gowtham, R. Sumathi, P. Chinnaasamy, and A. Kalaiarasi, "Plant monitoring system cum smart irrigation using bolt IoT," in *Proc. Int. Conf. Comput. Commun. Informat. (ICCCI)*, Jan. 2022, pp. 1–3.
- [266] B. Gupta, G. Madan, and A. Q. Md, "A smart agriculture framework for IoT based plant decay detection using smart croft algorithm," *Mater. Today, Proc.*, vol. 62, pp. 4758–4763, Jan. 2022.
- [267] J. Arshad, M. Aziz, A. A. Al-Huqail, M. H. U. Zaman, M. Husnain, A. U. Rehman, and M. Shafiq, "Implementation of a LoRaWAN based smart agriculture decision support system for optimum crop yield," *Sustainability*, vol. 14, no. 2, p. 827, Jan. 2022.
- [268] C. Suresh, M. Ravikanth, G. N. Reddy, K. B. S. Ranga, A. A. Rao, and K. Maheshwari, "Paddy crop monitoring system using IoT and deep learning," in *Innovative Data Communication Technologies and Application*. Cham, Switzerland: Springer, 2022, pp. 791–807.
- [269] M. Mudholkar, P. Mudholkar, V. H. R. Dornadula, K. Sreenivasulu, K. Joshi, and B. Pant, "A novel approach to IoT based plant health monitoring system in smart agriculture," in *Proc. 5th Int. Conf. Contemp. Comput. Informat. (IC3I)*, Dec. 2022, pp. 113–116.
- [270] S. N. M. Al-Faydi and H. N. Y. Al-Talb, "IoT and artificial neural network-based water control for farming irrigation system," in *Proc. 2nd Int. Conf. Comput. Mach. Intell. (ICMI)*, Jul. 2022, pp. 1–5.
- [271] L. Vijayaraja, R. Dhanasekar, R. Kesavan, D. Tamizhmalair, R. Premkumar, and N. Saravanan, "A cost effective agriculture system based on IoT using sustainable energy," in *Proc. 6th Int. Conf. Trends Electron. Informat. (ICOEI)*, Apr. 2022, pp. 546–549.
- [272] S. Postolache, P. Sebastião, V. Viegas, O. Postolache, and F. Cercas, "IoT-based systems for soil nutrients assessment in horticulture," *Sensors*, vol. 23, no. 1, p. 403, Dec. 2022.
- [273] P. Suanpang, P. Pothipassaa, K. Jermisittiparsert, and T. Netwong, "Integration of Kouprey-inspired optimization algorithms with smart energy nodes for sustainable energy management of agricultural orchards," *Energies*, vol. 15, no. 8, p. 2890, Apr. 2022.
- [274] B. Swaminathan, S. Palani, S. Vairavasundaram, K. Kotecha, and V. Kumar, "IoT-driven artificial intelligence technique for fertilizer recommendation model," *IEEE Consum. Electron. Mag.*, vol. 12, no. 2, pp. 109–117, Mar. 2023.
- [275] R. M. Ramli and W. A. Jabbar, "Design and implementation of solar-powered with IoT-enabled portable irrigation system," *Internet Things Cyber-Phys. Syst.*, vol. 2, pp. 212–225, Jan. 2022.
- [276] M. B. Tephila, R. A. Sri, R. Abinaya, J. A. Lakshmi, and V. Divya, "Automated smart irrigation system using IoT with sensor parameter," in *Proc. Int. Conf. Electron. Renew. Syst. (ICEARS)*, Mar. 2022, pp. 543–549.
- [277] R. K. Jain, "Experimental performance of smart IoT-enabled drip irrigation system using and controlled through web-based applications," *Smart Agricult. Technol.*, vol. 4, Aug. 2023, Art. no. 100215.
- [278] M. Guevara-Morales, L. Quiñones-Hermosa, S. Hernandez-Manrique, and W. Uccahuasi, "Implementation of and agriculture 4.0 oriented system for monitoring environmental variables: *Alstroemeria hybrida*—*Astromelia* case," in *Proc. Int. Conf. Sustain. Comput. Data Commun. Syst. (ICSCDS)*, Mar. 2023, pp. 1691–1696.
- [279] H. N. Jasim and L. A. Abdul-Rahaim, "Design and implementation of cloud computing smart irrigation system," *Majlesi J. Electr. Eng.*, vol. 17, no. 2, pp. 145–151, 2023.
- [280] F. Behzadipour, M. G. N. Raeni, S. A. Mehdizadeh, M. Taki, B. K. Moghadam, M. R. Z. Bavani, and J. Lloret, "A smart IoT-based irrigation system design using AI and prediction model," *Neural Comput. Appl.*, vol. 35, no. 35, pp. 24843–24857, Dec. 2023.
- [281] P. Nagaraj, V. Muneeswaran, C. Vasundhara, K. Koushik, B. S. Sumanth, and P. Naveen, "Automated horticulture for farmers using IoT," in *Proc. 2nd Int. Conf. Edge Comput. Appl. (ICECAA)*, Jul. 2023, pp. 1219–1222.
- [282] M. K. Kankara, A. Imtiaz, I. Chowdhury, M. K. M. Khan, and T. Ahmed, "Arduino and NodeMCU-based smart soil moisture balancer with IoT integration," in *Information Systems for Intelligent Systems: Proceedings of ISBM 2022*. Cham, Switzerland: Springer, 2023, pp. 621–636.
- [283] P. Vandôme, C. Leauthaud, S. Moinard, O. Sainlez, I. Mekki, A. Zairi, and G. Belaud, "Making technological innovations accessible to agricultural water management: Design of a low-cost wireless sensor network for drip irrigation monitoring in Tunisia," *Smart Agricult. Technol.*, vol. 4, Aug. 2023, Art. no. 100227.
- [284] T. E. Mathew, A. Sabu, S. Sengan, J. Sathiamoorthy, and A. Prasanth, "Microclimate monitoring system for irrigation water optimization using IoT," *Meas., Sensors*, vol. 27, Jun. 2023, Art. no. 100727.
- [285] M. Alaan, E. Manuel, J. Reyes, C. R. Mercado, and R. Concepcion, "Design of an optimized second-order soil-based drip fertigation control system with Internet-of-Things integration," in *Proc. 8th Int. Conf. Bus. Ind. Res. (ICBIR)*, May 2023, pp. 477–482.
- [286] M. Keerthana, D. Dhinakaran, M. Ananthi, R. Harish, S. M. U. Sankar, and M. S. Sree, "IoT based automated irrigation system for agricultural activities," in *Proc. 12th Int. Conf. Adv. Comput. (ICoAC)*, Aug. 2023, pp. 1–6.
- [287] D. B. A. Kumar, N. Doddabasappa, B. Bairwa, G. Raju, and Madhu, "IoT-based water harvesting, moisture monitoring, and crop monitoring system for precision agriculture," in *Proc. Int. Conf. Distrib. Comput. Electr. Circuits Electron. (ICDCECE)*, Apr. 2023, pp. 1–6.
- [288] H. Singh, N. Halder, B. Singh, J. Singh, S. Sharma, and Y. Shacham-Diamand, "Smart farming revolution: Portable and real-time soil nitrogen and phosphorus monitoring for sustainable agriculture," *Sensors*, vol. 23, no. 13, p. 5914, Jun. 2023.
- [289] S. Hassan, R. M. Mushinski, T. Amede, G. D. Bending, and J. A. Covington, "Integrated probe system for measuring soil carbon dioxide concentrations," *Sensors*, vol. 23, no. 5, p. 2580, Feb. 2023.
- [290] P. Di Gennaro, D. Lofù, D. Vitano, P. Tedeschi, and P. Boccadoro, "Waters: A Sigfox-compliant prototype for water monitoring," *Internet Technol. Lett.*, vol. 2, no. 1, p. e74, Jan. 2019.

- [291] L. M. Fernández-Ahumada, J. Ramírez-Faz, M. Torres-Romero, and R. López-Luque, "Proposal for the design of monitoring and operating irrigation networks based on IoT, cloud computing and free hardware technologies," *Sensors*, vol. 19, no. 10, p. 2318, May 2019.
- [292] X. Lei and Y. Liu, "The key technologies of intelligent urban drainage management," in *Proc. 2nd Int. Symp. Water Syst. Oper.* Beijing, China: IAHR Document Library, 2021.
- [293] M. M. Suwaid, M. H. Habaebi, and S. Khan, "Embedded LoRaWAN for agricultural sensing applications," in *Proc. IEEE 6th Int. Conf. Eng. Technol. Appl. Sci. (ICETAS)*, Dec. 2019, pp. 1–5.
- [294] T. S. Mohammed, O. F. Khan, and A. Al-Bazi, "A novel algorithm based on LoRa technology for open-field and protected agriculture smart irrigation system," in *Proc. 2nd IEEE Middle East North Afr. Commun. Conf. (MENACOMM)*, Nov. 2019, pp. 1–6.
- [295] T. A. Khoa, M. M. Man, T.-Y. Nguyen, V. Nguyen, and N. H. Nam, "Smart agriculture using IoT multi-sensors: A novel watering management system," *J. Sensor Actuator Netw.*, vol. 8, no. 3, p. 45, Aug. 2019.
- [296] P. Fraga-Lamas, M. Celaya-Echarri, L. Azpilicueta, P. Lopez-Iturri, F. Falcone, and T. M. Fernández-Caramés, "Design and empirical validation of a lorawan IoT smart irrigation system," in *Proceedings*, vol. 42, no. 1, p. 62, 2019.
- [297] E. Nigussie, T. Olwal, G. Musumba, T. Tegegne, A. Lemma, and F. Mekuria, "IoT-based irrigation management for smallholder farmers in rural sub-Saharan Africa," *Proc. Comput. Sci.*, vol. 177, pp. 86–93, Jan. 2020.
- [298] A. Nyéki, G. Teschner, B. Ambrus, M. Neményi, and A. J. Kovács, "Architecting farmer-centric Internet of Things for precision crop production," *Hung. Agricult. Eng.*, vol. 38, pp. 71–78, Jan. 2020.
- [299] J. D. Borrero and A. Zabalo, "An autonomous wireless device for real-time monitoring of water needs," *Sensors*, vol. 20, no. 7, p. 2078, Apr. 2020.
- [300] H. Zhang, L. He, F. Di Gioia, D. D. Choi, and P. Heinemann, "Internet of Things (IoT)-based precision irrigation with LoRaWAN technology applied to high tunnel vegetable production," in *Proc. ASABE Annu. Int. Virtual Meeting*, Jul. 2020, p. 1.
- [301] A. D. Boursianis, M. S. Papadopoulou, A. Gotsis, S. Wan, P. Sarigianidis, S. Nikolaidis, and S. K. Goudos, "Smart irrigation system for precision agriculture—The AREThOU5A IoT platform," *IEEE Sensors J.*, vol. 21, no. 16, pp. 17539–17547, Aug. 2021.
- [302] X. Jiang, L. He, and J. Tong, "Investigation of soil wetting pattern in drip irrigation using LoraWAN technology," in *Proc. ASABE Annu. Int. Virtual Meeting*, Jul. 2020, p. 1.
- [303] M. Emharraf, H. Taous, W. Benzekri, A. E. Moussati, and K. Aberkani, "Intelligent agriculture platform based on low energy and cost wireless sensors for efficient water irrigation," in *Proc. Int. Conf. Smart Inf. Commun. Technol. (SmartICT)*, Saidia, Morocco. Cham, Switzerland: Springer, Sep. 2020, pp. 460–469.
- [304] K. T. Mya, M. M. Sein, T. T. S. Nyunt, U. Lewlompaisarl, and Y. Owada, "A design for IoT based smart watering system using LoRa," in *Proc. IEEE 9th Global Conf. Consum. Electron. (GCCE)*, Oct. 2020, pp. 278–279.
- [305] R. Singh, M. Aernouts, M. De Meyer, M. Weyn, and R. Berkvens, "Leveraging LoRaWAN technology for precision agriculture in greenhouses," *Sensors*, vol. 20, no. 7, p. 1827, Mar. 2020.
- [306] G. Codeluppi, A. Cilfone, L. Davoli, and G. Ferrari, "LoRaFarM: A LoRaWAN-based smart farming modular IoT architecture," *Sensors*, vol. 20, no. 7, p. 2028, 2020.
- [307] I. Froiz-Míguez, P. Lopez-Iturri, P. Fraga-Lamas, M. Celaya-Echarri, Ó. Blanco-Novoa, L. Azpilicueta, F. Falcone, and T. M. Fernández-Caramés, "Design, implementation, and empirical validation of an IoT smart irrigation system for fog computing applications based on LoRa and LoRaWAN sensor nodes," *Sensors*, vol. 20, no. 23, p. 6865, Nov. 2020.
- [308] S. R. Peddinti, J. W. Hopmans, M. A. Najm, and I. Kisekka, "Assessing effects of salinity on the performance of a low-cost wireless soil water sensor," *Sensors*, vol. 20, no. 24, p. 7041, Dec. 2020.
- [309] A. Torre-Neto, J. R. Cotrim, J. H. Kleinschmidt, C. Kamienski, and M. C. Visoli, "Enhancing soil measurements with a multi-depth sensor for IoT-based smart irrigation," in *Proc. IEEE Int. Workshop Metrol. Agricult. Forestry (MetroAgriFor)*, Nov. 2020, pp. 78–82.
- [310] R. R. Shamshiri, I. A. Hameed, K. R. Thorp, S. K. Balasundram, S. Shafian, M. Fatemeh, M. Sultan, B. Mahns, and S. Samiei, "Greenhouse automation using wireless sensors and IoT instruments integrated with artificial intelligence," in *Next-Generation Greenhouses for Food Security*, 2021. [Online]. Available: <https://doi.org/10.5772/intechopen.97714>
- [311] J. D. F. Silveira, A. F. da S. Veloso, J. V. dos Reis Júnior, A. C. B. Soares, and R. A. L. Rabêlo, "A new low-cost LoRaWAN power switch for smart farm applications," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2021, pp. 3330–3335.
- [312] F. Sharofidinov, M. S. A. Muthanna, V. D. Pham, A. Khakimov, A. Muthanna, and K. Samouylov, "Agriculture management based on LoRa edge computing system," in *Proc. Int. Conf. Distrib. Comput. Commun. Netw.*, Moscow, Russia, Sep. 2020, pp. 113–125.
- [313] J. M. Jimenez, L. Parra, L. García, J. Lloret, P. V. Mauri, and P. Lorenz, "New protocol and architecture for a wastewater treatment system intended for irrigation," *Appl. Sci.*, vol. 11, no. 8, p. 3648, Apr. 2021.
- [314] M. Ahsan, M. A. Based, J. Haider, and E. M. G. Rodrigues, "Smart monitoring and controlling of appliances using LoRa based IoT system," *Designs*, vol. 5, no. 1, p. 17, Mar. 2021.
- [315] K. Varshney, S. Tripathi, and V. Purwar, "Expert system on smart irrigation using Internet of Things," in *Proc. Int. Conf. Commun. Artif. Intell.* Cham, Switzerland: Springer, 2021, pp. 181–188.
- [316] N. Kaur and G. Deep, "IoT-based Brinjal crop monitoring system," in *Smart Sensors for Industrial Internet of Things*. Cham, Switzerland: Springer, 2021, pp. 231–247.
- [317] S. Figorilli, F. Pallottino, G. Colle, D. Spada, C. Beni, F. Tocci, S. Vasta, F. Antonucci, M. Pagano, M. Fedrizzi, and C. Costa, "An open source low-cost device coupled with an adaptive time-lag time-series linear forecasting modeling for apple Trentino (Italy) precision irrigation," *Sensors*, vol. 21, no. 8, p. 2656, Apr. 2021.
- [318] A. Glória, J. Cardoso, and P. Sebastião, "Sustainable irrigation system for farming supported by machine learning and real-time sensor data," *Sensors*, vol. 21, no. 9, p. 3079, Apr. 2021.
- [319] P. Placidi, R. Morbidelli, D. Fortunati, N. Papini, F. Gobbi, and A. Scorzoni, "Monitoring soil and ambient parameters in the IoT precision agriculture scenario: An original modeling approach dedicated to low-cost soil water content sensors," *Sensors*, vol. 21, no. 15, p. 5110, Jul. 2021.
- [320] F. Sánchez-Sutil and A. Cano-Ortega, "Smart control and energy efficiency in irrigation systems using LoRaWAN," *Sensors*, vol. 21, no. 21, p. 7041, Oct. 2021.
- [321] T. Cao-Hoang, K. A. Su, T. T. Pham Van, V. T. Pham, D. C. Nguyen, and M. Mizoguchi, "Soil moisture monitoring system based on LoRa network to support agricultural cultivation in drought season," in *Soft Computing: Biomedical and Related Applications*. Cham, Switzerland: Springer, 2021, pp. 163–174.
- [322] H. Zhang, L. He, F. D. Gioia, D. Choi, A. Elia, and P. Heinemann, "LoRaWAN based Internet of Things (IoT) system for precision irrigation in plasticulture fresh-market tomato," *Smart Agricult. Technol.*, vol. 2, Dec. 2022, Art. no. 100053.
- [323] Z. Xu, "UAV surveying and mapping information collection method based on Internet of Things," *Internet Things Cyber-Phys. Syst.*, vol. 2, pp. 138–144, Jan. 2022.
- [324] A. Dahane, R. Benameur, and B. Kechar, "An IoT low-cost smart farming for enhancing irrigation efficiency of smallholders farmers," *Wireless Pers. Commun.*, vol. 127, no. 4, pp. 3173–3210, Dec. 2022.
- [325] C. Nicolas, B. Naila, and R.-C. Amar, "Energy efficient firmware over the air update for TinyML models in LoRaWAN agricultural networks," in *Proc. 32nd Int. Telecommun. Netw. Appl. Conf. (ITNAC)*, Nov. 2022, pp. 21–27.
- [326] L. Lyu, J. M. Caballero, and R. A. Juanatas, "Design of irrigation control system for vineyard based on LoRa wireless communication and dynamic neural network," in *Proc. 7th Int. Conf. Bus. Ind. Res. (ICBIR)*, May 2022, pp. 373–378.
- [327] R. Madhumathi, T. Arumuganathan, T. Vimal, and S. Vishnu, "A LoRa based wireless smart irrigation system," in *Proc. 6th Int. Conf. Electron., Commun. Aerosp. Technol.*, Dec. 2022, pp. 514–517.
- [328] M. Narasimharao, B. Swain, P. P. Nayak, and S. Bhuyan, "Cloud based automated low power long range smart farming modular IoT architecture," in *Proc. 2nd Odisha Int. Conf. Electr. Power Eng., Commun. Comput. Technol. (ODICON)*, Nov. 2022, pp. 1–5.

- [329] E. M. Olalla, H. D. Cadena-Lema, H. M. D. Limaico, J. C. Nogales-Romero, M. Zambrano, and C. V. Ayala, "Irrigation control system using machine learning techniques applied to precision agriculture (Internet of Farm Things IOFT)," in *I+D for Smart Cities Industry: Proc. RITAM 2021*. Cham, Switzerland: Springer, 2022, pp. 329–342.
- [330] C. Briciu-Burghina, J. Zhou, M. I. Ali, and F. Regan, "Demonstrating the potential of a low-cost soil moisture sensor network," *Sensors*, vol. 22, no. 3, p. 987, Jan. 2022.
- [331] V. Y. Chandrappa, B. Ray, N. Ashwatha, and P. Shrestha, "Spatiotemporal modelling to predict soil moisture for sustainable smart irrigation," *Internet Things*, vol. 21, 2023, Art. no. 100671, doi: [10.1016/j.iot.2022.100671](https://doi.org/10.1016/j.iot.2022.100671).
- [332] Z. Chang, F. Zhang, J. Xiong, J. Ma, B. Jin, and D. Zhang, "Sensor-free soil moisture sensing using LoRa signals," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 2, pp. 1–27, Jul. 2022.
- [333] F. F. Hossain, R. Messenger, G. L. Captain, S. Ekin, J. D. Jacob, S. Taghvaeian, and J. F. O'Hara, "Soil moisture monitoring through UAS-assisted Internet of Things LoRaWAN wireless underground sensors," *IEEE Access*, vol. 10, pp. 102107–102118, 2022.
- [334] M. Pramanik, M. Khanna, M. Singh, D. K. Singh, S. Sudhishri, A. Bhatia, and R. Ranjan, "Automation of soil moisture sensor-based basin irrigation system," *Smart Agricult. Technol.*, vol. 2, Dec. 2022, Art. no. 100032.
- [335] D. Kiv, G. Allabadi, B. Kaplan, and R. Kravets, "Smol: Sensing soil moisture using LoRa," in *Proc. 1st ACM Workshop Power Low Power Internet Things*, Jan. 2022, pp. 21–27.
- [336] S. R. Gopal and V. S. V. Prabhakar, "Intelligent edge based smart farming with LoRa and IoT," *Int. J. Syst. Assurance Eng. Manage.*, vol. 14, pp. 1–7, 2022, doi: [10.1007/s13198-021-01576-z](https://doi.org/10.1007/s13198-021-01576-z).
- [337] D. K. Singh, R. Sobti, P. Kumar Malik, S. Shrestha, P. K. Singh, and K. Z. Ghafoor, "IoT-driven model for weather and soil conditions based on precision irrigation using machine learning," *Secur. Commun. Netw.*, vol. 2022, pp. 1–10, May 2022.
- [338] A. Valente, C. Costa, L. Pereira, B. Soares, J. Lima, and S. Soares, "A LoRaWAN IoT system for smart agriculture for vine water status determination," *Agriculture*, vol. 12, no. 10, p. 1695, Oct. 2022.
- [339] C. O. Andrada, G. J. M. Tolentino, and J. C. Dela Cruz, "Application of LoRa technology in ET based irrigation of village parks," in *Proc. IEEE 18th Int. Colloq. Signal Process. Appl. (CSPA)*, May 2022, pp. 310–315.
- [340] Y. Wang, J. Liu, R. Li, X. Suo, and E. Lu, "Application of PSO-BPNN-PID controller in nutrient solution EC precise control system: Applied research," *Sensors*, vol. 22, no. 15, p. 5515, Jul. 2022.
- [341] L. Gong, J. Yan, Y. Chen, J. An, L. He, L. Zheng, and Z. Zou, "An IoT-based intelligent irrigation system with data fusion and a self-powered wide-area network," *J. Ind. Inf. Integr.*, vol. 29, Sep. 2022, Art. no. 100367.
- [342] M. Dillibabu, M. Arjun, M. Hariharan, and M. Lal, "LoRa based precision irrigation and monitoring system," in *Proc. 3rd Int. Conf. Pervasive Comput. Social Netw. (ICPCSN)*, Jun. 2023, pp. 1554–1562.
- [343] S. Gitahi, D. M. Kaburu, I. N. Oteyo, and S. Kimani, "A model implementation of Internet of Things (IoT)-based smart watering system for crops using LoRaWAN," in *Proc. IST-Africa Conf. (IST-Africa)*, May 2023, pp. p1–8.
- [344] R. Patten and A. Mungur, "Smart irrigation system using LoRaWAN," in *Proc. 3rd Int. Conf. Next Gener. Comput. Appl. (NextComp)*, Oct. 2022, pp. 1–6.
- [345] A. Hafian, M. Benbrahim, and M. N. Kabbaj, "IoT-based smart irrigation management system using real-time data," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 13, no. 6, p. 7078, Dec. 2023.
- [346] M. H. Seyar and T. Ahamed, "Development of an IoT-based precision irrigation system for tomato production from indoor seedling germination to outdoor field production," *Appl. Sci.*, vol. 13, no. 9, p. 5556, Apr. 2023.
- [347] T.-N. Doan, "A novel LoRa-based platform for remote monitoring of large-scale Rice fields," *Int. J. Membrane Sci. Technol.*, vol. 10, no. 2, pp. 1301–1322, Jun. 2023.
- [348] S. Bagwari, "Impact of Internet of Things based monitoring and prediction system in precision agriculture," *Think India J.*, vol. 22, no. 16, pp. 4599–4610, 2019.
- [349] D. S. Gangwar, S. Tyagi, and S. K. Soni, "A conceptual framework of agroecological resource management system for climate-smart agriculture," *Int. J. Environ. Sci. Technol.*, vol. 16, no. 8, pp. 4123–4132, Aug. 2019.
- [350] A. Sinha, G. Shrivastava, and P. Kumar, "Architecting user-centric Internet of Things for smart agriculture," *Sustain. Comput., Inform. Syst.*, vol. 23, pp. 88–102, Sep. 2019.
- [351] B. Keswani, A. G. Mohapatra, P. Keswani, A. Khanna, D. Gupta, and J. Rodrigues, "Improving weather dependent zone specific irrigation control scheme in IoT and big data enabled self driven precision agriculture mechanism," *Enterprise Inf. Syst.*, vol. 14, nos. 9–10, pp. 1494–1515, Nov. 2020.
- [352] H. Zhang, S. Shi, Y. Wu, and T. Feng, "Retracted article: Development of computer-based agricultural remote intelligent information monitoring system," *Int. J. Comput. Appl.*, vol. 45, no. 2, pp. 151–160, Feb. 2023.
- [353] B. Hu, W. Zhang, T. Ma, and Z. Zhao, "Optimization and simulation of farmland protection dynamic monitoring system based on Internet of Things technology," *Wireless Commun. Mobile Comput.*, vol. 2021, pp. 1–11, Jul. 2021.
- [354] G. Gagliardi, M. Lupia, G. Cario, F. C. Gaccio, V. D'Angelo, A. I. M. Cosma, and A. Casavola, "An Internet of Things solution for smart agriculture," *Agronomy*, vol. 11, no. 11, p. 2140, Oct. 2021.
- [355] O. Ayurzana and S. Tsagaanchuluun, "Monitoring system of agriculture fields using ZigBee modules," *Int. J. Adv. smart Conver.*, vol. 10, no. 1, pp. 89–96, 2021.
- [356] L. Zheng, "Study and application on big data information fusion system based on IoT," *Secur. Commun. Netw.*, vol. 2021, pp. 1–8, Dec. 2021.
- [357] K. L. Raju and V. Vijayaraghavan, "A self-powered, real-time, NRF24L01 IoT-based cloud-enabled service for smart agriculture decision-making system," *Wireless Pers. Commun.*, vol. 124, no. 1, pp. 207–236, May 2022.
- [358] A. Eraliev and U. Salomov, "Development of energy efficient WSN based smart monitoring system," in *Proc. 15th Int. Conf. Electron., Comput. Artif. Intell. (ECAI)*, Jun. 2023, pp. p. 1–5.
- [359] Y. Zhang and M. Chen, "An IoT-enabled energy-efficient approach for the detection of leaf curl disease in tomato crops," *Wireless Netw.*, vol. 29, no. 1, pp. 321–329, Jan. 2023.
- [360] D. R. Vincent, N. Deepa, D. Elavarasan, K. Srinivasan, S. H. Chauhdary, and C. Iwendi, "Sensors driven AI-based agriculture recommendation model for assessing land suitability," *Sensors*, vol. 19, no. 17, p. 3667, Aug. 2019.
- [361] Y. A. K. Utama, Y. Widiyanto, Y. Hari, and M. Habiburrahman, "Design of weather monitoring sensors and soil humidity in agriculture using Internet of Things (IoT)," *Trans. Mach. Learn. Artif. Intell.*, vol. 7, no. 1, p. 10, Feb. 2019.
- [362] A. R. Al-Ali, A. Al Nabulsi, S. Mukhopadhyay, M. S. Awal, S. Fernandes, and K. Ailabouni, "IoT-solar energy powered smart farm irrigation system," *J. Electron. Sci. Technol.*, vol. 17, no. 4, Dec. 2019, Art. no. 100017.
- [363] P. Kaur and M. Sharma, "Diagnosis of human psychological disorders using supervised learning and nature-inspired computing techniques: A meta-analysis," *J. Med. Syst.*, vol. 43, no. 7, pp. 1–30, Jul. 2019.
- [364] B. Sreeja, S. M. Kumar, P. Sherubha, and S. Sasirekha, "Crop monitoring using wireless sensor networks," *Mater. Today, Proc.*, 2020, doi: [10.1016/j.matpr.2020.10.373](https://doi.org/10.1016/j.matpr.2020.10.373).
- [365] L. G. R. C. and G. P., "An automated low cost IoT based fertilizer intimation system for smart agriculture," *Sustain. Comput., Informat. Syst.*, vol. 28, Dec. 2020, Art. no. 100300.
- [366] Y. Bhojwani, R. Singh, R. Reddy, and B. Perumal, "Crop selection and IoT based monitoring system for precision agriculture," in *Proc. Int. Conf. Emerg. Trends Inf. Technol. Eng. (IC-ETITE)*, Feb. 2020, pp. 1–11.
- [367] M. W. Rahman, M. E. Hossain, R. Islam, M. H. A. Rashid, M. N. A. Alam, and M. M. Hasan, "Real-time and low-cost IoT based farming using raspberry PI," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 17, no. 1, p. 197, Jan. 2020.
- [368] S. Kapse, S. Kale, S. Bhongade, S. Sangamnerkar, and Y. Gotmare, "IoT enable soil testing & NPK nutrient detection," *JAC J. Compos. Theory*, vol. 13, no. 5, pp. 310–318, 2020.
- [369] I. M. M. Fernando and S. Gunasekara, "Smart irrigation system for a small plant nursery based on soil moisture level," *CINEC Academic J.*, vol. 4, pp. 62–68, Dec. 2020.
- [370] R. Singh, S. Srivastava, and R. Mishra, "AI and IoT based monitoring system for increasing the yield in crop production," in *Proc. Int. Conf. Electr. Electron. Eng. (ICE)*, Feb. 2020, pp. 301–305.
- [371] R. K. Megalingam, G. Kishore Indukuri, D. S. K. Reddy, E. D. Vignesh, and V. K. Yarasuri, "Irrigation monitoring and prediction system using machine learning," in *Proc. Int. Conf. Emerg. Technol. (INCET)*, Jun. 2020, pp. 1–5.
- [372] A. Rehman, J. Liu, L. Keqiu, A. Mateen, and M. Q. Yasin, "Machine learning prediction analysis using IoT for smart farming," *Int. J.*, vol. 8, no. 9, pp. 1–6, 2020.

- [373] S. S. K. Pokala and A. Bini, "A low cost IoT enabled device for monitoring agriculture field and smart irrigation system," in *Inventive Communication and Computational Technologies*. Cham, Switzerland: Springer, 2021, pp. 923–932.
- [374] S. R. Dogiwal, P. Dadheech, A. Kumar, L. Raja, A. Kumar, and M. K. Beniwal, "An automated optimize utilization of water and crop monitoring in agriculture using IoT," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 1131, no. 1, Apr. 2021, Art. no. 012019.
- [375] E. Collado, E. Valdés, A. García, and Y. Sáez, "Design and implementation of a low-cost IoT-based agroclimatic monitoring system for greenhouses," *AIMS Electron. Elect. Eng.*, vol. 5, no. 4, pp. 251–283, 2021.
- [376] A. F. Suhaimi, N. Yaakob, S. A. Saad, K. A. Sidek, M. E. Elshaikh, A. K. Y. Dafhallah, O. B. Lynn, and M. Almashor, "IoT based smart agriculture monitoring, automation and intrusion detection system," *J. Phys., Conf. Ser.*, vol. 1962, no. 1, Jul. 2021, Art. no. 012016.
- [377] S. Mandal, I. Ali, and S. Saha, "IoT in agriculture: Smart farming using MQTT protocol through cost-effective heterogeneous sensors," in *Proc. Int. Conf. Frontiers Comput. Syst.* Cham, Switzerland: Springer, 2021, pp. 903–913.
- [378] O. A. Osanaiye, T. Mannan, and F. Aina, "An IoT-based soil moisture monitor," *Afr. J. Sci., Technol., Innov. Develop.*, vol. 14, no. 7, pp. 1908–1915, Nov. 2022.
- [379] G. Patrizi, A. Bartolini, L. Ciani, V. Gallo, P. Sommella, and M. Carratù, "A virtual soil moisture sensor for smart farming using deep learning," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.
- [380] P. Manikandan, G. Ramesh, P. Sivakumar, J. J. Kumar, R. L. Krishna, and G. Dinesh, "Soil nutrients monitoring and analyzing system using Internet of Things," in *Proc. 2nd Int. Conf. Advance Comput. Innov. Technol. Eng. (ICACITE)*, Apr. 2022, pp. 301–305.
- [381] N. Sindhwani, V. P. Maurya, A. Patel, R. K. Yadav, S. Krishna, and R. Anand, "Implementation of intelligent plantation system using virtual IoT," in *Internet of Things and Its Applications*. Cham, Switzerland: Springer, 2022, pp. 305–322.
- [382] R. Gill and P. Chawla, "Energy harvesting sensors based Internet of Things system for precision agriculture," in *Proc. 2nd Int. Conf. Innov. Practices Technol. Manage. (ICIPTM)*, vol. 2, Feb. 2022, pp. 270–273.
- [383] N. N. Thilakarathne, M. S. A. Bakar, P. E. Abas, and H. Yassin, "Towards making the fields talks: A real-time cloud enabled IoT crop management platform for smart agriculture," *Frontiers Plant Sci.*, vol. 13, Jan. 2023, Art. no. 1030168.
- [384] F. Pitu and N. C. Gaitan, "Surveillance of Sigfox technology integrated with environmental monitoring," in *Proc. Int. Conf. Develop. Appl. Syst. (DAS)*, May 2020, pp. 69–72.
- [385] A. Triantafyllou, P. Sarigiannidis, and S. Bibi, "Precision agriculture: A remote sensing monitoring system architecture," *Information*, vol. 10, no. 11, p. 348, Nov. 2019.
- [386] M. F. L. Pereira, P. E. Cruvinel, G. M. Alves, and J. M. G. Beraldo, "Parallel computational structure and semantics for soil quality analysis based on LoRa and Apache spark," in *Proc. IEEE 14th Int. Conf. Semantic Comput. (ICSC)*, Feb. 2020, pp. 332–336.
- [387] T. Syrový, R. Vik, S. Pretl, L. Syrová, J. Čengery, A. Hamáček, L. Kubáč, and L. Menšík, "Fully printed disposable IoT soil moisture sensors for precision agriculture," *Chemosensors*, vol. 8, no. 4, p. 125, Dec. 2020.
- [388] A. D. Coelho, B. G. Dias, W. de Oliveira Assis, F. de Almeida Martins, and R. C. Pires, "Monitoring of soil moisture and atmospheric sensors with Internet of Things (IoT) applied in precision agriculture," in *Proc. 14th Technol. Appl. Electron. Teaching Conf. (TAAE)*, Jul. 2020, pp. 1–8.
- [389] S. Yosep, "Implementation of fuzzy logic on Internet of Things-based greenhouse," *Internet Things Artif. Intell. J.*, vol. 1, no. 2, pp. 100–113, May 2021.
- [390] P. D. P. Adi and V. M. M. Siregar, "Soil moisture sensor based on Internet of Things LoRa," *Internet Things Artif. Intell. J.*, vol. 1, no. 2, pp. 120–132, May 2021.
- [391] S. Komkova, E. Kosolapova, V. Kosolapov, A. Chesnokov, and S. Stankovski, "Development of a system for operational monitoring of the soil agrochemical indicators," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 857, no. 1, Sep. 2021, Art. no. 012013.
- [392] Y.-W. Kuo, W.-L. Wen, X.-F. Hu, Y.-T. Shen, and S.-Y. Miao, "A LoRa-based multisensor IoT platform for agriculture monitoring and submersible pump control in a water bamboo field," *Processes*, vol. 9, no. 5, p. 813, May 2021.
- [393] F. A. Almalki, B. O. Soufiene, S. H. Alsamhi, and H. Sakli, "A low-cost platform for environmental smart farming monitoring system based on IoT and UAVs," *Sustainability*, vol. 13, no. 11, p. 5908, May 2021.
- [394] M. Abdallah, W. J. Lee, N. Raghunathan, C. Mousoulis, J. W. Sutherland, and S. Bagechi, "Anomaly detection through transfer learning in agriculture and manufacturing IoT systems," 2021, *arXiv:2102.05814*.
- [395] R. R. Shamshiri and C. Weltzien, "Development and field evaluation of a multichannel LoRa sensor for IoT monitoring in berry orchards," in *41. GIL-Jahrestagung, Informations-Und Kommunikationstechnologie Kritischen Zeiten*. Potsdam, Germany: Science and Information (SAI) Organization Limited, Leibniz Institute for Agricultural Engineering and Bioeconomy, 2021.
- [396] A. A. Ruslan, S. M. Salleh, S. F. W. M. Hatta, and A. A. B. Sajak, "IoT soil monitoring based on LoRa module for oil palm plantation," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 5, pp. 1–6, 2021.
- [397] G. Di Renzone, S. Parrino, G. Peruzzi, A. Pozzebon, and D. Bertoni, "LoRaWAN underground to aboveground data transmission performances for different soil compositions," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.
- [398] S. Parrino, G. Peruzzi, and A. Pozzebon, "Pilot analysis on soil moisture impact on underground to aboveground LoRaWAN transmissions for IoT contexts," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2021, pp. 1–6.
- [399] S. J. Suji Prasad, M. Thangatamilan, M. Suresh, H. Panchal, C. A. Rajan, C. Sagana, B. Gunapriya, A. Sharma, T. Panchal, and K. K. Sadasivuni, "An efficient LoRa-based smart agriculture management and monitoring system using wireless sensor networks," *Int. J. Ambient Energy*, vol. 43, no. 1, pp. 5447–5450, Dec. 2022.
- [400] M. Škiljo, Z. Blažević, L. Đujić-Rodić, T. Perković, and P. Šolić, "Self-sensing antenna for soil moisture: Beacon approach," *Sensors*, vol. 22, no. 24, p. 9863, Dec. 2022.
- [401] S. T. Fauziah, E. Mulyana, A. I. Nur, S. Uyun, T. Yusuf, and R. Mardiaty, "Prototype of smart garden system for monitoring horticulture plants based on LoRa technology," in *Proc. 8th Int. Conf. Wireless Telematics (ICWT)*, Jul. 2022, pp. 1–5.
- [402] A. Mishra, S. Singh, K. Verma, P. Bhatia, M. Ghosh, and Y. Shacham-Diamand, "Green energy-based efficient IoT sensor network for small farms," in *International Congress of Electrical and Computer Engineering*. Cham, Switzerland: Springer, 2022, pp. 15–27.
- [403] M. Cruz, S. Mafra, E. Teixeira, and F. Figueiredo, "Smart strawberry farming using edge computing and IoT," *Sensors*, vol. 22, no. 15, p. 5866, Aug. 2022.
- [404] U. C. Njoku, C. K. Agubor, and L. S. Ezema, "Development of a long-range weather and soil monitoring system for rural farmers," *Eximia*, vol. 4, no. 1, pp. 159–171, 2022.
- [405] A. Abdillah, M. N. Zakaria, and W. Waluyo, "Determination of quantity fertilizer for sugarcane based on wireless sensor network," *Jurnal Jartel Jurnal Jaringan Telekomunikasi*, vol. 12, no. 4, pp. 208–211, Dec. 2022.
- [406] Y. Wu, Z. Yang, and Y. Liu, "Internet-of-Things-based multiple-sensor monitoring system for soil information diagnosis using a smartphone," *Micromachines*, vol. 14, no. 7, p. 1395, Jul. 2023.
- [407] X. C. Pham, T. P. T. Nguyen, and M. T. Le, "Pathloss modelling and evaluation for a wireless underground soil moisture sensor network," in *Proc. Int. Conf. Intell. Syst. Netw. Cham*, Switzerland: Springer, 2023, pp. 335–345.
- [408] C. Cariou, L. Moiroux-Arvis, F. Pinet, and J.-P. Chanet, "Adaptive robot control based on wireless underground sensor network in agriculture 4.0," in *Proc. 5th Int. Conf. Comput. Commun. Internet (ICCCI)*, Jun. 2023, pp. 217–222.
- [409] M. F. Ibrahim, M. M. Isa, R. Hussin, S. M. M. S. Zakaria, N. Ahmad, C. K. J. Yuan, and A. F. T. Abdullah, "IoT monitoring system for fig in greenhouse plantation," *J. Adv. Res. Appl. Sci. Eng. Technol.*, vol. 31, no. 2, pp. 298–309, Jul. 2023.
- [410] M. Bertocco, S. Parrino, G. Peruzzi, and A. Pozzebon, "Estimating volumetric water content in soil for IoT contexts by exploiting RSSI-based augmented sensors via machine learning," *Sensors*, vol. 23, no. 4, p. 2033, 2023.
- [411] X. Xiuyun, X. Xufeng, Z. Zelong, Z. Bin, S. Shuran, L. Zhen, H. Tiansheng, and H. Huixian, "Variable rate liquid fertilizer applicator for deep-fertilization in precision farming based on ZigBee technology," *IFAC-PapersOnLine*, vol. 52, no. 30, pp. 43–50, 2019.
- [412] P. Zhou, Y. Zhang, W. Yang, M. Li, Z. Liu, and X. Liu, "Development and performance test of an in-situ soil total nitrogen-soil moisture detector based on near-infrared spectroscopy," *Comput. Electron. Agricult.*, vol. 160, pp. 51–58, May 2019.

- [413] A. Petre, I. Voicea, V. Vlăduț, and L. Vlăduțoiu, "Considerations on monitoring the state of soil and vegetation pollution in the affected areas," in *Proc. Int. Conf. Hydraul. Pneumatics*, 2019, pp. 301–307.
- [414] D. Jat, Y. A. Rajwade, N. S. Chandel, K. Dubey, and K. V. R. Rao, "Embedded system for regulating abiotic parameters for capsicum cultivation in a polyhouse with comparison to open-field cultivation," *Int. J. Vegetable Sci.*, vol. 26, no. 5, pp. 487–497, Sep. 2020.
- [415] P. Visconti, R. De Fazio, P. Primiceri, D. Cafagna, S. Strazzella, and N. I. Giannoccaro, "A solar-powered fertigation system based on low-cost wireless sensor network remotely controlled by farmer for irrigation cycles and crops growth optimization," *Int. J. Electron. Telecommun.*, vol. 66, no. 1, pp. 59–68, 2020.
- [416] A. Khanna and S. Kaur, "Wireless sensor and actuator network(S) and its significant impact on agricultural domain," in *Proc. 6th Int. Conf. Parallel, Distrib. Grid Comput. (PDGC)*, Nov. 2020, pp. 384–389.
- [417] N. S. Pezol, R. Adnan, and M. Tajjudin, "Design of an Internet of Things (IoT) based smart irrigation and fertilization system using fuzzy logic for chili plant," in *Proc. IEEE Int. Conf. Autom. Control Intell. Syst. (I2CACIS)*, Jun. 2020, pp. 69–73.
- [418] K. Karunanithy and B. Velusamy, "Energy efficient cluster and travelling salesman problem based data collection using WSNs for intelligent water irrigation and fertigation," *Measurement*, vol. 161, Sep. 2020, Art. no. 107835.
- [419] P. Thatipelli and R. Sujatha, "Smart agricultural robot with real-time data analysis using IBM Watson cloud platform," in *Advances in Clean Energy Technologies*. Cham, Switzerland: Springer, 2021, pp. 415–427.
- [420] C. Du, L. Zhang, X. Ma, X. Lou, Y. Shan, H. Li, and R. Zhou, "A cotton high-efficiency water-fertilizer control system using wireless sensor network for precision agriculture," *Processes*, vol. 9, no. 10, p. 1693, Sep. 2021.
- [421] M. Koushik, M. Srinivasan, R. Lavanya, S. Alfred, and S. Setty, "Design and development of wireless sensor network based data logger with ESP-NOW protocol," in *Proc. 6th Int. Conf. Conver. Technol. (I2CT)*, Apr. 2021, pp. 1–5.
- [422] R. Keote, P. Rewatkar, P. Durve, V. Domale, and S. Buradkar, "Pesticide and quality monitoring system for fruits and vegetables using IoT," in *Proc. Int. Conf. Intell. Vis. Comput.* Cham, Switzerland: Springer, 2022, pp. 522–531.
- [423] J. C. Guillermo, A. García-Cedeño, D. Rivas-Lalaleo, M. Huerta, and R. Clotet, "IoT architecture based on wireless sensor network applied to agricultural monitoring: A case of study of cacao crops in Ecuador," in *Proc. Int. Conf. ICT Adapting Agricult. Climate Change*, Cali, Colombia. Cham, Switzerland: Springer, 2019, pp. 42–57.
- [424] J. Ammouri, P. Minet, M. Boudiaf, S. Bouzeffane, and M. Yacoub, "Self-organizing maps applied to soil conservation in Mediterranean olive groves," in *Proc. 8th Int. Conf. Perform. Eval. Model. Wired Wireless Netw. (PEMWN)*, Nov. 2019, pp. 1–6.
- [425] K. Balamurugan and A. Sivakami, "LoRa-IoT based self-powered multi-sensors wireless network for next generation integrated farming," *Int. J. Sci. Technol. Res.*, vol. 8, pp. 1527–1533, Jan. 2019.
- [426] P. Angin, "AgriLoRa: A digital twin framework for smart agriculture," *J. Wireless Mobile Netw. Ubiquitous Comput. Depend. Appl.*, vol. 11, no. 4, pp. 77–96, 2020.
- [427] D. Taşkın and S. Yazar, "A long-range context-aware platform design for rural monitoring with IoT in precision agriculture," *Int. J. Comput. Commun. Control*, vol. 15, no. 2, pp. 1–11, Mar. 2020.
- [428] F. Akhter, H. Siddiquei, M. E. E. Alahi, and S. Mukhopadhyay, "Design and development of an IoT-enabled portable phosphate detection system in water for smart agriculture," *Sens. Actuators A, Phys.*, vol. 330, Art. no. 112861, 2021.
- [429] D. Perdana, W. R. P. Kusuma, and I. Alinursafa, "Developing of automatic fertilizer control system in soybean plant based on Internet of Things and LoRa networks," *Int. J. Electron. Telecommun.*, vol. 67, pp. 549–558, Jan. 2021.
- [430] D. P. Dahnil, Z. Hood, A. Adam, M. Z. A. Razak, and A. G. Ismail, "Drip irrigation detection for power outage-prone areas with Internet-of-Things smart fertigation management system," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 7, pp. 1–11, 2021.
- [431] P. K. Singh and A. Sharma, "An intelligent WSN-UAV-based IoT framework for precision agriculture application," *Comput. Electr. Eng.*, vol. 100, May 2022, Art. no. 107912.
- [432] S. Gibeaux, J. Pitti, and C. Gonzalez, "IoT for agriculture optimization: Preliminary results of a tropical precision farming project," in *Proc. 10th Jornadas de Cloud Comput., Big Data Emerg. Topics (La Plata)*, 2022, pp. 2–6.
- [433] M. A. de Oliveira Jr., G. Sedrez, G. de Souza, and G. G. H. Cavaleiro, "An application with Jetson nano for plant stress detection and on-field spray decision," in *Proc. Sensornets*, 2022, pp. 215–222.
- [434] N. Bristow, S. Rengaraj, D. R. Chadwick, J. Kettle, and D. L. Jones, "Development of a LoRaWAN IoT node with ion-selective electrode soil nitrate sensors for precision agriculture," *Sensors*, vol. 22, no. 23, p. 9100, Nov. 2022.
- [435] M. Á. Miranda, C. Barceló, F. Valdés, J. F. Feliu, D. Nestel, N. Papadopoulos, A. Sciarretta, M. Ruiz, and B. Alorda, "Developing and implementation of decision support system (DSS) for the control of olive fruit fly, *Bactrocera Oleae*, in Mediterranean olive orchards," *Agronomy*, vol. 9, no. 10, p. 620, Oct. 2019.
- [436] A. Tsipis, A. Papamichail, G. Koufoudakis, G. Tsoumanis, S. E. Polykalas, and K. Oikonomou, "Latency-adjustable cloud/fog computing architecture for time-sensitive environmental monitoring in olive groves," *AgriEngineering*, vol. 2, pp. 175–205, 2020.
- [437] Y. Li, "Research on precision planting management system based on agricultural big data," *J. Phys., Conf. Ser.*, vol. 1544, no. 1, May 2020, Art. no. 012174.
- [438] B. Zhang and L. Meng, "Energy efficiency analysis of wireless sensor networks in precision agriculture economy," *Sci. Program.*, vol. 2021, pp. 1–7, Aug. 2021.
- [439] X. Hua, X. Han, and H. Sun, "Design of environment monitoring system for greenhouse based on OneNET," in *Proc. IEEE Asia-Pacific Conf. Image Process., Electron. Comput. (IPEC)*, Apr. 2021, pp. 307–310.
- [440] S. Azfar, A. Nadeem, K. Ahsan, A. Mehmood, M. S. Siddiqui, M. Saeed, and M. Ashraf, "An IoT-based system for efficient detection of cotton pest," *Appl. Sci.*, vol. 13, no. 5, p. 2921, Feb. 2023.
- [441] A. L. Imoize, S. D. Odeyemi, and J. A. Adebiyi, "Development of a low-cost wireless bee-hive temperature and sound monitoring system," *Indonesian J. Electr. Eng. Informat. (IJEI)*, vol. 8, no. 3, pp. 476–485, Aug. 2020.
- [442] N. Fatima, S. A. Siddiqui, and A. Ahmad, "IoT-based smart greenhouse with disease prediction using deep learning," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 7, pp. 1–8, 2021.
- [443] K. Murali and B. Sridhar, "A smart agriculture irrigation system using sensor array based IoT," *J. Phys., Conf. Ser.*, vol. 2062, no. 1, Nov. 2021, Art. no. 012010.
- [444] M. S. Perumal, B. Manimozhi, H. Dandamudi, V. B. Durairaj, and A. Jawaharlalnehru, "Retraction note: Ultra-reliable low latency communication technique for agriculture wireless sensor networks," *Arabian J. Geosci.*, vol. 16, no. 12, pp. 1–9, Dec. 2023.
- [445] S. Wang, P. Qi, W. Zhang, and X. He, "Development and application of an intelligent plant protection monitoring system," *Agronomy*, vol. 12, no. 5, p. 1046, Apr. 2022.
- [446] M. Hnatiuc, S. Ghita, D. Alpetri, A. Ranca, V. Artem, I. Dina, M. Cosma, and M. A. Mohammed, "Intelligent grapevine disease detection using IoT sensor network," *Bioengineering*, vol. 10, no. 9, p. 1021, Aug. 2023.
- [447] G. S. Kuaban, P. Czekalski, E. L. Molua, and K. Grochla, "An architectural framework proposal for IoT driven agriculture," in *Proc. Int. Conf. Comput. Netw.* Cham, Switzerland: Springer, 2019, pp. 18–33.
- [448] M. Petrić, J. Vandendriessche, C. Marsboom, T. Matheussen, E. Ducheyne, and A. Touhafi, "Autonomous wireless sensor networks in an IPM spatial decision support system," *Computers*, vol. 8, no. 2, p. 43, May 2019.
- [449] A. García-Cedeño, J. C. Guillermo, B. Barzallo, C. Punín, A. Soto, D. Rivas, R. Clotet, and M. Huerta, "PLATANO: Intelligent technological support platform for Azuay province farmers in Ecuador," in *Proc. IEEE Int. Conf. Eng. Veracruz (ICEV)*, vol. 1, Oct. 2019, pp. 1–7.
- [450] D. Brunelli, A. Albanese, D. d'Acunto, and M. Nardello, "Energy neutral machine learning based IoT device for pest detection in precision agriculture," *IEEE Internet Things Mag.*, vol. 2, no. 4, pp. 10–13, Dec. 2019.
- [451] G. Kakamoukas, P. Sariciannidis, G. Livanos, M. Zervakis, D. Rammalis, V. Polychronos, T. Karamitsou, A. Folinias, and N. Tsitsiokas, "A multi-collective, IoT-enabled, adaptive smart farming architecture," in *Proc. IEEE Int. Conf. Imag. Syst. Techn. (IST)*, Dec. 2019, pp. 1–6.
- [452] A. Renehan, B. Rombach, A. Haikl, C. Nolan, W. Lupton, E. Timmons, and R. Bailey, "Low power wireless networks in vineyards," in *Proc. Syst. Inf. Eng. Design Symp. (SIEDS)*, Apr. 2020, pp. 1–6.
- [453] L. Varandas, J. Faria, P. Gaspar, and M. Aguiar, "Low-cost IoT remote sensor mesh for large-scale orchard monitorization," *J. Sensor Actuator Netw.*, vol. 9, no. 3, p. 44, Sep. 2020.

- [454] A. Albanese, D. d'Acunto, and D. Brunelli, "Pest detection for precision agriculture based on IoT machine learning," in *Applications in Electronics Pervading Industry, Environment and Society*. Cham, Switzerland: Springer, 2020, pp. 65–72.
- [455] R. Ross, L. Parsons, B. S. Thai, R. Hall, and M. Kaushik, "An IoT smart rodent bait station system utilizing computer vision," *Sensors*, vol. 20, no. 17, p. 4670, Aug. 2020.
- [456] R. K. Singh, P. P. Puluckul, R. Berkvens, and M. Weyn, "Energy consumption analysis of LPWAN technologies and lifetime estimation for IoT application," *Sensors*, vol. 20, no. 17, p. 4794, 2020.
- [457] R. Anzum and J. Naeem, "Leveraging LoRaWAN technology for smart agricultural monitoring of Malaysian palm oil plantation," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 756, no. 1, May 2021, Art. no. 012052.
- [458] N. J. Chapungo and O. Postolache, "Sensors and communication protocols for precision agriculture," in *Proc. 12th Int. Symp. Adv. Topics Electr. Eng. (ATEE)*, Mar. 2021, pp. 1–6.
- [459] P. Parameswari, N. Rajathi, and K. J. Harshanaa, "LoRa based framework to detect whitefly infestation in coconut trees," in *Proc. Int. Conf. Advancements Electr., Electron., Commun., Comput. Autom. (ICAECA)*, Oct. 2021, pp. 1–5.
- [460] M. Behjati, A. B. M. Noh, H. A. H. Alobaidy, M. A. Zulkifley, R. Nordin, and N. F. Abdullah, "LoRa communications as an enabler for Internet of Drones towards large-scale livestock monitoring in rural farms," *Sensors*, vol. 21, no. 15, p. 5044, Jul. 2021.
- [461] R. K. Singh, M. H. Rahmani, M. Weyn, and R. Berkvens, "Joint communication and sensing: A proof of concept and datasets for greenhouse monitoring using LoRaWAN," *Sensors*, vol. 22, no. 4, p. 1326, Feb. 2022.
- [462] G. Valecse, S. Strazzella, A. Radesca, and L. A. Grieco, "Solarfertigation: Internet of Things architecture for smart agriculture," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2019, pp. 1–6.
- [463] W.-Y. Chung, R.-H. Luo, C.-L. Chen, S. Heythem, C.-F. Chang, C.-C. Po, and Y. Li, "Solar powered monitoring system development for smart farming and Internet of Thing applications," in *Proc. Meeting Abstracts. Electrochem. Soc.*, vol. 28, 2019, pp. 1371–1375.
- [464] S. Sadowski and P. Spachos, "Wireless technologies for smart agricultural monitoring using Internet of Things devices with energy harvesting capabilities," *Comput. Electron. Agricult.*, vol. 172, May 2020, Art. no. 105338.
- [465] B. Wang, X. Zhang, and H. Wu, "A method of ZigBee automatic irrigation," *Int. J. Performability Eng.*, vol. 16, no. 4, p. 639, 2020.
- [466] J. Rodríguez-Robles, Á. Martín, S. Martín, J. A. Ruipérez-Valiente, and M. Castro, "Autonomous sensor network for rural agriculture environments, low cost, and energy self-charge," *Sustainability*, vol. 12, no. 15, p. 5913, Jul. 2020.
- [467] M. V. H. Deokar and R. S. Bindu, "Real-time controlling and monitoring of solar drying and water pumping system using IoT," *Mukt Shabd J.*, vol. 9, no. 5, pp. 1242–1248, 2020.
- [468] R. Morais, J. Mendes, R. Silva, N. Silva, J. J. Sousa, and E. Peres, "A versatile, low-power and low-cost IoT device for field data gathering in precision agriculture practices," *Agriculture*, vol. 11, no. 7, p. 619, Jun. 2021.
- [469] D. Dilmurod, K. Khujamatov, S. Norkobilov, and I. Jamshid, "Features of using the energy-saving leach protocol to control the temperature of stored cotton piles via a wireless network of sensors," *Int. J. Discoveries Innov. Appl. Sci.*, vol. 1, pp. 278–283, 2021.
- [470] Y. H. Reddy, A. Ali, A. Z. Sha, P. Madhulaya, P. Madhulatha, G. Varahi, P. Lakshmisravva, and R. Varaprasad, "Photovoltaic, Internet-of-Things-enabled intelligent agricultural surveillance system," *South Asian Res. J. Eng. Technol.*, vol. 4, no. 5, pp. 78–85, Sep. 2022.
- [471] J. Wu, "Crop growth monitoring system based on agricultural Internet of Things technology," *J. Electr. Comput. Eng.*, vol. 2022, pp. 1–10, May 2022.
- [472] S. K. Gharghan, R. D. Al-Kafaji, S. Q. Mahdi, S. L. Zubaidi, and H. M. Ridha, "Indoor localization for the blind based on the fusion of a metaheuristic algorithm with a neural network using energy-efficient WSN," *Arabian J. Sci. Eng.*, vol. 48, no. 5, pp. 6025–6052, May 2023.
- [473] H. Wang, X. Meng, Z. Chen, X. Zhang, R. Cheng, Y. Zhang, W. Li, W. Song, and Y. Zhang, "A feedback control method for plant factory environment based on photosynthetic rate prediction model," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 108007.
- [474] V. Udutalappally, S. P. Mohanty, V. Pallagani, and V. Khandelwal, "sCrop: A Internet-of-Agro-Things (IoAT) enabled solar powered smart device for automatic plant disease prediction," 2020, *arXiv:2005.06342*.
- [475] J. A. Hassan and B. H. Jasim, "Design and implementation of Internet of Things-based electrical monitoring system," *Bull. Electr. Eng. Informat.*, vol. 10, no. 6, pp. 3052–3063, Dec. 2021.
- [476] E.-T. Bouali, M. R. Abid, E.-M. Boufounas, T. A. Hamed, and D. Benhaddou, "Renewable energy integration into cloud & IoT-based smart agriculture," *IEEE Access*, vol. 10, pp. 1175–1191, 2022.
- [477] D. Loukatos and K. G. Arvanitis, "Multi-modal sensor nodes in experimental scalable agricultural IoT application scenarios," in *IoT-based Intelligent Modelling for Environmental and Ecological Engineering*. Cham, Switzerland: Springer, 2021, pp. 101–128.
- [478] M. Mabon, M. Gautier, B. Vrigneau, M. Le Gentil, and O. Berder, "The smaller the better: Designing solar energy harvesting sensor nodes for long-range monitoring," *Wireless Commun. Mobile Comput.*, vol. 2019, pp. 1–11, Jul. 2019.
- [479] F. Benkhelifa, Z. Qin, and J. McCann, "Minimum throughput maximization in LoRa networks powered by ambient energy harvesting," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–7.
- [480] S. Escobar, F. Rincón, X. del Toro, J. Barba, F. J. Villanueva, M. J. Santofimia, D. Villa, and J. C. López, "The PLATINO experience: A LoRa-based network of energy-harvesting devices for smart farming," in *Proc. 34th Conf. Design Circuits Integr. Syst. (DCIS)*, Nov. 2019, pp. 1–6.
- [481] A. Valente, S. Silva, D. Duarte, F. Cabral Pinto, and S. Soares, "Low-cost LoRaWAN node for agro-intelligence IoT," *Electronics*, vol. 9, no. 6, p. 987, Jun. 2020.
- [482] V.-P. Hoang, M.-H. Nguyen, T. Q. Do, D.-N. Le, and D. D. Bui, "A long range, energy efficient Internet of Things based drought monitoring system," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 10, no. 2, p. 1278, Apr. 2020.
- [483] A. Joseph, "Design of LoRa-wan powered using renewable energy," *Int. J. Multidisciplinary Res. Sci., Eng. Technol. (IJMRSET)*, vol. 1, pp. 9–15, Sep. 2020.
- [484] D. Gao, Q. Sun, B. Hu, and S. Zhang, "A framework for agricultural pest and disease monitoring based on Internet-of-Things and unmanned aerial vehicles," *Sensors*, vol. 20, no. 5, p. 1487, Mar. 2020.
- [485] F. Maita and L. Maiolo, "Low power wireless sensor network for precision agriculture: A battery-less operation scenario," in *Proc. IEEE Int. Workshop Metrol. Agricult. Forestry (MetroAgriFor)*, Nov. 2021, pp. 75–79.
- [486] M. Capuzzo, C. Delgado, A. K. Sultania, J. Famaey, and A. Zanella, "Enabling green IoT: Energy-aware communication protocols for battery-less lorawan devices," in *Proc. 24th Int. ACM Conf. Modeling, Anal. IoT Wireless Mobile Syst.*, 2021, pp. 95–98.
- [487] S. R. J. Ramson, W. D. León-Salas, Z. Brecheisen, E. J. Foster, C. T. Johnston, D. G. Schulze, T. Filley, R. Rahimi, M. J. C. V. Soto, J. A. L. Bolivar, and M. P. Málaga, "A self-powered, real-time, LoRaWAN IoT-based soil health monitoring system," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 9278–9293, Jun. 2021.
- [488] M. A. Ahmed, "LoRa based IoT platform for remote monitoring of large-scale agriculture farms in Chile," *Sensors*, vol. 22, no. 8, p. 2824, Apr. 2022.
- [489] J. Spisic, J. Balen, D. Zagar, and V. Galić, "IoT based network model and sensor node prototype for precision agriculture application," in *Proc. IEEE 8th World Forum Internet of Things (WF-IoT)*, Sep. 2022, pp. 1–8.
- [490] C. Nicolas, B. Naila, and R.-C. Amar, "TinyML smart sensor for energy saving in Internet of Things precision agriculture platform," in *Proc. 13th Int. Conf. Ubiquitous Future Netw. (ICUFN)*, Jul. 2022, pp. 256–259.
- [491] M. R. Cruz, S. Mafra, and F. A. P. D. Figueiredo, "Design, application, and validation of an IoT wireless sensor network based on LoRa for strawberry farming," in *Proc. Anais do XL Simpósio Brasileiro de Telecomunicações E Processamento de Sinais*, 2022, pp. 1–5.
- [492] K. Balakrishna, N. Rakshith, R. Shashank, Y. Nithish, and K. Praveen, "Implementation of LoRa and Bluetooth technology in farming application with performance analysis," *Int. J. Comput. Artif. Intell.*, vol. 4, no. 1, pp. 45–50, Jan. 2023.
- [493] S.-G. Zhu, Z.-G. Cheng, J. Wang, D.-S. Gong, F. Ullah, H.-Y. Tao, H. Zhu, H.-X. Duan, Y.-M. Yang, and Y.-C. Xiong, "Soil phosphorus availability and utilization are mediated by plant facilitation via rhizosphere interactions in an intercropping system," *Eur. J. Agronomy*, vol. 142, Jan. 2023, Art. no. 126679.
- [494] Y. Huang, Y. Huang, J. Hou, L. Wu, P. Christie, and W. Liu, "Microbial community assembly of the hyperaccumulator plant sedum plumbizincicola in two contrasting soil types with three levels of cadmium contamination," *Sci. Total Environ.*, vol. 863, Mar. 2023, Art. no. 160917.

- [495] W. Li, M. Migliavacca, M. Forkel, J. M. C. Denissen, M. Reichstein, H. Yang, G. Duveiller, U. Weber, and R. Orth, "Widespread increasing vegetation sensitivity to soil moisture," *Nature Commun.*, vol. 13, no. 1, p. 3959, Jul. 2022.
- [496] H. T. Khiavi and R. Mostafazadeh, "The spatiotemporal dependencies of terrain indices with soil characteristics in a steep hillslope mountainous area," *Arabian J. Geosci.*, vol. 15, no. 10, pp. 1–18, May 2022.
- [497] P. Kumar, A. Udayakumar, A. A. Kumar, K. S. Kannan, and N. Krishnan, "Multiparameter optimization system with DCNN in precision agriculture for advanced irrigation planning and scheduling based on soil moisture estimation," *Environ. Monitor. Assessment*, vol. 195, no. 1, p. 13, Jan. 2023.
- [498] K. Obaideen, B. A. A. Yousef, M. N. AlMallahi, Y. C. Tan, M. Mahmoud, H. Jaber, and M. Ramadan, "An overview of smart irrigation systems using IoT," *Energy Nexus*, vol. 7, Sep. 2022, Art. no. 100124.
- [499] W. Purcell and T. Neubauer, "Digital twins in agriculture: A state-of-the-art review," *Smart Agricult. Technol.*, vol. 3, Feb. 2023, Art. no. 100094.
- [500] Y. Wang, B.-Q. Liang, H. Bao, Q. Chen, Y.-L. Cao, Y.-Q. He, and L.-Z. Li, "Potential of crop straw incorporation for replacing chemical fertilizer and reducing nutrient loss in Sichuan province, China," *Environ. Pollut.*, vol. 320, Mar. 2023, Art. no. 121034.
- [501] M. Allam, E. Radicetti, V. Quintarelli, V. Petroselli, S. Marinari, and R. Mancinelli, "Influence of organic and mineral fertilizers on soil organic carbon and crop productivity under different tillage systems: A meta-analysis," *Agriculture*, vol. 12, no. 4, p. 464, Mar. 2022.
- [502] S. Kulkarni, S. Budhavale, and V. Langote, "Astute farm monitoring using WSN and AI-A solution for optimally monitoring environmental conditions," in *Intelligent Systems and Applications: Select Proceedings of ICISA 2022*. Cham, Switzerland: Springer, 2023, pp. 61–72.
- [503] A. Rehman, T. Saba, M. Kashif, S. M. Fati, S. A. Bahaj, and H. Chaudhry, "A revisit of Internet of Things technologies for monitoring and control strategies in smart agriculture," *Agronomy*, vol. 12, no. 1, p. 127, Jan. 2022.
- [504] P. N. V. Pessolano, P. De Hegedüs, R. G. Miller, and E. Vicente, "Characterization and functioning of the strawberry innovation system in Salto," *Agrociencia Uruguay*, vol. 25, no. 1, pp. 1–16, Apr. 2021.
- [505] S. Zhao, J. Liu, and S. Wu, "Multiple disease detection method for greenhouse-cultivated strawberry based on multiscale feature fusion faster R-CNN," *Comput. Electron. Agricult.*, vol. 199, Aug. 2022, Art. no. 107176.
- [506] A. Kochhar, N. Kumar, and S. Aneja, "Variance adaptive sporadic sampling for greenhouse monitoring," *Sustain. Comput., Informat. Syst.*, vol. 37, Jan. 2023, Art. no. 100825.
- [507] K. Goel and A. K. Bindal, "Regulated energy harvesting scheme for self-sustaining WSN in precision agriculture," in *Proceedings of Data Analytics and Management*. Cham, Switzerland: Springer, 2022, pp. 367–385.
- [508] H. K. Sharma, "Intelligent IoT for precision agriculture," in *Computational Intelligence for Wireless Sensor Networks*. London, U.K.: Chapman & Hall, 2023, pp. 77–92.
- [509] P. Movva, K. K. Kamarajugadda, and T. R. Polipalli, "An energy aware cluster-based routing and adaptive semi-synchronized MAC for energy harvesting WSN," *Int. J. Commun. Syst.*, vol. 35, no. 12, Aug. 2022, Art. no. e5202.
- [510] A. Kumar, A. Mehbodniya, J. L. Webber, M. A. Haq, K. K. Gola, P. Singh, S. Karupusamy, and M. B. Alazzam, "Optimal cluster head selection for energy efficient wireless sensor network using hybrid competitive swarm optimization and harmony search algorithm," *Sustain. Energy Technol. Assessments*, vol. 52, Aug. 2022, Art. no. 102243.
- [511] S. M. Hussein, J. A. L. Ramos, and A. M. Ashir, "A secure and efficient method to protect communications and energy consumption in IoT wireless sensor networks," *Electronics*, vol. 11, no. 17, p. 2721, Aug. 2022.
- [512] A. S. Sadeq, R. Hassan, H. Sallehudin, A. H. M. Aman, and A. H. Ibrahim, "Conceptual framework for future WSN-MAC protocol to achieve energy consumption enhancement," *Sensors*, vol. 22, no. 6, p. 2129, Mar. 2022.
- [513] M. S. Farooq, O. O. Sohail, A. Abid, and S. Rasheed, "A survey on the role of IoT in agriculture for the implementation of smart livestock environment," *IEEE Access*, vol. 10, pp. 9483–9505, 2022.
- [514] M. Dhanaraju, P. Chenniappan, K. Ramalingam, S. Pazhanivelan, and R. Kaliaperumal, "Smart farming: Internet of Things (IoT)-based sustainable agriculture," *Agriculture*, vol. 12, no. 10, p. 1745, 2022.
- [515] E. Elbasi, N. Mostafa, Z. AlArnaout, A. I. Zreikat, E. Cina, G. Varghese, A. Shdefat, A. E. Topcu, W. Abdelbaki, S. Mathew, and C. Zaki, "Artificial intelligence technology in the agricultural sector: A systematic literature review," *IEEE Access*, vol. 11, pp. 171–202, 2023.
- [516] K. Srinivasan and V. K. Yadav, "An integrated literature review on urban and peri-urban farming: Exploring research themes and future directions," *Sustain. Cities Soc.*, vol. 99, Dec. 2023, Art. no. 104878.
- [517] S. Benhamaid, A. Bouabdallah, and H. Lakhlef, "Recent advances in energy management for green-IoT: An up-to-date and comprehensive survey," *J. Netw. Comput. Appl.*, vol. 198, Feb. 2022, Art. no. 103257.
- [518] A. D. Dhruva, B. Prasad, S. Kamepalli, and S. Kuniseti, "An efficient mechanism using IoT and wireless communication for smart farming," *Mater. Today, Proc.*, vol. 80, pp. 3691–3696, Jan. 2023.
- [519] Y. Edan, G. Adamides, and R. Oberti, "Agriculture automation," in *Springer Handbook of Automation*, 2nd ed. Cham, Switzerland: Springer, 2023, pp. 1055–1078.
- [520] R. R. Lamsal, P. Karthikeyan, P. Otero, and A. Ariza, "Design and implementation of Internet of Things (IoT) platform targeted for smallholder farmers: From Nepal perspective," *Agriculture*, vol. 13, no. 10, p. 1900, Sep. 2023.
- [521] P. M. Rao and B. D. Deebak, "A comprehensive survey on authentication and secure key management in Internet of Things: Challenges, countermeasures, and future directions," *Ad Hoc Netw.*, vol. 146, Jul. 2023, Art. no. 103159.
- [522] M. Javaid, A. Haleem, R. P. Singh, and R. Suman, "Enhancing smart farming through the applications of agriculture 4.0 technologies," *Int. J. Intell. Netw.*, vol. 3, pp. 150–164, Jan. 2022.
- [523] S. Ziesche, S. Agarwal, U. Nagaraju, E. Prestes, and N. Singha, "Role of artificial intelligence in advancing sustainable development goals in the agriculture sector," in *The Ethics of Artificial Intelligence for the Sustainable Development Goals*. Cham, Switzerland: Springer, 2023, pp. 379–397.
- [524] A. Alabdulatif, N. N. Thilakarathne, Z. K. Lawal, K. E. Fahim, and R. Y. Zakari, "Internet of Nano-Things (IoNT): A comprehensive review from architecture to security and privacy challenges," *Sensors*, vol. 23, no. 5, p. 2807, 2023.
- [525] S. Padhy, M. Alowaidi, S. Dash, M. Alshehri, P. P. Malla, S. Routray, and H. Alhumyani, "AgriSecure: A fog computing-based security framework for agriculture 4.0 via blockchain," *Processes*, vol. 11, no. 3, p. 757, Mar. 2023.
- [526] K. Kethineni and P. Gera, "IoT-based privacy-preserving anomaly detection model for smart agriculture," *Systems*, vol. 11, no. 6, p. 304, Jun. 2023.
- [527] M. Amiri-Zarandi, R. A. Dara, E. Duncan, and E. D. G. Fraser, "Big data privacy in smart farming: A review," *Sustainability*, vol. 14, no. 15, p. 9120, Jul. 2022.
- [528] M. A. Ferrag, L. Shu, O. Friha, and X. Yang, "Cyber security intrusion detection for agriculture 4.0: Machine learning-based solutions, datasets, and future directions," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 3, pp. 407–436, Mar. 2022.
- [529] O. Friha, M. A. Ferrag, L. Shu, L. Maglaras, K.-K. R. Choo, and M. Nafaa, "FELIDS: Federated learning-based intrusion detection system for agricultural Internet of Things," *J. Parallel Distrib. Comput.*, vol. 165, pp. 17–31, Jul. 2022.
- [530] G. J. Rosline, P. Rani, and D. Gnana Rajesh, "Comprehensive analysis on security threats prevalent in IoT-based smart farming systems," in *Ubiquitous Intelligent Systems: Proceedings of ICUIS 2021*. Cham, Switzerland: Springer, 2022, pp. 185–194.
- [531] A. N. Alahmadi, S. U. Rehman, H. S. Alhazmi, D. G. Glynn, H. Shoaib, and P. Solé, "Cyber-security threats and side-channel attacks for digital agriculture," *Sensors*, vol. 22, no. 9, p. 3520, May 2022.
- [532] R. A. Ahmed, E. E. Hemdan, W. El-Shafai, Z. A. Ahmed, E. M. El-Rabaie, and F. E. A. El-Samie, "Climate-smart agriculture using intelligent techniques, blockchain and Internet of Things: Concepts, challenges, and opportunities," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 11, Nov. 2022, Art. no. e4607.
- [533] A. Faid, M. Sadik, and E. Sabir, "An agile AI and IoT-augmented smart farming: A cost-effective cognitive weather station," *Agriculture*, vol. 12, no. 1, p. 35, Dec. 2021.
- [534] G. Idoje, T. Dagiuklas, and M. Iqbal, "Survey for smart farming technologies: Challenges and issues," *Comput. Electr. Eng.*, vol. 92, Jun. 2021, Art. no. 107104.

- [535] A. Rehman, I. Abunadi, K. Haseeb, T. Saba, and J. Lloret, "Intelligent and trusted metaheuristic optimization model for reliable agricultural network," *Comput. Standards Interface*, vol. 87, Jan. 2024, Art. no. 103768.
- [536] R. R. Irshad, S. Hussain, I. Hussain, J. A. Nasir, A. Zeb, K. M. Alalayah, A. A. Alattab, A. Yousif, and I. M. Alwayle, "IoT-enabled secure and scalable cloud architecture for multi-user systems: A hybrid post-quantum cryptographic and blockchain-based approach toward a trustworthy cloud computing," *IEEE Access*, vol. 11, pp. 105479–105498, 2023.
- [537] A. V. Shvetsov, S. H. Alsamhi, A. Hawbani, S. Kumar, S. Srivastava, S. Agarwal, N. S. Rajput, A. A. Alammari, and F. M. A. Nashwan, "Federated learning meets intelligence reflection surface in drones for enabling 6G networks: Challenges and opportunities," *IEEE Access*, vol. 11, pp. 130860–130887, 2023.
- [538] P. Majumdar, D. Bhattacharya, S. Mitra, and B. Bhushan, "Application of green IoT in agriculture 4.0 and beyond: Requirements, challenges and research trends in the era of 5G, LPWANs and Internet of UAV Things," *Wireless Pers. Commun.*, vol. 131, no. 3, pp. 1767–1816, Aug. 2023.
- [539] A. El-Ghamry, A. Darwish, and A. E. Hassanien, "An optimized CNN-based intrusion detection system for reducing risks in smart farming," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100709.
- [540] T. H. H. Aldhyani and H. Alkahtani, "Cyber security for detecting distributed denial of service attacks in agriculture 4.0: Deep learning model," *Mathematics*, vol. 11, no. 1, p. 233, Jan. 2023.
- [541] C. Prakash, A. Barthwal, S. Avikal, and G. K. Singh, "FSAS: An IoT-based security system for crop field storage," *Int. J. Distrib. Sensor Netw.*, vol. 2023, pp. 1–13, Jul. 2023.
- [542] F.-J. Ferrández-Pastor, J. Mora-Pascual, and D. Díaz-Lajara, "Agricultural traceability model based on IoT and blockchain: Application in industrial hemp production," *J. Ind. Inf. Integr.*, vol. 29, Sep. 2022, Art. no. 100381.
- [543] R. Chaganti, V. Varadarajan, V. S. Gorantla, T. R. Gadekallu, and V. Ravi, "Blockchain-based cloud-enabled security monitoring using Internet of Things in smart agriculture," *Future Internet*, vol. 14, no. 9, p. 250, Aug. 2022.
- [544] S. Ali, Q. Li, and A. Yousafzai, "Blockchain and federated learning-based intrusion detection approaches for edge-enabled industrial IoT networks: A survey," *Ad Hoc Netw.*, vol. 152, Jan. 2024, Art. no. 103320.
- [545] L. Yogarajan, M. Masukujjaman, M. H. Ali, N. Khalid, L. H. Osman, and S. S. Alam, "Exploring the hype of blockchain adoption in Agri-food supply chain: A systematic literature review," *Agriculture*, vol. 13, no. 6, p. 1173, May 2023.
- [546] O. Friha, M. A. Ferrag, L. Maglaras, and L. Shu, "Digital agriculture security: Aspects, threats, mitigation strategies, and future trends," *IEEE Internet Things Mag.*, vol. 5, no. 3, pp. 82–90, Sep. 2022.
- [547] A. R. D. A. Zanella, E. da Silva, and L. C. P. Albini, "CEIFA: A multi-level anomaly detector for smart farming," *Comput. Electron. Agricult.*, vol. 202, Nov. 2022, Art. no. 107279.
- [548] T. Manoj, K. Makthaya, and V. Narendra, "A trusted IoT data sharing and secure Oracle based access for agricultural production risk management," *Comput. Electron. Agricult.*, vol. 204, Jan. 2023, Art. no. 107544.
- [549] W. Ma, G. Jiang, T. Zhou, and Y. Qu, "Do decaying rural communities have an incentive to maintain large-scale farming? A comparative analysis of farming systems for peri-urban agriculture in China," *J. Cleaner Prod.*, vol. 397, Apr. 2023, Art. no. 136590.
- [550] Y. Zhao, Q. Li, W. Yi, and H. Xiong, "Agricultural IoT data storage optimization and information security method based on blockchain," *Agriculture*, vol. 13, no. 2, p. 274, Jan. 2023.
- [551] S. Lee and J. S. Shin, "A new location verification protocol and blockchain-based drone rental mechanism in smart farming," *Comput. Electron. Agricult.*, vol. 214, Nov. 2023, Art. no. 108267.
- [552] Z. A. Abduljabbar, V. O. Nyangaresi, H. M. Jasim, J. Ma, M. A. Hussain, Z. A. Hussien, and A. J. Y. Aldarwish, "Elliptic curve cryptography-based scheme for secure signaling and data exchanges in precision agriculture," *Sustainability*, vol. 15, no. 13, p. 10264, Jun. 2023.
- [553] T. Feng, R. Xiong, and P. Huan, "Productive use of natural resources in agriculture: The main policy lessons," *Resour. Policy*, vol. 85, Aug. 2023, Art. no. 103793.
- [554] A. Reyana, S. Kautish, P. M. S. Karthik, I. A. Al-Baltah, M. B. Jasser, and A. W. Mohamed, "Accelerating crop yield: Multisensor data fusion and machine learning for agriculture text classification," *IEEE Access*, vol. 11, pp. 20795–20805, 2023.
- [555] K. Banti, M. Louta, and P. Baziana, "Data quality in human-centric sensing-based next-generation IoT systems: A comprehensive survey of models, issues, and challenges," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 2286–2317, 2023.
- [556] M. McCaig, R. Dara, and D. Rezaia, "Farmer-centric design thinking principles for smart farming technologies," *Internet Things*, vol. 23, Oct. 2023, Art. no. 100898.
- [557] M. Amiri-Zarandi, M. H. Fard, S. Yousefinaghani, M. Kaviani, and R. Dara, "A platform approach to smart farm information processing," *Agriculture*, vol. 12, no. 6, p. 838, Jun. 2022.
- [558] A. Rejeb, K. Rejeb, A. Abdollahi, F. Al-Turjman, and H. Treiblmaier, "The interplay between the Internet of Things and agriculture: A bibliometric analysis and research agenda," *Internet Things*, vol. 19, Aug. 2022, Art. no. 100580.
- [559] L. Zhu and F. Li, "Agricultural data sharing and sustainable development of ecosystem based on block chain," *J. Cleaner Prod.*, vol. 315, Sep. 2021, Art. no. 127869.
- [560] A. Z. Bayih, J. Morales, Y. Assabie, and R. A. de By, "Utilization of Internet of Things and wireless sensor networks for sustainable smallholder agriculture," *Sensors*, vol. 22, no. 9, p. 3273, Apr. 2022.
- [561] D. Silva, A. Heideker, I. D. Zyrianoff, J. H. Kleinschmidt, L. Roffia, J.-P. Soininen, and C. A. Kamienski, "A management architecture for IoT smart solutions: Design and implementation," *J. Netw. Syst. Manage.*, vol. 30, no. 2, p. 35, Apr. 2022.
- [562] S. Bera, T. Dey, S. Ghosh, and A. Mukherjee, "Internet of Things and dew computing-based system for smart agriculture," in *Dew Computing: The Sustainable IoT Perspectives*. Cham, Switzerland: Springer, 2023, pp. 289–316.
- [563] P. Saranya and R. Maheswari, "Proof of transaction (PoTx) based traceability system for an agriculture supply chain," *IEEE Access*, vol. 11, pp. 10623–10638, 2023.
- [564] S. Bökle, D. S. Paraforos, D. Reiser, and H. W. Griepentrog, "Conceptual framework of a decentral digital farming system for resilient and safe data management," *Smart Agricult. Technol.*, vol. 2, Dec. 2022, Art. no. 100039.
- [565] R. Chandra, M. Swaminathan, T. Chakraborty, J. Ding, Z. Kapetanovic, P. Kumar, and D. Vasisht, "Democratizing data-driven agriculture using affordable hardware," *IEEE Micro*, vol. 42, no. 1, pp. 69–77, Jan. 2022.
- [566] N. Gilmore, T. Britz, E. Maartensson, C. Orbegoso-Jordan, S. Schroder, and M. Malerba, "Continental-scale assessment of micro-pumped hydro energy storage using agricultural reservoirs," *Appl. Energy*, vol. 349, Nov. 2023, Art. no. 121715.
- [567] A. Poonia, D. Lakshmi, T. Garg, and G. Vishnuvarthanan, "A comprehensive study on smart farming for transforming agriculture through cloud and IoT," in *Convergence of Cloud Computing, AI, and Agricultural Science*. Hershey, PA, USA: IGI Global, 2023, pp. 67–99.
- [568] Y. Yao, X. Li, Y. Cui, J. Wang, and C. Wang, "Energy-efficient routing protocol based on multi-threshold segmentation in wireless sensors networks for precision agriculture," *IEEE Sensors J.*, vol. 22, no. 7, pp. 6216–6231, Apr. 2022.
- [569] S. Hao, Y. Hong, and Y. He, "An energy-efficient routing algorithm based on greedy strategy for energy harvesting wireless sensor networks," *Sensors*, vol. 22, no. 4, p. 1645, Feb. 2022.
- [570] R. Tarife, Y. Nakanishi, Y. Chen, Y. Zhou, N. Estoperez, and A. Tahud, "Optimization of hybrid renewable energy microgrid for rural agricultural area in Southern Philippines," *Energies*, vol. 15, no. 6, p. 2251, Mar. 2022.
- [571] M. Doshi and A. Varghese, "Smart agriculture using renewable energy and ai-powered IoT," in *AI, Edge IoT-Based Smart Agriculture*. Amsterdam, The Netherlands: Elsevier, 2022, pp. 205–225.
- [572] S. K. Jadhav and R. Shreelavaniya, "Energy harvesting systems for agricultural needs," in *Energy Harvesting Trends for Low Power Compact Electron. Devices*. Cham, Switzerland: Springer, 2023, pp. 101–127.
- [573] R. R. Kovvuri, A. Kaushik, and S. Yadav, "Disruptive technologies for smart farming in developing countries: Tomato leaf disease recognition systems based on machine learning," *Electron. J. Inf. Syst. Developing Countries*, vol. 89, no. 6, Nov. 2023, Art. no. e12276.
- [574] J. L. Vilas-Boas, J. J. P. C. Rodrigues, and A. M. Alberti, "Convergence of distributed ledger technologies with digital twins, IoT, and AI for fresh food logistics: Challenges and opportunities," *J. Ind. Inf. Integr.*, vol. 31, Feb. 2023, Art. no. 100393.

- [575] V. S. Narwane, A. Gunasekaran, and B. B. Gardas, "Unlocking adoption challenges of IoT in Indian agricultural and food supply chain," *Smart Agricult. Technol.*, vol. 2, Dec. 2022, Art. no. 100035.
- [576] K. Zkik, A. Belhadi, S. A. Rehman Khan, S. S. Kamble, M. Oudani, and F. E. Touriki, "Exploration of barriers and enablers of blockchain adoption for sustainable performance: Implications for E-enabled agriculture supply chains," *Int. J. Logistics Res. Appl.*, vol. 26, no. 11, pp. 1498–1535, Nov. 2023.
- [577] C. Giagnocavo, M. de Cara-García, M. González, M. Juan, J. I. Marín-Guirao, S. Mehrabi, E. Rodríguez, J. van der Blom, and E. Crisol-Martínez, "Reconnecting farmers with nature through agroecological transitions: Interacting niches and experimentation and the role of agricultural knowledge and innovation systems," *Agriculture*, vol. 12, no. 2, p. 137, Jan. 2022.
- [578] K. Valujeva, E. K. Freed, A. Nipers, J. Jauhiainen, and R. P. O. Schulte, "Pathways for governance opportunities: Social network analysis to create targeted and effective policies for agricultural and environmental development," *J. Environ. Manage.*, vol. 325, Jan. 2023, Art. no. 116563.
- [579] G. K. Akella, S. Wibowo, S. Grandhi, and S. Mubarak, "A systematic review of blockchain technology adoption barriers and enablers for smart and sustainable agriculture," *Big Data Cognit. Comput.*, vol. 7, no. 2, p. 86, May 2023.
- [580] H. B. Mahajan, A. Badarla, and A. A. Junnarkar, "CL-IoT: Cross-layer Internet of Things protocol for intelligent manufacturing of smart farming," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 7, pp. 7777–7791, Jul. 2021.
- [581] P. Antwi-Agyei and L. C. Stringer, "Improving the effectiveness of agricultural extension services in supporting farmers to adapt to climate change: Insights from northeastern Ghana," *Climate Risk Manage.*, vol. 32, 2021, Art. no. 100304.
- [582] N. Chandnani and C. N. Khairnar, "An analysis of architecture, framework, security and challenging aspects for data aggregation and routing techniques in IoT WSNs," *Theor. Comput. Sci.*, vol. 929, pp. 95–113, Sep. 2022.
- [583] M. A. Ali, R. K. Dhanaraj, and A. Nayyar, "A high performance-oriented AI-enabled IoT-based pest detection system using sound analytics in large agricultural field," *Microprocessors Microsyst.*, vol. 103, Nov. 2023, Art. no. 104946.
- [584] A. V. Prabu, G. S. Kumar, S. Rajasoundaran, P. P. Malla, S. Routray, and A. Mukherjee, "Internet of Things-based deeply proficient monitoring and protection system for crop field," *Expert Syst.*, vol. 39, no. 5, Jun. 2022, Art. no. e12876.
- [585] S. Muruganandam, R. Joshi, P. Suresh, N. Balakrishna, K. H. Kishore, and S. V. Manikanthan, "A deep learning based feed forward artificial neural network to predict the K-barriers for intrusion detection using a wireless sensor network," *Meas., Sensors*, vol. 25, Feb. 2023, Art. no. 100613.
- [586] H. D. Praveena, V. Srilakshmi, S. Rajini, R. Kolluri, and M. Manohar, "Balancing module in evolutionary optimization and deep reinforcement learning for multi-path selection in software defined networks," *Phys. Commun.*, vol. 56, Feb. 2023, Art. no. 101956.
- [587] I. Moutsinas, A. Kalkanof, J. Mavridis, V. Zafeiris, F. Oikonomou, G. Tziokas, K. Themelis, K. Kyriakou, C. Theologou, A. Serafeim, A. Apostolaras, P. Maletsika, G. D. Nanos, and T. Korakis, "AgroNIT: Innovating precision agriculture," in *Proc. Global Inf. Infrastruct. Netw. Symp. (GIIS)*, Sep. 2022, pp. 6–12.
- [588] Q. Zeng, Z. Lv, C. Li, Y. Shi, Z. Lin, C. Liu, and G. Song, "FedProLs: Federated learning for IoT perception data prediction," *Appl. Intell.*, vol. 53, no. 3, pp. 3563–3575, Feb. 2023.
- [589] M. Venkatasubramanian, A. H. Lashkari, and S. Hakak, "IoT malware analysis using federated learning: A comprehensive survey," *IEEE Access*, vol. 11, pp. 5004–5018, 2023.
- [590] M. Aggarwal, V. Khullar, N. Goyal, A. Alammari, M. A. Albahar, and A. Singh, "Lightweight federated learning for Rice leaf disease classification using non independent and identically distributed images," *Sustainability*, vol. 15, no. 16, p. 12149, Aug. 2023.
- [591] G. Idoje, T. Dagiklas, and M. Iqbal, "Federated learning: Crop classification in a smart farm decentralised network," *Smart Agricult. Technol.*, vol. 5, Oct. 2023, Art. no. 100277.
- [592] B. B. Sezer, H. Turkmen, and U. Nuriyev, "PPFchain: A novel framework privacy-preserving blockchain-based federated learning method for sensor networks," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100781.
- [593] R. Dara, S. M. H. Fard, and J. Kaur, "Recommendations for ethical and responsible use of artificial intelligence in digital agriculture," *Frontiers Artif. Intell.*, vol. 5, Jul. 2022, Art. no. 884192.
- [594] M. de Benito Fernández, D. L. Martínez, A. González-Briones, P. Chamoso, and E. S. Corchado, "Evaluation of XAI models for interpretation of deep learning techniques' results in automated plant disease diagnosis," in *Proc. Sustain. Smart Cities Territories Int. Conf. Cham, Switzerland: Springer*, 2023, pp. 417–428.
- [595] I. Kök, F. Y. Okay, Ö. Muyanli, and S. Özdemir, "Explainable artificial intelligence (XAI) for Internet of Things: A survey," *IEEE Internet Things J.*, vol. 10, no. 16, pp. 14764–14779, 2023, doi: 10.1109/JIOT.2023.3287678.
- [596] N. Moustafa, N. Koroniotis, M. Keshk, A. Y. Zomaya, and Z. Tari, "Explainable intrusion detection for cyber defences in the Internet of Things: Opportunities and solutions," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 3, pp. 1775–1807, 3rd Quart., 2023.
- [597] M. Ryo, "Explainable artificial intelligence and interpretable machine learning for agricultural data analysis," *Artif. Intell. Agricult.*, vol. 6, pp. 257–265, Jan. 2022.
- [598] M. Bouni, B. Hssina, K. Douzi, and S. Douzi, "Towards an efficient recommender systems in smart agriculture: A deep reinforcement learning approach," *Proc. Comput. Sci.*, vol. 203, pp. 825–830, Jan. 2022.
- [599] Y. Chen, G. Li, X. Zhang, J. Jia, K. Zhou, and C. Wu, "Identifying field and road modes of agricultural machinery based on GNSS recordings: A graph convolutional neural network approach," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107082.
- [600] Y. Chen, G. Li, K. Zhou, and C. Wu, "Field-road operation classification of agricultural machine GNSS trajectories using spatio-temporal neural network," *Agronomy*, vol. 13, no. 5, p. 1415, May 2023.
- [601] P. Li, H. Hao, Z. Zhang, X. Mao, J. Xu, Y. Lv, W. Chen, and D. Ge, "A field study to estimate heavy metal concentrations in a soil-rice system: Application of graph neural networks," *Sci. Total Environ.*, vol. 832, Aug. 2022, Art. no. 155099.
- [602] Y. Lu, D. Chen, E. Olaniyi, and Y. Huang, "Generative adversarial networks (GANs) for image augmentation in agriculture: A systematic review," *Comput. Electron. Agricult.*, vol. 200, Sep. 2022, Art. no. 107208.
- [603] L. F. V. Ferrão, R. Dhakal, R. Dias, D. Tieman, V. Whitaker, M. A. Gore, C. Messina, and M. F. R. Resende, "Machine learning applications to improve flavor and nutritional content of horticultural crops through breeding and genetics," *Current Opinion Biotechnol.*, vol. 83, Oct. 2023, Art. no. 102968.
- [604] Z. Zhang, Q. Gao, L. Liu, and Y. He, "A high-quality Rice leaf disease image data augmentation method based on a dual GAN," *IEEE Access*, vol. 11, pp. 21176–21191, 2023.
- [605] E. Olaniyi, D. Chen, Y. Lu, and Y. Huang, "Generative adversarial networks for image augmentation in agriculture: A systematic review," 2022, *arXiv:2204.04707*.
- [606] L. G. Divyanth, D. S. Guru, P. Soni, R. Machavaram, M. Nadimi, and J. Paliwal, "Image-to-image translation-based data augmentation for improving crop/weed classification models for precision agriculture applications," *Algorithms*, vol. 15, no. 11, p. 401, Oct. 2022.
- [607] R. Zhao, Y. Zhu, and Y. Li, "CLA: A self-supervised contrastive learning method for leaf disease identification with domain adaptation," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 107967.
- [608] Y. Yang, S. Sun, M. Huang, and Q. Zhu, "PEAMATL: A strategy for developing near-infrared spectral prediction models under domain shift using self-supervised transfer learning," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [609] B. Hurst, N. Bellotto, and P. Bosilj, "An assessment of self-supervised learning for data efficient potato instance segmentation," in *Proc. Annu. Conf. Towards Auto. Robotic Syst. Cham, Switzerland: Springer*, 2023, pp. 267–278.
- [610] F. Dou, J. Ye, G. Yuan, Q. Lu, W. Niu, H. Sun, L. Guan, G. Lu, G. Mai, and N. Liu, "Towards artificial general intelligence (Agi) in the Internet of Things (IoT): Opportunities and challenges," 2023, *arXiv:2309.07438*.
- [611] G. Lu, S. Li, G. Mai, J. Sun, D. Zhu, L. Chai, H. Sun, X. Wang, H. Dai, and N. Liu, "Agi for agriculture," 2023, *arXiv:2304.06136*.
- [612] S. K. Jha, "Need for an orchestration platform to unlock the potential of remote sensing data for agriculture," in *Digital Ecosystem for Innovation in Agriculture*. Cham, Switzerland: Springer, 2023, pp. 61–73.
- [613] A. Holzinger, K. Keiblinger, P. Holub, K. Zatloukal, and H. Müller, "AI for life: Trends in artificial intelligence for biotechnology," *New Biotechnol.*, vol. 74, pp. 16–24, May 2023.

- [614] I. Laktionov, L. Rutkowski, O. Vovna, A. Byrski, and M. Kabanets, "A novel approach to intelligent monitoring of gas composition and light mode of greenhouse crop growing zone on the basis of fuzzy modelling and human-in-the-loop techniques," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 106938.
- [615] S. Biswas, "Importance of chat GPT in agriculture: According to chat GPT," *SSRN Electron. J.*, 2023, Art. no. 4405391, doi: 10.2139/ssrn.4405391.
- [616] Y. Wang, "Agricultural products price prediction based on improved RBF neural network model," *Appl. Artif. Intell.*, vol. 37, no. 1, Dec. 2023, Art. no. 2204600.
- [617] A. L. Duguma and X. Bai, "Contribution of Internet of Things (IoT) in improving agricultural systems," *Int. J. Environ. Sci. Technol.*, vol. 20, pp. 1–14, 2023.
- [618] T. Wang, B. Chen, Z. Zhang, H. Li, and M. Zhang, "Applications of machine vision in agricultural robot navigation: A review," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107085.
- [619] A. T. Meshram, A. V. Vanalkar, K. B. Kalambe, and A. M. Badar, "Pesticide spraying robot for precision agriculture: A categorical literature review and future trends," *J. Field Robot.*, vol. 39, no. 2, pp. 153–171, Mar. 2022.
- [620] A. D. Boursianis, M. S. Papadopoulou, P. Diamantoulakis, A. Liopa-Tsakalidi, P. Barouchas, G. Salahas, G. Karagiannidis, S. Wan, and S. K. Goudos, "Internet of Things (IoT) and agricultural unmanned aerial vehicles (UAVs) in smart farming: A comprehensive review," *Internet Things*, vol. 18, May 2022, Art. no. 100187.
- [621] A. Rejeb, A. Abdollahi, K. Rejeb, and H. Treiblmaier, "Drones in agriculture: A review and bibliometric analysis," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107017.
- [622] A. Fascista, "Toward integrated large-scale environmental monitoring using WSN/UAV/crowdsensing: A review of applications, signal processing, and future perspectives," *Sensors*, vol. 22, no. 5, p. 1824, 2022.
- [623] S. Sathiy, C. Antony, and P. K. Ghodke, "Smart agriculture: Emerging data in agriculture: Between opportunity and solution," *Agricult. Syst.*, vol. 195, Jan. 2022, Art. no. 103298.
- [624] S. A. Osinga, D. Paudel, S. A. Mouzakitis, and I. N. Athanasiadis, "Big data in agriculture: Between opportunity and solution," *Agricult. Syst.*, vol. 195, Jan. 2022, Art. no. 103298.
- [625] A. Shrivastava, C. K. Nayak, R. Dilip, S. R. Samal, S. Rout, and S. M. Ashfaq, "Automatic robotic system design and development for vertical hydroponic farming using IoT and big data analysis," *Mater. Today, Proc.*, vol. 80, pp. 3546–3553, Jan. 2023.
- [626] D. Garg and M. Alam, "Smart agriculture: A literature review," *J. Manage. Analytics*, vol. 10, no. 2, pp. 359–415, Apr. 2023.
- [627] T. Alahmad, M. Neményi, and A. Nyéki, "Applying IoT sensors and big data to improve precision crop production: A review," *Agronomy*, vol. 13, no. 10, p. 2603, Oct. 2023.
- [628] R. Sitharthan, M. Rajesh, S. Vimal, S. Kumar, S. Yuvaraj, A. Kumar, J. Raglend, and K. Vengatesan, "A novel autonomous irrigation system for smart agriculture using AI and 6G enabled IoT network," *Microprocessors Microsyst.*, vol. 101, Sep. 2023, Art. no. 104905.
- [629] S. Polymeni, S. Plastras, D. N. Skoutas, G. Kormentzas, and C. Skianis, "The impact of 6G-IoT technologies on the development of agriculture 5.0: a review," *Electronics*, vol. 12, no. 12, p. 2651, Jun. 2023.
- [630] J. Liu, L. Shu, X. Lu, and Y. Liu, "Survey of intelligent agricultural IoT based on 5G," *Electronics*, vol. 12, no. 10, p. 2336, May 2023.
- [631] M. Alhafnawi, H. A. Bany Salameh, A. Masadeh, H. Al-Obiedollah, M. Ayyash, R. El-Khazali, and H. Elgala, "A survey of indoor and outdoor UAV-based target tracking systems: Current status, challenges, technologies, and future directions," *IEEE Access*, vol. 11, pp. 68324–68339, 2023.
- [632] M. A. Kachouei, A. Kaushik, and M. A. Ali, "Internet of Things-enabled food and plant sensors to empower sustainability," *Adv. Intell. Syst.*, vol. 5, no. 12, 2023, Art. no. 2300321. [Online]. Available: <https://doi.org/10.1002/aisy.202300321>
- [633] S. K. Das, F. Benkhelifa, Y. Sun, H. Abumarshoud, Q. H. Abbasi, M. A. Imran, and L. Mohjazi, "Comprehensive review on ML-based RIS-enhanced IoT systems: Basics, research progress and future challenges," *Comput. Netw.*, vol. 224, Apr. 2023, Art. no. 109581.
- [634] N. K. Jadav, T. Rathod, R. Gupta, S. Tanwar, N. Kumar, and A. Alkhayyat, "Blockchain and artificial intelligence-empowered smart agriculture framework for maximizing human life expectancy," *Comput. Electr. Eng.*, vol. 105, Jan. 2023, Art. no. 108486.
- [635] K. Chatterjee and A. Singh, "A blockchain-enabled security framework for smart agriculture," *Comput. Electr. Eng.*, vol. 106, Mar. 2023, Art. no. 108594.
- [636] G. Kaur and M. Bhattacharya, "Intelligent fault diagnosis for AIT-based smart farming applications," *IEEE Sensors J.*, vol. 23, no. 22, pp. 28261–28269, Nov. 2023.
- [637] A. K. Koshariya, D. Kalaiyarasi, A. A. Jovith, T. Sivakami, D. S. Hasan, and S. Boopathi, "Ai-enabled IoT and WSN-integrated smart agriculture system," in *Artificial Intelligence Tools and Technologies for Smart Farming and Agriculture Practices*. Hershey, PA, USA: IGI Global, 2023, pp. 200–218.
- [638] M. M. Hossain, M. A. Rahman, S. Chaki, H. Ahmed, A. Haque, I. Tamanna, S. Lima, M. J. Ferdous, and M. S. Rahman, "Smart-agri: A smart agricultural management with IoT-ML-blockchain integrated framework," *Int. J. Adv. Comput. Sci. Appl.*, vol. 14, no. 7, pp. 1–12, 2023.
- [639] A. Din, M. Y. Ismail, B. Shah, M. Babar, F. Ali, and S. U. Baig, "A deep reinforcement learning-based multi-agent area coverage control for smart agriculture," *Comput. Electr. Eng.*, vol. 101, Jul. 2022, Art. no. 108089.
- [640] N. S. Redhu, Z. Thakur, S. Yashveer, and P. Mor, "Artificial intelligence: A way forward for agricultural sciences," in *Bioinformatics in Agriculture*. Amsterdam, The Netherlands: Elsevier, 2022, pp. 641–668.
- [641] S. Parez, N. Dilshad, N. S. Alghamdi, T. M. Alanazi, and J. W. Lee, "Visual intelligence in precision agriculture: Exploring plant disease detection via efficient vision transformers," *Sensors*, vol. 23, no. 15, p. 6949, Aug. 2023.
- [642] B. Zhao, W. Jin, J. Del Ser, and G. Yang, "ChatAgri: Exploring potentials of ChatGPT on cross-linguistic agricultural text classification," *Neurocomputing*, vol. 557, Nov. 2023, Art. no. 126708.
- [643] B. Swathi, H. Mohan, B. Srujan Reddy, V. Sandeep, T. Vigneshwara, and G. V. Jadhav, "Green WorkLine—An AI based assistant for self-farming," in *Proc. Int. Conf. Sustain. Comput. Data Commun. Syst. (ICSCDS)*, Mar. 2023, pp. 401–406.
- [644] A. Gupta, P. Immadisetty, P. Rajesh, and S. G., "Joint intent classification and slot tagging on agricultural dataset for indic languages," in *Proc. 9th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, vol. 1, Mar. 2023, pp. 288–293.
- [645] M. Javaid, A. Haleem, I. H. Khan, and R. Suman, "Understanding the potential applications of artificial intelligence in agriculture sector," *Adv. Agrochem*, vol. 2, no. 1, pp. 15–30, Mar. 2023.
- [646] M. Gardezi, B. Joshi, D. M. Rizzo, M. Ryan, E. Prutzer, S. Brugler, and A. Dadkhah, "Artificial intelligence in farming: Challenges and opportunities for building trust," *Agronomy J.*, vol. 6, no. 1, pp. 52138–52160, 2023, doi: 10.1002/agj.21353.



MD. NAJMUL MOWLA received the B.Sc. degree from the Department of Electrical and Electronic Engineering, International Islamic University Chittagong, Bangladesh, in 2016, and the first M.Sc. degree from the Department of Electrical and Electronics Engineering, Cukurova University, Adana, Turkey. He is currently pursuing the second M.Sc. as a Research Assistant (under Türkiye Bilimsel ve Teknolojik Araştırma Kurumu, TÜBİTAK) with the Department of Aerospace Engineering, Adana Alparslan Türkeş Science and Technology University, Adana, Turkey. His academic interests revolve around the innovative application of leading technologies, specifically machine learning, deep learning, and computer vision, focusing on optimizing smart urban spaces to enhance the quality of life. His research interests include target smart agriculture, healthcare informatics, and smart grids, utilizing predictive analysis, and advanced visualization techniques to drive progress and innovation in these vital sectors. During the B.Sc. degree, his academic achievements marked by notable contributions, were duly acknowledged and supported through a merit scholarship, from 2013 to 2016. He received the Honour Award by the Turkish Government for the M.Sc. degree.



NEAZMUL MOWLA is currently pursuing the B.Sc. degree with the Department of Computer Science and Engineering, North South University. His academic pursuits and enthusiasm are primarily directed toward artificial intelligence, encompassing machine learning and deep learning. He is deeply engrossed in research about smart agriculture and smart health applications.



A. F. M. SHAHEN SHAH (Senior Member, IEEE) received the B.Sc. degree in electronics and telecommunication engineering from Daffodil International University, Bangladesh, in 2009, the M.Sc. degree in information technology from the University of Dhaka, Bangladesh, in 2011, and the Ph.D. degree in electronics and communication engineering from Yildiz Technical University, Turkey, in 2020. He is currently an Associate Professor with the Department of Electronics and Communication Engineering, Yildiz Technical University. He is the author of a book. He has published many research papers in international conferences and journals. His research interests include wireless communication, artificial intelligence, 6G, and blockchain. He is a Life Member of the Institution of Engineers, Bangladesh (IEB). He has been a TPC member of several IEEE conferences and a regular reviewer for various IEEE journals. For the Ph.D. work, he received a Gold Medal from the 32nd International Invention, Innovation and Technology Exhibition (ITEX 2021). He is currently serving as an Editor for the *Open Transportation Journal* (Bentham) and an Associate Editor for the *Journal of Cyber Security Technology* (Taylor & Francis).



KHALED M. RABIE (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees in electrical and electronic engineering from the University of Manchester, in 2011 and 2015, respectively. He is currently a Reader with the Department of Engineering, Manchester Metropolitan University (MMU), U.K. He has a part in several large-scale industrial projects and has published more than 200 journals and conference articles (mostly IEEE). He is a fellow of the U.K. Higher Education Academy (FHEA) and a fellow of the European Alliance for Innovation (EAI). He regularly serves on the Technical Program Committee (TPC) for several major IEEE conferences, such as GLOBECOM, ICC, and VTC. He has received many awards over the past few years in recognition of the research contributions, including the Best Paper Awards from the 2021 IEEE CITS and the 2015 IEEE ISPLC and the IEEE ACCESS Editor of the Month Award, in August 2019. He is currently serving as an Editor for IEEE COMMUNICATIONS LETTERS, an Editor for *IEEE Internet of Things Magazine*.



THOKOZANI SHONGWE (Senior Member, IEEE) received the B.Eng. degree in electronic engineering from the University of Swaziland, Swaziland, the M.Eng. degree in telecommunications engineering from the University of the Witwatersrand, South Africa, and the D.Eng. degree from the University of Johannesburg, South Africa. He is currently an Associate Professor of telecommunications and the Head of the School of Electrical and Electronic Engineering, University of Johannesburg. His research interests include digital communications, visible light communications (VLC), and security.

...